

Sustainability

Net-Zero Carbon

Circular Economy

Nature Positive

Sustainability by Design

Ethical Considerations

Life Cycle Assessment



CO₂

CO₂



CO₂



CO₂



ESG



ESG





GUIDE: Sustainability

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Preface

While the core mission of the pharmaceutical industry remains the development of innovative therapies that improve and save lives, the industry must also embrace its responsibility to minimize environmental impact and use resources wisely.

This Guide was conceived as a response to the increasing need for a unified, practical framework that supports sustainable practices across the pharmaceutical lifecycle. From early-stage product development to manufacturing, supply chain operations, and beyond, sustainability must be embedded not as an afterthought but as a guiding principle. Recognizing the diversity of organizations within the industry, from small and medium-sized enterprises to global corporations, this Guide offers scalable insights and strategies that can be adapted to various contexts and maturity levels.

Drawing on global and local guidance, existing regulations, and industry best practices, this document fosters a shared understanding of key sustainability concepts and provides actionable direction, serving as a foundational resource to inspire collaboration, informed decision-making, and drive meaningful change.

We invite readers to engage with this Guide as a reference and a catalyst for transformation, toward a future where pharmaceutical innovation and environmental considerations go hand in hand.

Discussion of proprietary technologies is included to convey practical information to the readers. In no way does this Guide endorse or promote the use of these items.

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1 Introduction

The pharmaceutical industry dedicates substantial resources to developing new medicines and treatment options to combat various illnesses and diseases globally. While financial investment in novel therapies provides patients with improved quality of care, the industry also recognizes its responsibility to transition toward more sustainable development, manufacturing, and stewardship of medicines throughout their lifecycle. This commitment includes integrating eco-design, green chemistry and circular economy principles into the lifecycle management of medications and manufacturing processes.

1.1 Purpose and Objectives

While sustainability practices vary across regions of the world, the primary purpose of this Guide is to enable an industry-wide alignment on sustainability, ensuring a unified understanding that fosters commitment to sustainable practices.

The objectives are:

- To provide an overview of key environmental sustainability definitions, forming the basis for developing common sustainability knowledge – *be an informative reference*
- To develop a cohesive industry approach that integrates sustainability into existing pharmaceutical development, operations, quality, and supply chain processes, thereby facilitating a shared vision and practices – *support an integrated approach to sustainability*
- To provide guidance on the management of sustainability programs, including the use of metrics to measure performance – *support the deployment and management oversight of sustainability*

These objectives serve to guide the pharmaceutical industry toward a future where sustainable practices are not just aspirational but are embedded in every lifecycle stage of pharmaceutical development and production.

This Guide is a foundational document presenting key principles and concepts that reinforces sustainable practices in the pharmaceutical industry, based on available global and local guides, including existing regulations when available.

1.2 Scope and Audience

This Guide covers pharmaceutical, biopharmaceutical, vaccine, and medical device industries, collectively referred to as the pharmaceutical industry. It includes a high-level view of sustainability concepts and principles and the supporting interactions between different functions and disciplines. It covers the entire lifecycle of sustainability considerations, from product development through launch and ongoing supply of finished products. This Guide excludes specific, in-depth examples, detailed technical solutions, and direct regulatory guidance.

This Guide is intended for Small and Medium Sized Enterprises (SME) at the start of their sustainability journey. Larger organizations may also use this Guide to support their sustainability strategy.

1.3 Structure of the Guide

This *ISPE Guide: Sustainability*, is presented in the following sections:

- Chapter 1 – Overview of key environmental sustainability concepts and definitions

- Chapter 2 – Environmental Sustainability Enterprise Goals
- Chapter 3 – Environmental Sustainability in Manufacturing Operations
- Chapter 4 – Environmental Sustainability of Pharmaceutical Products
- Chapter 5 – The Alignment of Sustainability, Quality, and Regulatory Aspects
- Chapter 6 – Sustainability within the Supply Chain

1.4 Key Concepts and Terms

The following key concepts are relevant for all chapters in this Guide:

- Product and Environmental Sustainability
- Net-Zero (Greenhouse Gas Emissions)
- Nature Positive
- Sustainability by Design
- Circular Economy
- Ethical Considerations

1.4.1 *Product and Environmental Sustainability*

This Guide uses the following definitions for sustainability concepts relevant to pharmaceuticals as integrated in the wide context of overall sustainability [1]:

- **Environmental Sustainability:** Responsible interaction with the environment to avoid depletion or degradation of natural resources and allow for long-term environmental quality, and support health and well-being. It is also important to consider preservation and enhancement of biodiversity within the environmental scope.
- **Product/Chemical/Material Environmental Sustainability:** The ability of a product, chemical, material to deliver its function without exceeding environmental and ecological boundaries along its entire lifecycle, while providing welfare, socio-economic benefits and minimizing unintended environmental and social impacts.
- **Facility/Operation/Process Environmental Sustainability:** Design of a facility, operation, or process to conserve resources, use renewable energy sources, and deliver its function without exceeding environmental and ecological boundaries.

1.4.2 *Net-Zero (Greenhouse Gas Emissions)*

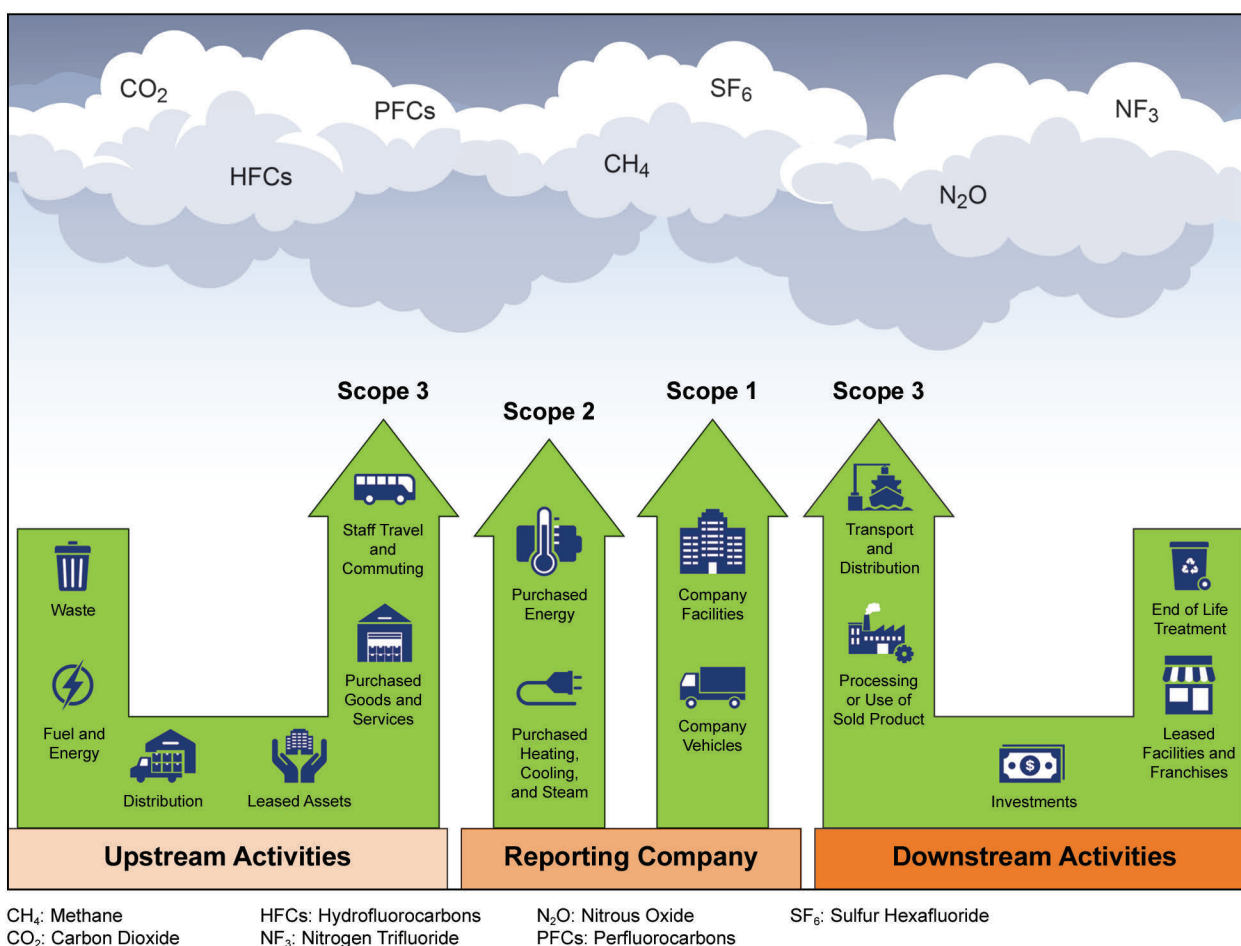
Net-zero refers to Greenhouse Gas (GHG) emissions reduction across the following three types of emissions:

- **Scope 1:** Direct GHG emissions emitted on the premises of a particular company (including company vehicles and mobile equipment). These could be process related (e.g., using CO₂ as an inerting gas) or not (e.g., providing building heating). In the pharmaceutical industry, typically HVAC and the generation of steam are the largest contributors, although special processes such as inhalers might also contribute significantly.

- **Scope 2:** Indirect (owned) GHG emissions related to energy generated off-premise but consumed on company premises. These off-premise generators are typically for electricity, but may also consist of purchased heat, steam, or cooling. Sourcing green electricity is a typical example of Scope 2 reductions.
- **Scope 3:** Indirect (not owned) GHG emissions generated by the upstream and downstream value chain needed to ensure proper operations. These may include emissions associated with the production of manufacturing equipment. There are typically 15 types of Scope 3 emissions ranging from purchased goods and services (e.g., raw materials, clinical services in Research and Development (R&D)), capital goods, distribution, use, and end-of-life for the given product. For pharmaceutical companies, purchased goods and services, followed by the use of sold products and distribution, typically compose a significant share of Scope 3 GHG emissions.

There is a scientific consensus that emissions of a set of so-called GHG, of which CO₂ is the best-known example, contribute to man-made climate change. However, methane (a typical refrigerant cooling agent) and other industrial gases also contribute, in some cases with a significantly higher factor per equivalent relative to CO₂ [2]. The cumulative effect of all GHG-relevant emissions is expressed in CO₂ equivalents (CO₂e), i.e., the equivalent amount of CO₂ that would generate the same effect. All reporting and comparisons are done on the basis of CO₂ equivalents.

Figure 1.1: Greenhouse Gas Emissions



1.4.3 Nature Positive

Nature positive refers to the ambition to halt and reverse the decline of nature by enhancing biodiversity and restoring ecosystems. Pharmaceutical companies should integrate biodiversity considerations into their operations spanning the development of new products, manufacturing, and supply operations, thereby ensuring that their activities contribute positively to the natural environment. Nature positive includes assessing internal and external service providers. Being nature positive, a pharmaceutical company not only strives for overall environmental improvement but also actively ensures the restoration and protection of endangered species and protected ecosystems impacted by its activities and products.

1.4.4 Sustainability by Design

Sustainability by Design (SbD), also referred to as eco-design, involves integrating environmental considerations into the design and development of pharmaceutical products and their packaging. The goal is to minimize the environmental impact throughout the product's lifecycle, from raw material extraction to end-of-life disposal. This includes measuring the effectiveness of SbD initiatives through specific metrics and assessments. Additionally, conducting a Life Cycle Assessment (LCA) (also known as life cycle analysis) helps identify and quantify the environmental impacts at each stage of the product's lifecycle. Key aspects include:

- **Material Selection:** Choosing materials that are renewable, biodegradable, or recyclable for both the product and its packaging
- **Energy Efficiency:** Designing manufacturing processes that consume less energy and generate fewer emissions
- **Waste Reduction:** Developing processes that produce less waste and finding ways to reuse or recycle by-products
- **Product Lifecycle Management:** Ensuring that products are designed for longer use, with considerations for reuse, refurbishment, and recycling
- **Green Chemistry:** Applying principles of green chemistry to reduce the use of hazardous substances and improve the efficiency of chemical reactions
- **Supply Chain Optimization:** Working with suppliers to source sustainable raw materials and reduce the environmental footprint of the supply chain
- **Regulatory Compliance and Beyond:** Ensuring that products meet environmental regulations and standards, and striving to exceed these requirements by adopting best practices in sustainability
- **Stakeholder Engagement:** Collaborating with stakeholders to promote sustainable practices and ensure the proper disposal and recycling of pharmaceutical products

1.4.5 Circular Economy

The circular economy [3] is a resource-conservative approach to tackle the issue of limited planetary resources. It is based on three principles: eliminate waste and pollution, circulate products and materials at their highest value, and regenerate nature.

In the pharmaceutical industry this involves several key practices:

- **Resource Efficiency:** Minimizing waste and optimizing material use
- **Product Lifecycle Management:** Extending product lifecycles through reuse, refurbishment, and recycling

- **Waste Minimization:** Preventing waste generation throughout the supply chain
- **Eco-design:** Designing pharmaceutical products and their packaging to be more sustainable. This can involve synthesis and yield optimization using biodegradable materials and designing for easier disassembly and recycling.
- **Collaboration Across the Value Chain:** Engaging with suppliers, healthcare providers, and waste management companies to create a closed-loop system where materials are continually reused and recycled
- **Regenerative Practices:** Supporting the regeneration of natural systems affected by pharmaceutical production, such as through the restoration of ecosystems impacted by raw material extraction.

Circular economy aligns with net-zero, SbD, and nature positive concepts.

1.4.6 Ethical Considerations

Ethical considerations in sustainability are critical for fostering trust, protecting vulnerable populations, and ensuring long-term business viability in an interconnected world. These considerations ensure that business practices respect human rights, promote fair-trade, and adhere to regulatory requirements. This includes:

- **Human Rights:** Upholding fundamental rights through freely chosen employment, prohibiting child labor and modern slavery, fostering nondiscrimination, and ensuring freedom of association. These principles protect vulnerable populations and promote dignified work environments.
- **Fair-Trade Practices:** Integrating continuous human rights due diligence, identifying and mitigating risks in the supply chain, and embedding clauses into supplier contracts that address modern slavery, fair wages, and working conditions and practices.
- **Regulatory Compliance:** Adhering to evolving global and local regulatory requirements that shape human rights and fair-trade standards.

2 Environmental Sustainability Enterprise Goals

2.1 Introduction

Companies have a duty of service to their shareholders to proactively identify and manage key risks that could impact their operations. These expectations are defined in various national governance codes [4, 5, 6]. Failure to properly consider environmental sustainability will impede wider industry efforts to address climate change and could have an adverse material impact on a company's business. The European Union's Corporate Sustainability Reporting Directive (CSRD) [7], for example, places a focus on double materiality, meaning companies not only need to report on the environmental and social impact of their operations and medicines (spans end-to-end business activities, including the residual impact of used and unused medicines on the environment), but also how climate change could influence the company's future performance. While the EU's CSRD focuses on double materiality, the US Environmental Protection Agency (EPA) emphasizes carbon footprint reduction [8]. Japan's Ministry of Health, Labour and Welfare (MHLW) [9] and China's National Medical Products Administration (NMPA) [10], as well as Australia's National Greenhouse and Energy Reporting (NGER) scheme [11] have their own sustainability standards reflecting regional priorities.

For example, in the future, companies may struggle to meet new and evolving sustainability requirements, risking lost sales to competitors with stronger sustainability performance, or facing financial penalties or fines for noncompliance. Beyond these tangible risks, there are significant reputational concerns. Failing to take sustainability seriously can damage a company's image, making it harder to attract and retain top talent especially as potential new recruits become more value driven in their choice of employer, and loss of public goodwill and government support, which can adversely impact revenue and investment decisions.

The risks apply equally to privately held and publicly traded companies. Alternative sources of supply may be sought if companies and their suppliers cannot meet their government and public expectations. For pharmaceutical companies, together with their suppliers of goods and services, environmental sustainability must be prioritized alongside rather than behind revenue generation.

2.2 Scope and Scale of Corporate Ambition

- The public and investors expect ambitious and clearly defined sustainability goals (e.g., a time horizon of 2030 to be much more engaging than 2050). Tactical plans with measurable milestones must be developed to make meaningful progress toward long-term objectives.

Future market access for medicines will increasingly depend on meeting sustainability requirements before being able to supply medicinal products. These include, for example, environmental risk assessments needed for EU product authorization [12], and performance data accompanying commercial tenders for UK National Health Service.

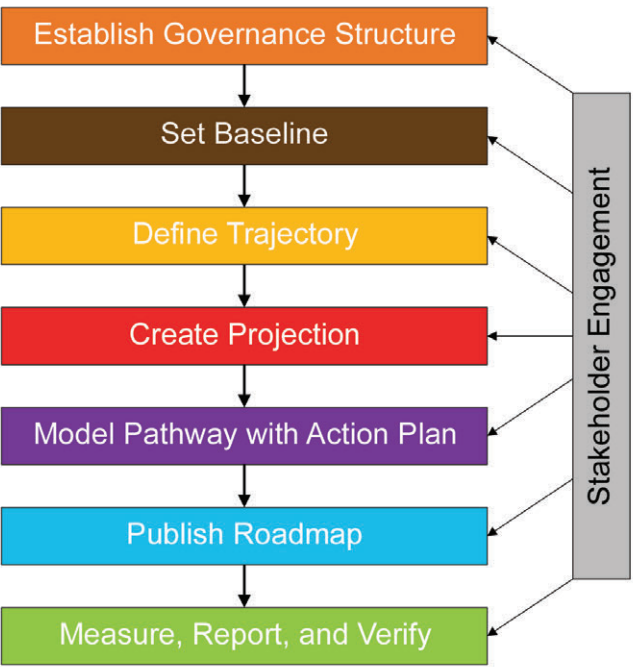
- Environmental sustainability goes beyond continuing conventional energy reduction and waste elimination programs. It encompasses broader concerns such as the impact on water systems (e.g., antimicrobial resistance), air quality, and biodiversity, including the preservation and restoration of ecosystems. However, environmental initiatives must never compromise patient safety or product quality. A robust internal dialogue is essential to define a level of ambition for an organization that is both challenging and achievable in terms of reducing GHG (carbon) emissions, minimizing negative impacts on nature, and enabling sustainable business growth.
- In response, many companies are developing a sustainability strategy supported by dedicated roles and practices. A key component to these strategies is the introduction of a new dedicated Environment, Social and Governance (ESG) scoreboard, which is published as part of, or alongside, the annual shareholder report.

The ESG scoreboard should include both the company's Corporate Carbon Footprint (CCF) and Product Carbon Footprint (PCF) for its highest impact products. The PCF represents the total amount of carbon emissions generated by a product or a service over the different stages of its lifecycle. Organizations are expected to confirm that their sustainability goals are both credible and achievable before they are published. Many companies rely on the Science Based Targets initiative (SBTi) [15] to establish, verify and report on their sustainability targets. Additionally, independent third-party certification schemes are available to support transparency and accountability.

- A foundational sequence of activities for establishing a corporate sustainability program is illustrated in Figure 2.1. Governance structures should be developed with the involvement of senior stakeholders, emphasizing the importance and strategic value of environmental sustainability. Clear individual accountabilities must be assigned to ensure effective stakeholder engagement.
- One of the initial steps should be to conduct an environmental impact assessment of the company's operations, products, and customer services, if this has not already been done. These impact assessments serve as a baseline for measuring future improvements in environmental sustainability. To drive these improvements, a portfolio of projects is required. The sequence and rate of progress of these initiatives will shape the overall trajectory of improvements made. Change pathways for facilities, products, and customer services can be modeled and supported with detailed action plans.
- Performance improvements need to be measured, verified, and reported as discussed in later sections. A high-level summary of the overall approach should be used to create a company-wide sustainability roadmap, complemented by specific roadmaps for facilities and products.

While common metrics for environmental sustainability are emerging, there is no universal set of standard metrics for measuring a company's performance in this area. As a result, companies, therefore, need to define their own set of metrics to measure and report progress based on applicable standards and guidance relevant to their operations and the geographic jurisdictions. Many companies adopt a set of metrics for use over several years, anticipating they will be replaced by more relevant metrics. Metrics may be updated as more effective measurement tools become available or as the focus of the sustainability program shifts to new priority areas that require different indicators to track progress.

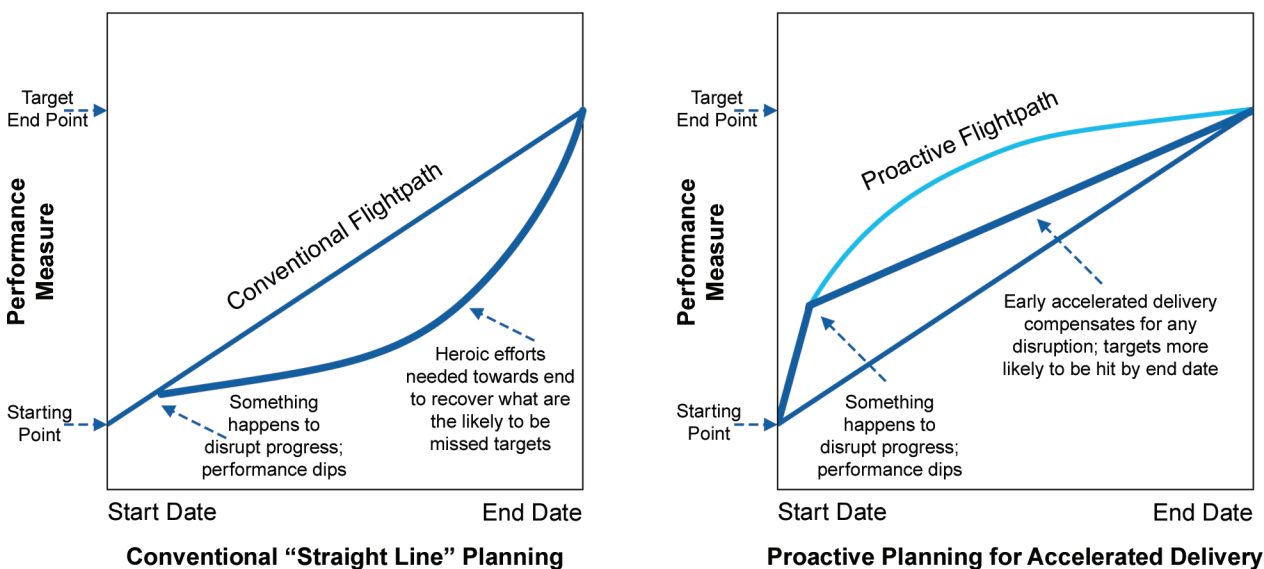
Figure 2.1: Establishing a Corporate Sustainability Program



Company boards and senior leadership teams should anticipate some serious internal executive debate when setting corporate goals [13]. Company commitments made in response to stakeholder pressure may override conventional cost of ownership assessments. It is important to anticipate challenges coming from all levels across an organization. Both strategic and operational challenges need to be addressed to engage the whole organization in the sustainability goal. Being bold requires business model trade offs such as key investments in new facilities or reformulated products, closing or repurposing facilities that can no longer meet expectations, and changing suppliers and service providers where needed. Companies are expected to take responsibility for ensuring their suppliers and service providers are aligned with their goals. They cannot abdicate corporate accountability for environmental sustainability by handing over responsibilities to a third-party.

Trustworthy and responsible organizations must ensure their actions are appropriate and proportionate in both the short and long-term. Pharmaceutical companies and their suppliers will face increasing pressure from the public, shareholders, governments, and employees to act more decisively and with greater urgency. To meet these expectations, interim targets should be established to challenge conventional “straight-line” planning approaches (as shown in Figure 2.2). Such linear models risk missing the end goals when unforeseen disruptions occur unless extraordinary efforts are made to recover. Instead, it is recommended to pursue an early, tangible step-change in performance. This approach, an “inverted hockey stick” performance profile, accelerates progress early on, making it easier to stay on track even if unforeseen events arise (see Figure 2.2).

Figure 2.2: Planning for Accelerated Delivery [13]



This change in mindset will increase pressure on the business. Senior leaders can expect some pushback from managers who will already be consumed with existing business objectives. It is vital to be clear on priorities to mitigate the squeeze on middle management and ensure creative thinking is employed to seek win-win solutions that collectively address improved sustainability performance, quality and safety alongside other business objectives. Creating a line-of-sight from strategic corporate objectives through an individual's annual targets is key to harnessing the good will, focus, and drive of the workforce. Development of a clear decision-making framework will also help establish and maintain consistent and ethical priorities.

Environmental sustainability programs are long-term (ongoing) initiatives and cannot be reported the same way as shorter-term new product developments, factory builds, or major IT deployments. Bridging the gap between the teams that set the goals and those that implement them is crucial for success and team support and participation. A meaningful method of reporting tangible progress should therefore be developed that can be consistently applied over the years ahead.

A quantifiable roadmap with clear milestones set against projects over a timeframe should be developed to understand and prioritize key projects. Progress with implementing these projects and measuring their achieved impact will feed into ESG/CSRD reporting. It is crucial to ensure that capital investment is available for these projects. It is also important to calculate the operating costs of solutions to be implemented and thus ensure that procurement and financial teams see the overall CaPEX and OpEX costs. Decisions, targets and roadmaps are typically agreed at executive leadership level and implemented at site level.

Roadmaps may take many years to realize, for example, moving to 100% renewable energy (generated on and off-site) may not be feasible in the short-term. During the transition, there are alternative options such as Power Purchase Agreements (PPAs), which provide a long-term contract between a company and a renewable energy developer or supplier to buy electricity at a set price. PPAs can be a key tool for companies to help their suppliers to achieve their sustainability goals.

PPAs must be third-party verified to ensure they are viable, ideally in the same geographic region, and address the shortfall in a particular goal, e.g., investment in forestry developments or water preservation. As technologies improve and renewable power becomes more accessible, transitioning to net-zero emissions/net-zero carbon will become more commonplace. The use of interim solutions while awaiting technology improvements is important.

The use of offsets has come under some criticism because it has been perceived as an excuse for not doing more within the business. Therefore, before deploying offsets, companies should be able to demonstrate that they did as much as reasonably practical to reduce their carbon footprint by addressing parameters under their control. Additionally, some offset projects may take many years to come to fruition before a company can claim credits for benefits realized. For example, new woodland or meadows may need to mature before they can realize their full potential for carbon reduction, and then these ecosystems must be maintained over the long-term to secure and preserve their carbon reduction potential as ongoing benefits.

Once the company's strategic goals and the resulting financial implications are understood and agreed upon, a transitional program with supporting governance and reporting can be put in place. The financial implications of the strategy may include cost of investment as well as potential operational savings (through reduced waste, for example). The progress of the strategy may be followed using a scorecard.

A suitable scorecard comprising a mix of both leading and lagging measures should be developed to show progress against strategic objectives. Senior leaders may regularly review progress with company boards being updated at least annually, and more frequently if there are decision points or escalation items requiring their attention.

Companies should consider having external verification of progress on their environmental sustainability goals rather than relying on internal reporting to avoid claims of "greenwashing" and the damage that this can do to their corporate reputation. Independent certifications and statements of assurance can be used to provide evidence of company standards. External lobby groups and industry benchmarking organizations such as Dow Jones Sustainability Index [90] and Sustainalytics' ESG [91] ratings can use these and other information released by the company to assess progress. Key milestone achievements can also be shared with the public as part of a wider communication campaign, taking care to ensure there is a steady stream of meaningful milestones that the public can relate to which will build trust in the company's commitment and delivery.

2.3 Materiality

Double materiality is a concept that is crucial for sustainability reporting, particularly in industries such as pharmaceuticals where the impact on both society and the environment is significant. It consists of two dimensions: impact materiality and financial materiality.

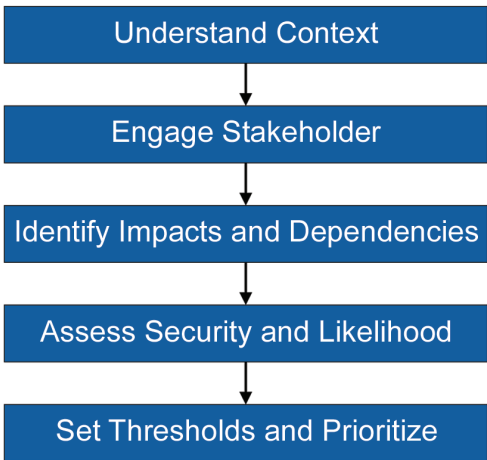
- **Impact Materiality:** This dimension focuses on the actual or potential effects of a company's activities on people and the environment. For the pharmaceutical industry, this could involve the impact of drug manufacturing processes on local water supplies, the ethical implications of clinical trials, or the accessibility of essential medicines to underserved populations. It considers both positive and negative impacts, whether they are directly caused by the company or linked through its business relationships, such as suppliers or distributors. The severity of these impacts is assessed based on their scale, scope, and whether they can be remedied.
- **Financial Materiality:** This dimension looks at how sustainability matters can affect a company's financial performance. In the pharmaceutical industry, this could mean evaluating how environmental regulations, public health policies, or social expectations might influence the company's costs, revenues, and overall financial health. For example, a company might face financial risks from stricter environmental laws that increase manufacturing costs or opportunities for developing drugs that address emerging health issues.

In essence, double materiality requires pharmaceutical companies to consider not only how their operations impact the world but also how external sustainability factors can influence their financial success. This dual perspective ensures a comprehensive approach to sustainability that aligns with both ethical responsibilities and business interests.

2.3.1 Steps to Create a Double Materiality Assessment

The process for creating a double materiality assessment is illustrated in Figure 2.3.

Figure 2.3: Process for Creating a Double Materiality Assessment



2.3.1.1 Understanding the Context

- **Activities and Business Relationships:** Map out the company's operations, including R&D, manufacturing, distribution, and sales. Identify key business relationships, such as suppliers of raw materials, contract manufacturers, and healthcare providers.
- **Sustainability Context:** Analyze the broader sustainability landscape, including regulatory requirements, industry standards, and societal expectations. This might involve reviewing guidelines from bodies such as the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH) [92] or the World Health Organization (WHO) [93].

2.3.1.2 Engaging the Stakeholders

- **Internal Stakeholders:** Consult employees, management, and board members to understand internal perspectives on sustainability.
- **External Stakeholders:** Engage with patients, healthcare professionals, regulators, and local communities to gather insights on how the company's operations impact them. For example, feedback from clinical trial participants or communities near manufacturing plants can be invaluable.

2.3.1.3 Identifying Impacts and Dependencies

- **Impact Materiality:** Identify actual and potential impacts on people and the environment. This could include the environmental footprint of manufacturing processes, the ethical considerations of clinical trials, and the accessibility of medicines.
- **Financial Materiality:** Determine how these sustainability matters could affect the company's financial performance. For example, assess the risks of noncompliance with environmental regulations or the financial benefits of developing drugs for emerging health threats.

2.3.1.4 Assessing Severity and Likelihood

- **Severity:** Evaluate the scale, scope, and irremediable nature of identified impacts. For example, consider the long-term environmental damage from pharmaceutical waste or the widespread health benefits of a new vaccine.
- **Likelihood:** Estimate the probability of these impacts occurring and their potential financial effects. Use scenario analysis and forecasting to understand possible future outcomes.

2.3.1.5 Setting Thresholds and Prioritizing

- **Materiality Thresholds:** Establish thresholds to determine which impacts and financial risks are material enough to report. This involves deciding what level of impact or financial effect warrants inclusion in sustainability statements.
- **Prioritization:** Prioritize the most significant impacts and risks to focus on reporting and strategic planning.

2.3.2 Utilizing the Double Materiality Assessment

2.3.2.1 Reporting

- **Sustainability Statements:** Integrate the findings into the company's sustainability statements, ensuring they meet the requirements of relevant standards such as the European Sustainability Reporting Standards (ESRS) [26].
- **Transparency:** Provide clear, balanced, and detailed disclosures on both impact and financial materiality, ensuring stakeholders understand the company's sustainability performance and future risks.

2.3.2.2 Strategic Planning

- **Risk Management:** Develop strategies to mitigate identified risks, such as investing in cleaner technologies or enhancing supply chain transparency.
- **Opportunities:** Leverage opportunities identified through the assessment, such as entering new markets with high demand for sustainable healthcare solutions.

2.3.2.3 Continual Improvement

- **Monitoring and Evaluation:** Implement metrics and targets to track progress on sustainability goals. Regularly review and update the materiality assessment to reflect changing conditions and new information.
- **Stakeholder Engagement:** Maintain ongoing dialogue with stakeholders to refine understanding of impacts and financial implications, ensuring the company remains responsive and responsible.

2.4 Setting Strategic Objectives

2.4.1 Net-Zero Carbon

Net-zero carbon is one of the 17 United Nations (UN) sustainable development goals [14], companies should establish science-based carbon reduction programs that are verifiable as part of a public commitment to performance targets.

There are a variety of definitions for carbon-neutral and net-zero carbon. This Guide uses the SBTi framework definition as this is the most widely used and accepted definition across the industry sector [15]. Within this framework, at least 90% reduction of Scope 1, 2, and 3 each and compensation measures for the remaining emissions (max. 10% allowed) are required to be considered carbon net-zero [16].

While net-zero targets are becoming more common, line-by-line reduction plans to reach net-zero are often still considered to require a leap of faith, where alternative carbon-free technology is not yet commercially available at industrial scale. However, new technology and innovation in this area is progressing at a pace with viable economic pricing if evolving equivalent carbon price increases are taken into account. Chapter 3 explores the available technology in more detail.

Key Considerations for the Pharmaceutical Industry

- **Setting Strategic Objectives:** Strategic planning to integrate sustainability into the product development process to guide, facilitate, and accelerate the implementation of environmental sustainability.
- **Product Lifecycle:** From R&D to the end-of-life disposal of pharmaceutical products, every stage must be evaluated for its carbon footprint. This includes considering the emissions from clinical trials, production, packaging, and disposal of unused medicines.
- **R&D Activities:** During the R&D phase, consideration should be given to the positive benefits of using green chemistry and evaluating certain ingredients and manufacturing processes to be implemented during later commercial production.
- **Manufacturing Processes:** Pharmaceutical manufacturing often involves energy-intensive processes. Achieving net-zero requires significant improvements in energy efficiency, adoption of renewable energy sources, and possibly redesigning manufacturing processes to minimize emissions.
- **Innovation and Technology:** Investing in new technologies such as green chemistry, which minimizes the use and generation of hazardous substances, can contribute significantly to achieving net-zero targets. Additionally, developing biodegradable or recyclable packaging can help reduce the carbon footprint.
- **Regulatory Compliance and Reputation:** Meeting net-zero targets aligns with global regulatory trends and enhances the company's reputation among stakeholders, including investors, customers, and regulatory bodies. It demonstrates a commitment to sustainability and can lead to competitive advantages.

- **Health and Environment:** Reducing emissions contributes to better environmental health, which is intrinsically linked to public health. As a sector dedicated to improving health outcomes, the pharmaceutical industry has a vested interest in mitigating climate-change impacts.
- **Supply Chain:** The pharmaceutical supply chain is complex, involving raw material extraction, transportation, and optimized distribution channels. Cold-chain requirements for some products add further complexity. Each step needs to be optimized to reduce emissions. Collaborating with suppliers to ensure they also follow sustainable practices is crucial.

There are a variety of implementation strategies such as:

- Upgrading facilities with more energy-efficient equipment and processes as part of new or existing technology investment plans
- Transitioning to renewable energy sources for manufacturing and operations either as individual corporate entities or as part of collective purchasing agreements (e.g., PPA for renewable energy are being made available to pharmaceutical companies [17])
- Implementing sustainable practices on key topics across the supply chain, including logistics and waste management

The Pharmaceutical Supply Chain Initiatives (PSCI) is one example of pharmaceutical companies working together on key topic areas, such as their decarbonization playbook [18]. These and other key considerations and implementation strategies are discussed in more detail in this Guide and other supporting ISPE Guides [94].

2.4.2 *Nature Positive*

2.4.2.1 *Biodiversity and Nature Protection*

Biodiversity protection and nature restoration programs are key components of any environmental sustainability program. In the past, many companies, including those in the pharmaceutical sector, primarily focused on developing climate strategies to address their role in reducing GHG emissions and mitigating global warming. However, it has become evident that focusing solely on climate change is not sufficient. The degradation of biodiversity and ecosystems directly threatens the stability of supply chains and public health, which are core priorities for pharmaceutical companies. Efforts to protect biodiversity, restore ecosystems and manage natural resources sustainably are just as important as carbon reduction. The UK's Biodiversity Net Gain (BNG) [95] provides a recognized approach to development, or land management, that aims to leave the natural environment in a measurably better state than it was beforehand. The UK's BNG is complemented by the US Endangered Species Act [19], Brazil's Amazon conservation efforts [20], and India's National Biodiversity Authority [21].

2.4.2.2 *Impact of Nature-Derived Sourcing*

Pharmaceutical companies are deeply reliant on ecosystems for plant and animal-based ingredients when developing new medicines. It has been reported that approximately 80% of registered medicines either come from or have been inspired by the natural world [22]. The manufacturing of pharmaceutical products may also impact biodiversity. Some products rely on certain ingredients from rare and endangered species. Other products have an adverse impact on the environment, such as damaging protected habitats to source ingredients or using propellants that damage the ozone layer. Production waste and use of water resources are other considerations. The reformulation of pharmaceutical products should be considered to examine the possibility of using alternative ingredients and formulations that are more acceptable. Addressing such challenges can take many years to resolve, especially if products need new regulatory registrations. A transition path to ensure continuity of supply for essential medicines is needed so that patient medical care is not adversely impacted.

2.4.2.3 *Impact of Facility Design and Waste Management Design*

Maintaining Good Manufacturing Practice (GMP) standards, especially to reduce cross-contamination risks in pharmaceutical manufacturing, is also crucial for patient safety and product quality. Internal operations of buildings must not be adversely impacted by their surrounding natural environment (e.g., airborne contamination and animals such as insects and rodents). On the other hand, the design of new facilities can have unintentional impacts on the local environment. Extensive hard surfacing and limited landscaping can have a negative impact on the natural environment surrounding the site. Waste by-products from manufacturing may require landfill disposal or incineration, both of which impact the environment.

2.4.2.4 *Packaging Design*

Excessive packaging requires extra wood pulp or the use of plastics. Single-use equipment in laboratories and disposal of sterile parts can create additional biohazards. Therefore, it is necessary to balance the need for GMP standards with environmental considerations, both prospectively when constructing new buildings and retrospectively for existing facilities.

2.4.2.5 *Use of Green Chemistry, Biodiversity Friendly Landscaping and Runoff Water Mitigation*

Companies should adopt sustainable practices such as green chemistry to lessen environmental impact. Features including green walls, rain gardens, and permeable pavements can reduce runoff and support biodiversity; however, as these measures are adopted, impact to the risk of insects getting into controlled manufacturing and packing buildings should be considered. Landscaping and wildlife corridors can enhance ecological connectivity and site biodiversity. Optimizing site design with sustainable elements allows GMP compliance while safeguarding the environment. This strategy promotes sustainable development, ensuring operational efficiency and environmental stewardship.

2.4.3 *Regulatory Compliance*

New regulations and initiatives to consult depending on the geographic region include:

- **Taskforce on Nature-related Financial Disclosures (TNFD) [23]:** This taskforce makes recommendations providing companies and financial institutions of all sizes with a risk management and disclosure framework to identify, assess, manage and, where appropriate, disclose nature-related issues.
- **Global Reporting Initiative (GRI) 101:Biodiversity 2024 [24]:** This standard contains disclosures for organizations to report information about their biodiversity-related impacts, and how they manage these impacts.
- **CSRD [25]:** The EU requires companies to report on the impact of corporate activities on the environment and society and requires the audit (assurance) of reported information.
- **Biodiversity and Ecosystems (ESRS E4) [26]:** A standard under the ESRS that focuses on how companies impact biodiversity and ecosystems. It sets out expectations for companies to disclose their impacts on biodiversity, both positive and negative, across their operations and value chains.

The expectations described in these documents underline that both governments and investors are increasingly requiring biodiversity to be a priority in any business strategy. Integrating biodiversity considerations into business strategies and reporting frameworks can demonstrate a company's commitment to sustainable practices and contribute to a more transparent and accountable approach to biodiversity management.

Implementation strategies should consider an appropriate selection of activities relevant and proportionate to the impact the pharmaceutical company has on biodiversity. Some examples include but are not limited to:

- Enhancing **biodiversity conservation** by setting up biodiversity assessment and action plans at their facilities. This includes creating green spaces, restoring natural habitats, and integrating biodiversity considerations into the facility management practices.
- Ensuring **sustainable sourcing of materials in the value chains**, and regarding certain material that has been positively characterized as being hazardous to the environment, even avoiding such material through process efficiencies and synthetic alternatives. Therefore, the following hierarchy is applied: avoid, reduce, and practice nature derived sourcing.
- Implementing strategies to **reduce, reuse, and recycle waste and materials** such as chemical by-products, packaging, and plastics. This can include reducing packaging waste by redesigning packaging with minimal environmental impact and responsible product end-of-life management (e.g., encouraging take-back programs).
- **Tackling Antimicrobial Resistance (AMR)**: This may include better control over the disposal and handling of antibiotics.

Engaging with a biodiversity consultant can support a company define the baseline biodiversity of a site, explore design options to enhance biodiversity, quantify impacts, calculate biodiversity credits, evaluate improvement ambitions, and introduce biodiversity quantification in stakeholder interactions. This helps identify the efforts that are most needed in the areas of management, mitigation, or wildlife restoration and track the progress of these actions.

2.5 The Role of Corporate Governance

Overall governance should be kept as simple as possible. Given the significance of environmental sustainability, it is appropriate to have a dedicated high-level forum to provide strategic direction and to oversee implementation and delivery of targets. The seniority of the appointed chair of the high-level forum will indicate how seriously the company is taking environmental sustainability (e.g., having a senior leader to chair the group sets a clear tone from the top of an organization).

Representatives from R&D, manufacturing and supply, and commercial should be included in the top-level forum, along with the main functions needed for delivery such as the sustainability group, engineering, procurement, legal, and corporate communications. The quality function must also be engaged where sustainability changes impact product registrations, analytical testing, manufacturing processes, etc., to ensure that the quality, safety, and efficacy of medicines and devices are not compromised in any way. Care must be taken to ensure quality is not used as a change barrier to avoid sustainability improvements where there is no impact on product quality. Stakeholders need to be fully informed so that they can have robust conversations and agree on the best solution in what are sometimes difficult and challenging situations.

Responsibilities for each member of the governance team should be assigned. Decision-making expectations and meeting cadence should be clear. More frequent governance meetings should be considered when initially setting up the transformation program and its supporting activities compared to later routine oversight of established work items.

Existing company governance structures and processes can be used to support a new high-level governance, assuming these meetings give sufficient priority and agenda time to environmental sustainability. Amendments will be needed to the terms of reference and membership of governance meetings and new dedicated oversight is required where existing governance cannot be leveraged.

Financial impact assessments that test various business scenarios should be refreshed each year, with their output linked into routine business planning. The first intent should be to avoid environmental impacts (e.g., applying new technology or perhaps simplifying a process to remove a problematic step). Consideration can then be given to reducing the environmental impacts that remain. For example, a company could consider dramatically improving internal circularity to reuse what might otherwise be waste materials.

Another example involves developing symbiotic relationships with other companies for them to use what might otherwise be waste (a “circular economy”), e.g., reusing preheated water between neighboring firms or enabling a critical mass of plastic packaging materials to be collected for reprocessing. Firms co-located in a shared building or on a shared campus could work together to reduce the environmental impact of communal energy streams. An example of this in practice is the Kalundborg Symbiosis, a public-private partnership between pharmaceutical and other companies in Denmark in which they proactively design their businesses operations to exchange material, water, and energy streams to reduce their expenses as well as their collective environmental impact [27].

A repeating cycle of risk scenarios can be spread over several years – for example, key products followed by internal operations and then external supply chain – to build up a comprehensive picture that can be shared with the company board. Guidance with supporting templates and tools has been issued by both the Task Force on Climate-Related Financial Disclosures (TCFD) [28] and the TNFD [29]. These assessments provide a vital link that aggregates various company risk mitigation activities into a consolidated report.

2.6 Program Management

A holistic and comprehensive program plan needs to be developed to coordinate the various workstreams that span the company [30]. Each workstream may contribute to more than one of the company’s sustainability goals. Detailed plans for each workstream need to be developed with critical paths to capture the rate-determining activities that must be successfully completed. Potential bottlenecks between parts of the organization that should work together will need to be addressed. Organizational silos that impede progress should not be allowed. Toward this goal, individual workstream plans should be aligned with business plans, as well as with interdependencies identified for proactive management. Roles need to be defined to make sure that the split of responsibilities between the central transformation team and other functions and local business locations is clear. Agreements on roles and responsibilities between central and business governance need to be unambiguous to avoid a silo mentality undermining collective delivery.

Waterfall charts are recommended to identify a series of improvement opportunities for each end goal (e.g., reduction in carbon, wastewater reduction, improved air quality, increased biodiversity index). The charts can be used to prioritize opportunities to be implemented within the workstreams and to give confidence that goals will be achieved. Implementation timelines for each goal can then be developed for conventional project management.

Supporting Key Performance Indicators (KPI) dashboards should be developed with lagging and leading indicators to track current achieved performance and prospective performance improvements, respectively. Care must be taken to avoid the mix of workstream plans and waterfall charts becoming too complex, as this impacts the ability to effectively maintain them. It is important to be able to drive delivery and maintain momentum of the overall program.

The central transformation team should also put into place effective risk management processes that acknowledge and leverage local processes and procedures. Introducing a rigid central standard that overrides local practices can cause confusion and lead to a lack of ownership. Guidance on risk assessment should be provided to promote consistent understanding of the significance of risks. Risk logs should be maintained to track progress with risk mitigation. Risk mitigation actions need to be timely and proportionate to the characteristics of the risk posed, with local risks managed according to local practices.

In order to avoid emerging challenges becoming problematic, timely interventions are needed whenever warning signals for these are identified. Identified problems should receive an effective root cause analysis with on-time closure of remedial actions reviewed by appropriate governance. Solutions for thematic issues and problems should

be shared across the business, recognizing that global plans require approval by high-level governance and should be tracked to completion by the central transformation team. Escalation items should only be referred to high-level governance if they cannot be resolved at a local level.

Communicating keynote successes helps encourage engagement and support. It is important to select meaningful news items rather than rely on KPI metrics. Stories that are easily shared are best for describing a tangible achievement: for example, zero to landfill for a region or globally, formulations changed to remove impact on endangered species or habitats, external awards from accredited authorities such as Carbon Disclosure Project certification [96].

Companies should ensure transparent reporting against established public standards, such as the Greenhouse Gases (GHG) Protocol Corporate Standard [97], which categorizes carbon emissions associated with a CCF, and high-visibility metrics such as the Product Carbon Footprint (PCF). Targets and achievements for both CCF and PCF, together with progress against any nature goals, should be published for full transparency [31].

Further guidance on program management can be found in separate ISPE Guides and other internationally recognized program management standards [32, 33, 34].

2.7 Monitoring and Reporting on Sustainability Performance

Monitoring and reporting on sustainability performance are critical components of a pharmaceutical company's strategy to ensure it meets both its ethical responsibilities and business objectives. Both net-zero carbon and nature positive targets should be monitored and reported.

Relevant compliance requirements and sustainability initiatives will depend on the geographic region in scope. International standards include:

- ISO 14001 Environmental management systems – Requirements with guidance for use [98]
- ISO 26000 Guidance on social responsibility [99]
- GRI – Global Reporting Initiative [24]
- CDP – Carbon Disclosure Project [100]
- GHG – Greenhouse Gas Protocol [31]
- EcoVadis toolset for sustainability risk management and compliance across supply chains [101]

In addition, there are applicable legal requirements such as:

- EU Directive 2022/2464 [7]
- EU Directive 2004/109/EC [25]
- EU Directive 2006/43/EC [102]
- EU Directive 2013/34/EU setting out expectations for sustainability reporting [103]

Other national regulations are relevant in jurisdictions outside of the EU.

Measuring the impact of a nature positive strategy can be particularly challenging given the lack of standardized metrics. The Science Based Target Network (SBTN) provides various example metrics and methods to measure performance [15]. In order to be successful, nature positive initiatives also require biodiversity improvements to be maintained over the long-term. It is important that activities for monitoring and reporting company initiatives are not paralyzed by the lack of defined metrics. If metrics have not been defined, then reporting should be focused on available and relevant information describing what has been achieved (e.g., narratives on what has been achieved). In the early years of a sustainability program, it may only be possible to report on activity milestones rather than on baseline performance improvements.

It is important to agree on who is responsible for the collection and management of supporting data. There are no specific requirements stating which departments must own sustainability data. Consequently, the division of responsibilities for sustainability data may vary between different organizations.

2.7.1 Steps to Monitor Sustainability Performance

2.7.1.1 Establishing Metrics and Targets

Relevant Metrics: Identify qualitative and quantitative indicators that align with the company's sustainability goals. For example, metrics could include energy usage, carbon emissions, water usage, waste generation, and use of substances of concern.

SMART Targets: Set Specific, Measurable, Achievable, Relevant, and Time-bound (SMART) targets for each metric. Science-Based Targets (SBTs) are specific types of SMART targets that align a company's goals with the latest climate science to meet the objectives of the Paris Agreement [104], which aims to limit global warming to well below 2°C above pre-industrial levels and pursue efforts to limit warming to 1.5°C. Another metric for nature positive impact is Mean Species Abundance (MSA), which measures species abundance at a location today, compared to the original species abundance in untouched nature.

2.7.1.2 Data Collection and Management

Data Sources: Utilize various data sources such as internal reports, third-party audits, and stakeholder feedback. For example, data on environmental impacts might come from manufacturing plants, while supplier data could be gathered from outreach to suppliers. The quality and reliability of any data used should be verified.

Data Volume: The volume of data can be enormous and challenging to work with. Setting clear frameworks and dividing data into related batches are essential steps in managing and analyzing large volumes of data effectively. By establishing these frameworks early on, companies can prioritize relevant data, make informed decisions, and optimize their operations.

Technology and Tools: Implement robust data management systems and software to collect, store, and analyze sustainability data. Technologies such as the Internet of Things (IoT) sensors can provide real-time data on environmental metrics, while digital platforms can facilitate stakeholder surveys.

Existing Systems: Maintaining, optimizing, and replacing existing systems pose unique challenges compared to constructing new ones, emphasizing the importance of the management strategy. Existing systems require considerations such as infrastructure compatibility, operational continuity, and seamless technology integration. Leveraging tools tailored to collect real-time performance data from existing systems is essential. An integrated Operating System (OS) platform can utilize this data to pinpoint areas for optimization, improvements, and guide targeted replacements when necessary. By creating a roadmap aligning capital investment, operational savings, and emission reductions with sustainability objectives, organizations can strategically enhance their environmental performance. This process not only optimizes operations but also showcases a strong dedication to environmental responsibility and the journey toward a net-zero future.

2.7.1.3 Regular Monitoring

Continual Tracking: Regularly track performance against the established metrics and targets. This might involve monthly or quarterly reviews of KPIs to ensure the company stays on track.

Internal Audits: Conduct internal audits to verify the accuracy and reliability of sustainability data. This helps in identifying any discrepancies and areas needing improvement.

2.7.2 Reporting on Sustainability Performance

In the context of environmental sustainability, pharmaceutical companies should prepare detailed sustainability statements as part of their annual management report. These statements need to comprehensively cover all material impacts, risks, and opportunities identified through the double materiality assessment. Ensuring regulatory compliance is crucial, and the sustainability statements should adhere to relevant standards and regulations, such as the ESRS [26] or the GRI [24].

Transparency and accountability are key components of effective sustainability reporting. Companies should provide a balanced view of both positive and negative performance, such as reporting on successes in reducing environmental impacts as well as challenges faced in meeting these goals. It is essential to communicate the sustainability performance to all stakeholders, including investors, employees, regulators, and the public. This can be achieved through various channels such as annual reports, sustainability reports, and digital platforms.

Continual improvement is vital for maintaining and enhancing sustainability performance. Establishing mechanisms for receiving feedback from stakeholders on the content of sustainability reports is important. This could involve public consultations, surveys, or stakeholder meetings. Based on the feedback and performance data, companies should develop action plans. For example, if the company falls short on a target, it might implement new initiatives or adjust existing strategies to improve performance.

2.7.3 Utilizing the Monitoring and Reporting Data

In the realm of strategic decision-making, pharmaceutical companies should leverage data from monitoring and reporting to make informed strategic decisions. For example, if data reveals a significant environmental impact from a particular manufacturing process, the company might decide to invest in cleaner technologies or alternative processes. Early identification and mitigation of risks through the analysis of trends and patterns in sustainability data are crucial. Machine Learning (ML) and Artificial Intelligence (AI) may be able to assist with such a proactive approach to help avoid potential regulatory fines, reputational damage, and operational disruptions.

Performance improvement is another critical area. Companies should compare their performance against industry benchmarks and best practices, which helps in identifying areas for improvement and adopting more effective sustainability practices. Innovation should be fostered by using sustainability performance data to explore new opportunities. For example, developing new products that meet emerging sustainability standards or entering markets with high demand for sustainable healthcare solutions can be highly beneficial.

Reporting to stakeholders is essential for maintaining transparency and accountability. Companies should provide clear and detailed sustainability performance reports to investors, highlighting how sustainability initiatives contribute to long-term financial performance and risk management. Additionally, publishing sustainability reports that are accessible to the public demonstrates the company's commitment to transparency and accountability.

2.8 Culture Change

A shift in organizational culture will almost certainly be required to support corporate transformation [13]. There will likely be a strong pull from the broader workforce in support of company efforts toward environmental sustainability. Indeed, many employees want to be proactively involved in initiatives. While a company does not want to distract employees from their immediate duties, it is also important not to dampen their enthusiasm and commitment for sustainability.

- **Engagement through Tangible Activities**

Many companies are engaging their workforce using tangible activities through which employees can feel that they are contributing. A good example is the removal of unnecessary single-use plastics in the laboratories and on-site cafeterias. Another example is restoring land in the immediate vicinity of company buildings to promote biodiversity and to create walking routes for employees to relax in their breaks (with the added benefit of the well-being these spaces can provide).

- **Engagement through Seamless Integration**

Middle managers may be more reluctant to fully engage with a corporate transformation program. They may feel squeezed with too many changing priorities, and environmental sustainability can come across as another item on their growing list. In order to address these engagement challenges, the corporate transformation program needs to ensure business priorities are clearly aligned with sustainability goals, rather than trying to make space for separate new targets. An almost seamless integration of objectives will make a big difference to getting things done. Even though priorities may change from one year to the next, focusing on what really matters each year by means of this seamless integration may be the key to success.

- **Mentoring by Senior Leadership**

The manner in which senior leadership leads by example should not be underestimated. Senior leaders should lead from the front and not just assign a member of their respective management teams to take accountability for sustainability on their behalf. The corporate transformation office should take time to map out senior stakeholders and consider how to best support them in their company's journey. However, the reality is that not all leaders champion environmental sustainability to the same extent; therefore, the team should focus on the design of the transformation program as well.

Keeping the company's sustainability objectives and transformation program simple and straightforward will help engage leaders while establishing a clear roadmap of projected achievements that can be used to illustrate their commitment and success.

Incentive schemes are worth considering for senior leaders, middle managers, and the broader workforce to foster and reward the good governance and practical execution needed for a successful environmental sustainability transformation. Experience suggests that selecting a couple of current KPIs from the ESG scoreboard works well for shared bonuses and can be supplemented with personal objectives used in annual performance reviews relevant to an individual's role. Pharmaceutical companies can also use recognition events with accompanying publicity to encourage the participation and performance improvements of suppliers. Transparent targets and progress reporting across the supply chain will promote engagement and wider confidence in achieving commitments.

3 Environmental Sustainability in Manufacturing Operations

3.1 Introduction

Environmental sustainability in pharmaceutical manufacturing operations is a critical and evolving topic that addresses the industry's impact on the environment. As the pharmaceutical sector continues to grow, it faces increasing pressure to minimize its ecological footprint. Key areas of focus for manufacturing include **reducing GHG emissions, managing waste, conserving water and sourcing (and generating) renewable energy**.

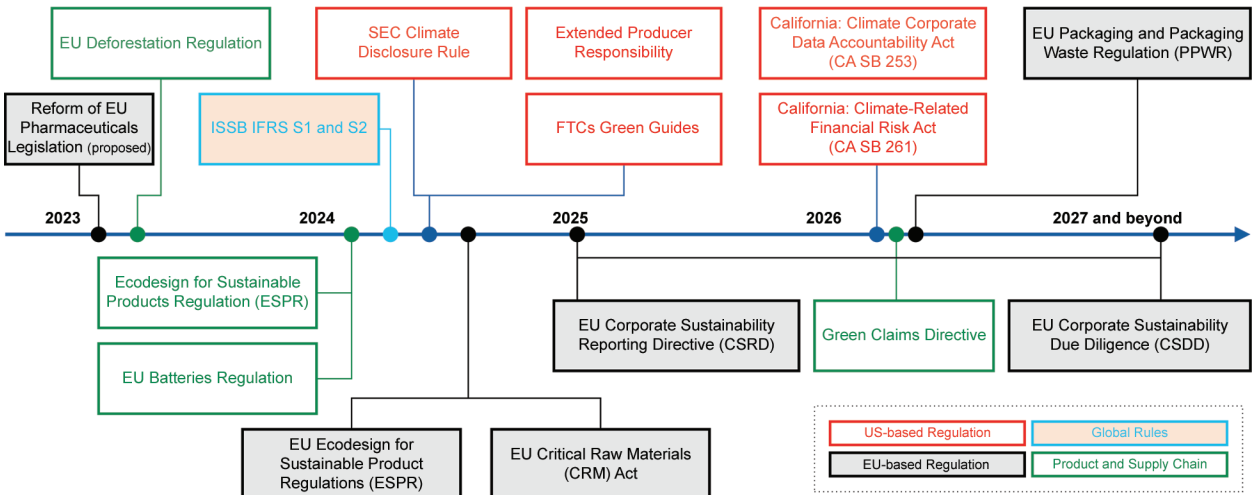
Protection of biodiversity and habitat is also a key driver to ensure all new development and processes cause a biodiversity net gain. A biodiversity net gain is achieved by an approach to development, or land management, that aims to leave the natural environment in a measurably better state than it was beforehand. Sustainable practices such as green chemistry, which aims to design products and processes that reduce or eliminate hazardous substances, are gaining traction but can take decades to reach the manufacturing plant due to the long development timelines.

Additionally, the implementation of circular economy principles, where resources are reused and recycled, is essential for minimizing waste, although there is currently no common agreement on the extent to which recycled materials, including water, can be reused in processes. Regardless of the element, product, or process, the approach to material sourcing, design, manufacturing, and construction must take account of the entire system lifecycle including embodied and operational energy and carbon.

Another significant aspect is the responsible sourcing of raw materials, ensuring that they are sourced, manufactured, and disposed of in an environmentally friendly and ethical manner. Applying these parameters to any project can drive improvement and innovation. Furthermore, regulatory compliance and adherence to environmental standards play a crucial role in driving sustainability initiatives and increasing expectations on managing corporate, social, and environmental impacts. Figure 3.1 shows a snapshot of the regulations that impact this sector.

Figure 3.1: Environmental Regulations Impacting Manufacturing Operations

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Continuous R&D are necessary to advance sustainable technologies and practices within the manufacturing scope. Most of the solutions needed exist; the challenge is to fully apply them across the industry and throughout the supply chain.

For a number of problems, innovation has been made, perhaps at a lower scale, and only scaling up may be needed. By addressing these main topics with already identified solutions, the pharmaceutical industry can make significant strides toward reducing its environmental impact and promoting a healthier planet. However, significant challenges may arise.

The pharmaceutical industry faces several significant challenges in achieving environmental sustainability:

- **High Energy Consumption:** Pharmaceutical manufacturing processes are energy-intensive, requiring substantial amounts of electricity and heat, often generated from non-renewable sources.
- **Waste Management:** The industry produces a large volume of hazardous waste, including chemical solvents and by-products, which require careful handling and disposal to prevent environmental contamination.
- **Water Usage:** Pharmaceuticals require large quantities of water for production and cleaning processes. Managing water use and ensuring it is treated properly before discharge is a critical challenge.
- **Facility and Site Management:** Where existing facilities are at risk of becoming a stranded asset (an asset that has lost value or become unusable due to changes in the market or risk profile), remediation works will be required. Carrying out a detailed audit of the existing facility and utilities is crucial to set a clear baseline from which improvements and efficiencies can be made. New facilities are to be designed as efficiently as possible. The EU has placed increasing emphasis on minimizing both operational and embodied energy use in new buildings. It is actively pursuing policies to ensure that newly constructed buildings are designed to achieve high levels of energy efficiency and significantly reduced carbon emissions throughout their lifecycle [105]. This is quite a transition from traditional forms of construction and takes a coordinated team effort (client, design and construction teams) based on clear targets and deliverables.
- **Supply Chain Complexity:** The global supply chains in the pharmaceutical industry involve multiple stages and partners, making an integrated approach to environmental system management challenging.
- **Regulatory Compliance:** Navigating the complex regulatory landscape to meet environmental standards while maintaining product quality and safety can be challenging.
- **Packaging Waste:** Pharmaceuticals often come with extensive packaging to ensure product safety and integrity, leading to significant amounts of process packaging waste.
- **Innovation and Technology:** Developing and implementing new, sustainable technologies and practices can be time-consuming and impactful on operations.

Addressing these challenges requires a concerted effort from all stakeholders, including companies, regulators, and suppliers, to drive innovation and adopt more sustainable practices across the industry.

3.2 Lifecycle of Sustainability in Pharmaceutical Operations (Design, Construction, Operation, and Decommission)

3.2.1 Design and Construction of Pharmaceutical Operations

It is thought that a significant portion of a product or project's environmental impact is influenced by decisions made in the early part of the design phase. At the early stages of development if all stakeholders, advisers, specialist designers, and vendors are engaged in all aspects of the project's lifecycle, synergies between one or more systems can emerge and lead progress toward innovation, significant environmental benefits, and long-term cost savings. The Integrated Design Process (IDP) is a collaborative approach to design that involves multiple stakeholders working together from the beginning of a project.

The goal of IDP is to avoid suboptimal design solutions and create highly efficient designs based on decisions made with a holistic view, considering various interconnected factors to optimize environmental impact and efficiency with minimal capital and operating costs. Since all systems are linked in one form or another, a change to one area will impact another area, and as a result, when specialists are only brought in the later stages of the project, opportunities to take these into account early in the design phase may be missed. Although it can add time to the initial design phase, it is the recognized approach to create innovative and sustainable solutions that address not only the immediate requirements but also the long-term implications to a product or project.

Challenging the status quo with regards to manufacturing pharmaceuticals can lead to various benefits in terms of efficiency, sustainability, and cost-effectiveness. By exploring alternatives and critiquing the accepted process from a sustainability perspective during the design phase, a more streamlined and resource-efficient process can be envisioned. Some examples include:

- **Air Change Reduction:** In the pharmaceutical industry, HVAC systems account for a substantial portion of overall energy consumption, typically ranging from 40% to 60%. This high percentage reflects the critical role that HVAC systems play in maintaining the precise environmental conditions required for cleanrooms and other controlled areas. The energy-intensive nature of HVAC operations stems from the need to frequently filter and condition large volumes of air to stringent cleanliness, temperature, and humidity standards. Consequently, optimizing HVAC performance, such as through reduced air change rates, offers a significant opportunity for energy savings without compromising product safety or regulatory compliance. By implementing strategies to improve HVAC efficiency, pharmaceutical facilities can achieve substantial reductions in their energy footprint.
- **Transitioning to Ambient Water for Injection (WFI) Systems:** (refer to Section 4.6.2 for further details)
Traditional WFI generation systems typically rely on distillation processes, which consume significant amounts of energy. Ambient WFI systems, on the other hand, could produce water of equivalent quality without requiring distillation using membrane-based techniques including Reverse Osmosis (RO), Electrodeionization (EDI), and Ultrafiltration (UF) [35]. Transitioning to ambient WFI generation and distribution systems can result in a substantial reduction in energy consumption within a facility. When contemplating this change, it is crucial to assess the overall impact by involving a broad team of stakeholders and regulators. Thoroughly reviewing all benefits and drawbacks collectively enables informed decision-making that considers various perspectives and ensures alignment with operational and regulatory requirements for a successful transition.
- **Open to Closed Processing:** Transitioning to a closed processing system provides several significant advantages. By segregating the process, the overall cleanroom classification can be lowered, leading to reduced air changes and decreased energy consumption by the supporting HVAC system. With fewer gowning stages to pass through, there is a decrease in waste associated with consumables and cleaning efforts. This approach enhances protection for both products and operators by mitigating contamination risks. Furthermore, this setup allows for increased flexibility to accommodate future changes and technological advancements efficiently.
- **Direct Material Flows to Reduce Classified Areas:** Using robotics can reduce cross-contamination by transferring product straight into a packaging area, omitting classified material circulation corridors. The footprint of a facility can be reduced – stacking the functions is possible – and associated energy reduction from HVAC, lighting, and heating may be achieved.
- **Continuous Manufacturing:** Shifting from batch to continuous manufacturing offers a significant benefit to sustainability by optimizing resource usage and reducing waste. Continuous manufacturing enhances process efficiency by maintaining a steady state of production, which minimizes the start-up and shutdown phases associated with traditional batch processes. This leads to optimized energy consumption and lower raw material waste. Moreover, continuous processes typically require smaller equipment footprints and can operate with less environmental impact, as they produce fewer emissions and by-products.

Any change within the industry, whether related to sustainability initiatives in product, process, or facility improvements, necessitates a thorough evaluation of its impact on various critical factors including product quality, process efficiency, and regulatory compliance. Considering how a system, material, or space can be used now and adapted in the future will aid decision-making. Additionally, it is important to understand that a system that can be adapted to suit an alternative product still may require additional elements to be integrated. When making changes in the production area, consideration should be given to assess whether the existing space will be able to facilitate the change, for example, by allowing the transport of additional equipment into the required space.

On a broader scale, sustainability can be achieved through either active or passive approaches. Active measures involve the creation or utilization of energy, such as procuring renewable power or installing renewable energy sources on site for power generation. On the other hand, passive measures aim to reduce reliance on active strategies by harnessing ambient energy sources such as natural light, ventilation, and passive heating and cooling methods. For example, optimizing building orientation and layout to maximize natural lighting in office spaces or situating warehouse and utility areas on the northern side of a facility to prevent overheating (southern side in the southern hemisphere) are examples of passive measures. It is essential to evaluate these solutions early in the project development phase to prevent rework or costly modifications and ensure alignment with the overall project goals effectively.

When considering the impact of a product or facility over its lifecycle, it is essential to understand the difference between embodied and operational carbon.

- **Embodied Carbon:** This refers to the carbon emissions associated with the extraction, manufacturing, transportation, and installation of materials used in the production of equipment or construction of buildings. It represents the carbon footprint incurred before the product or facility is put into use. The amount of embodied carbon in a production facility can range from 20–40% of its total carbon footprint. This is expected to rise to 50% between 2020 and 2050. Of this range 20–40% is produced from the HVAC system alone. In process equipment, there is approximately 30% embodied carbon.
- **Operational Carbon:** This relates to the carbon emissions generated during the operation and use of a product or facility. It includes energy consumption, maintenance, and end-of-life processes. Operational carbon accounts for the ongoing carbon footprint during the lifecycle.
- **EPD:** An Environmental Product Declaration is a standardized and globally available document that offers transparent and comparable information on a product's environmental impact over its lifecycle. Also known as a material passport, it covers areas such as operational and embodied carbon, energy usage, water consumption, and other environmental factors. By providing detailed sustainability data, EPDs enable consumers, engineers, architects, and builders to make eco-conscious choices, helping toward a more sustainable future. While EPDs are readily accessible for construction materials such as cladding panels and glazing and partially for utilities such as HVAC systems, they are currently unavailable for process equipment. EU regulation increasingly emphasizes requiring suppliers to provide transparent information about the environmental impact of construction products, often through Environmental Product Declarations (EPDs) in accordance with standards such as EN 15804. While formal EPDs are not universally mandatory for all product categories – including most process equipment – their use is recognized as a best practice and is often recommended by project teams to facilitate assessment and comparison of environmental impacts. [106]. Although it is not obligatory for process equipment, it is recommended that these documents are requested from client and design teams to normalize their usage. For utilities lacking EPDs, alternatives such as CIBSE TM65 [107] (a calculation methodology for calculating embodied carbon in building services and utilities) can be utilized to guide the assessment of embodied and operational carbon emissions. Methods to encourage suppliers to provide this data include adding clauses in project payment terms.

Understanding and managing the total carbon footprint (embodied and operational) of a system is crucial for developing sustainable products and facilities. Using a steam sterilizer as an example, the breakdown is 98% operational energy and 2% embodied energy, highlighting the significant impact of operational processes on overall energy consumption. By integrating both operational and embodied carbon into metrics and setting goals that encompass the entire carbon lifecycle, organizations can make informed decisions to reduce the overall carbon impact across a system, site, or company-wide.

Building construction contributes to approximately 35–40% of global GHG emissions, encompassing both embodied and operational carbon. It is important to note that these figures are subject to change as emission patterns evolve over time.

When planning a new product, manufacturing area, or overall facility design for renovation, extension, or new construction, it is crucial to account for the carbon footprint of not only the new elements but also the existing components. In the context of existing facilities and utilities, carbon is embedded in the materials used during their construction or assembly. Upon demolition or dismantling, this embodied carbon is released back into the environment, affecting a company's carbon footprint.

3.2.2 Early Engagement of Operational Teams

Whether it is about early integration of a sustainability approach across the company or executing the corporate decarbonization roadmap for a specific site, involving the operational teams from the outset is crucial for seamlessly integrating sustainability practices across the company. By engaging the team early in projects related to sustainability initiatives such as corporate decarbonization, organizations can ensure that team members not only understand the “why” behind sustainability efforts but also see the tangible benefits these initiatives can bring.

When the operational team comprehends the rationale behind sustainability measures and recognizes the positive impacts they can have, it becomes easier to translate sustainability objectives into practical actions within day-to-day operations. This shared understanding and buy-in from the team members not only enhance the implementation of sustainable practices but also foster a culture of continual improvement and innovation toward meeting sustainability goals at every level of the organization. When the operational team is actively involved in the planning and decision-making stages, they can provide valuable insights, identify operational inefficiencies, and suggest sustainable solutions that might otherwise be overlooked.

This collaborative approach not only enhances the overall sustainability performance but also empowers employees to take pride in contributing to a greener and more efficient workplace. Conducting audits to assess current energy, water, and waste usage is crucial for understanding the baseline of a product, system, or facility's environmental footprint. Analyzing manufacturing processes can reveal significant opportunities for efficiency improvements. Optimizing running times, implementing shutdown protocols for equipment and building services during non-operational hours, and leveraging automation for smart energy management are common strategies to enhance resource efficiency in manufacturing facilities. It is crucial to consider the knock-on effects of operational changes, such as on material capacity (incoming and outgoing), logistics, and workforce requirements. By carefully evaluating these factors and implementing efficiency measures, organizations can not only reduce costs but also lessen their environmental impact, showcasing a commitment to sustainability and responsible resource management.

3.2.3 Reuse and Rebuild versus Decommission

Embodied carbon of equipment can account for up to approximately 30% of the total carbon and needs to be considered in the whole LCA of a selected system. Retrofitting and repair options can significantly impact sustainability. For example, retrofitting a steam sterilizer can save approximately 26,000 lb of CO₂ emissions and extend the system's lifespan by more than 10 years [108].

Before a system, product, or space is removed and replaced, its current use and function should be reviewed to explore whether they could be adapted to a future use. For example, conducting a thorough assessment of the existing system (function, age, and efficiency), identifying potential modifications needed, and ensuring that the production space can accommodate the changes, creative ways can be found to repurpose it in a cost-efficient manner. Retrofitting or repurposing existing systems can not only be more sustainable but also offer opportunities to optimize resources and improve overall performance. This approach aligns with the principles of circular economy and sustainable development, where maximizing the utility of existing resources is prioritized before opting for new replacements.

3.2.4 Facility Design

Pharmaceutical facilities, although potentially subject to periodic upgrades or renovations for expansion or to comply with new regulations and technological advancements, tend to have a longer lifespan compared to the equipment within them. Therefore, it is crucial for new facilities to be designed with future flexibility, expandability, and ease of repurposing in mind to minimize the impact of future changes on sustainability.

3.2.4.1 Leadership and Cross-Functional Collaboration

Leadership plays a critical role in fostering a culture of innovation and sustainability within an organization. By driving the mindset from the top down, companies can empower employees at all levels to think creatively, challenge traditional practices of automatically replacing with new, and actively seek out sustainable solutions. By emphasizing the importance of retaining, adapting, and optimizing existing systems before considering replacements, companies can cultivate a culture of continual improvement, efficiency, and environmental responsibility throughout their operations, this setting the tone for the entire organization to follow suit. Reward programs could be introduced to encourage operational and engineering teams to develop new solutions to existing challenges. This, in turn, can lead to tangible benefits such as cost savings, improved stakeholder relationships, and a stronger competitive edge in the market.

Considering factors such as ensuring compliance to regulatory requirements and GMP standards, pursuing energy efficiency from the outset of an upgrade project, and engaging with all relevant stakeholders ensures that their expertise, concerns, and requirements are considered in the planning and implementation process. This allows the team to proactively address potential obstacles, ensure alignment with industry best practices, and streamline the project timeline.

3.3 Reduction of Energy Consumption

The International Energy Agency's World Energy Outlook predicts a reduction in global carbon emissions by 2030 [36]. To align with net-zero goals, a significant decrease is required by that time. As electricity demand continues to grow, along with the sales of solar and battery technologies, improvements in energy efficiency could lead to substantial emissions reductions by 2030 more than any other measure. A key strategy for achieving net-zero goals aligned with the SBTi [15] is to minimize energy consumption in pharmaceutical operations. Scope 1 and Scope 2 GHG emissions are directly tied to the energy usage in industrial plants. Enhancing energy efficiency in manufacturing involves implementing energy-efficient technology, optimizing production processes, and conducting regular energy audits and monitoring to identify continual improvement opportunities. Energy consumption reduction should be prioritized both during the initial facility design and throughout the lifespan of the facility. Live monitoring of all systems is as important as Energy-Efficient Design (EED) since it provides an operational baseline for the facility. Adjustments can be made to optimize efficiency and alert operators in real time where a system is not performing or wasting energy. Establishing an energy balance during process and facility design can serve as a valuable tool, providing a baseline target for future performance optimization.

New technologies such as digital twins, AI, and other advanced tools offer substantial opportunities for reducing energy consumption in the pharmaceutical industry. AI enhances process optimization, allowing for precise control and greater efficiency in manufacturing, while predictive maintenance ensures equipment runs smoothly, minimizing energy waste. AI systems could be utilized to monitor and manage energy use in real time, providing insights that facilitate more efficient facility operations. Any AI-driven decision must adhere to regulatory requirements and industry standards, serving as a supportive tool within the industry's framework.

Considering how energy is used and in what capacity is important. An interesting example can be seen with a 1-ton stainless steel container and 1-ton of water. It takes approximately 8.3 times more energy to heat the water than the stainless steel container. Therefore, consideration should be given to the heat-capacity of a system of material and the required energy to heat it.

The benefits of the reduction in energy consumption include:

- Electric grid stability through reduced peak loading
- Economic benefits: operational cost savings through lower utility bills and potential utility incentives
- Cost savings on electrical infrastructure upgrades needed for decarbonization
- Resilience and risk management as energy-efficient buildings are less vulnerable to demand spikes that can lead to outages. Renewable energy systems can maintain critical functions during outages.

Examples of energy-efficient technology to be considered include:

- **Optimized Cleanroom Operations:** The use of advanced airflow management systems and High-Efficiency Particulate Air (HEPA) filters should be used to maintain stringent contamination control. Minimize air change rates while maximizing air recirculation to maintain low contaminant levels efficiently. Use real-time monitoring systems to adjust airflow based on occupancy and process needs, improving performance and energy efficiency.
- **High-Efficiency Equipment Selection** such as heat pumps, chillers, pumps, and fans.
- **HVAC Air Handling Unit (AHU) Optimization:** AHUs should be appropriately sized to meet the required space loads while minimizing ventilation air requirements. Sizing should minimize pressure drop through unit components, such as high-efficiency filters and coils. Variable speed drives, paired with high-efficiency motors or electrically commutated motors, reduces energy demands. Fan arrays should be considered for their high-efficiency at part-load conditions and ease of maintenance. Incorporating energy recovery systems, such as heat pipes or heat wheels, is essential. Demand control ventilation, combined with distribution zone controls, minimizes the heating and cooling loads needed for ventilation. Economizers can be used to precondition air, reducing heating and cooling needs when outdoor air conditions are favorable.
- **HVAC Distribution Optimization:** HVAC distribution should be appropriately sized using precise engineering space load calculations. Incorporating accurate space load data helps avoid overcooling and reheating. Ventilation loads should be calculated to minimize energy demand, with recirculation or passive heating and cooling strategies employed to maintain space temperatures. HVAC systems should be zoned individually with dedicated controls for temperature and occupancy. This allows widening temperature and humidity deadbands when spaces are unoccupied. Ductwork should be sized to prevent high pressure drops, and Variable Air Volume (VAV) control devices should be specified to match airflow and temperature with space demands.

For more detailed information on HVAC system design, operation, and their impact on facilities' energy consumption, refer to the *ISPE Good Practice Guide: Heating, Ventilation, and Air Conditioning (Second Edition)* [85].

- **Utilities System Optimization:** Utility optimization should include a combination of operational enhancements, technological advancements, and system optimization. This involves the recovery of waste heat for pretreating other processes and demand matching to ensure efficiency. Technological improvements may involve upgrading to more energy-efficient equipment with advanced control systems, such as smart energy management systems, to further enhance energy efficiency. Regular maintenance and performance monitoring are crucial to sustaining these improvements, ensuring that the overall utility system operates at peak efficiency.
- **Efficient Sizing of Equipment and Utility Loads:** Properly sized equipment operates more efficiently because it runs at optimal performance levels. Rightsizing equipment prevents it from cycling on and off frequently, which can lead to increased energy consumption and higher energy bills. Rightsized equipment typically has a longer life span, reducing the embodied carbon of the equipment over its lifetime.

- **Implement Waste Heat Recovery Systems:** Install heat exchangers to capture excess heat from processes and reuse it for heating or power generation. This can significantly reduce energy costs and improve overall efficiency. Consider systems such as heat recovery wheels, heat pipes, and run-around loop. Waste heat should be considered as a source for heat pumps.
- **Optimize Water System Operations:** Implementing membrane-based technologies, such as RO, can reduce the energy required for water purification compared to traditional distillation methods. Heat recovery systems can capture and reuse waste heat; by using heat exchangers, hot wastewater can be used to preheat incoming cold water. Water storage and distribution temperatures should be considered, matching end-use requirements. Cold storage coupled with ozone disinfection in ambient systems can minimize GHG emissions, energy demand, and meet regulations. For more information, refer to the *ISPE Baseline® Guide: Water and Steam Systems (Third Edition)* [86], and the *ISPE Good Practice Guide: Ozone Sanitization of Pharmaceutical Water Storage and Distribution Systems (Second Edition)* [87].
- **Lighting System Controls and Optimized Lighting Design:** Low-energy LED lighting should be specified to maximize energy efficiency. Lighting design plans should integrate layered lighting approaches, including natural light, task lighting, and zone controls. Proper fixture placement should be calculated to avoid over lighting, and optical design efficiency should be employed to minimize wasted light. Lighting control strategies should incorporate occupancy sensors, vacancy sensors, timers, and daylight sensors where appropriate and safe, to reduce lighting energy consumption.
- **Building Envelope Design:** Enhancing the building shell – including walls, fenestration, foundations, and roofs – can significantly reduce heating and cooling loads, as well as ventilation airflow. High-efficiency insulation and windows, combined with minimizing thermal breaks and air leakage in the building shell, support decreased HVAC demand. Additionally, strategic building orientation and the use of both interior and exterior window shading can further optimize HVAC space requirements. Using vertical living walls can also further insulate a building by reducing internal temperature variances while protecting a building envelope from solar gain and driving wind. The vegetation supports biodiversity, reduces the heat island effect in built up areas, filters the air, and reduces rainwater volume entering the surface water system.

The embodied carbon of these systems must be considered in the overall selection process.

Conducting an LCA of both the equipment and the building structure can provide insights for making informed choices. While the initial cost of a system may be higher than expected, a comprehensive evaluation over the entire lifecycle may reveal it to be more cost-effective in terms of operational expenses, thereby reducing the overall system cost. Utilizing a Life Cycle Cost Analysis (LCCA) in conjunction with system energy modeling is essential for optimizing system selection and ensuring sustainable performance.

Regarding energy reuse, prioritizing local reuse can enhance efficiency and sustainability by reducing transmission losses and maximizing the utilization of available resources. Strategic placement of systems within utility spaces to facilitate local energy reuse can contribute to overall energy efficiency and cost savings. When evaluating energy recovery systems, it is important to consider the impact of parasitic loads, which can offset energy savings. These auxiliary energy demands, such as those from pumps and fans, should be minimized to ensure the overall efficiency of the system.

Improvement in Energy Management Processes can be achieved through the following activities (list not exhaustive):

- Energy benchmarking against similar buildings
- Comprehensive metering strategies for identification of significant energy users. Consider both utility data as well as data that correlates and scales, such as production data, weather data, operations data, and occupancy data
- ISO 50001 certification [109]

- Annual energy audits
- Setting targets and creating systems for accountability
- Continual improvement and identification of opportunities; maintaining a list of energy conservation measures that support continual improvement
- Lifecycle cost analysis and LCA to support decision-making

3.4 Renewable Energy Sources

Energy-intensive manufacturing and production processes are a major source of emissions for pharmaceutical companies, particularly those reliant on local grid electricity. Once energy consumption has been optimized within the facility, renewable energy sources offer a further opportunity to decarbonize operations. Often the terms renewable and sustainable are used interchangeably in relation to energy sources. However, there is no such thing as a completely sustainable and renewable energy source. Even renewable energy sources (primarily electricity) generated from wind, solar, and hydroelectric power rely on commodities such as copper, cobalt, silicon, and other minerals. The extraction and processing of these materials involve environmental and sustainability practices that can vary significantly and may have detrimental impacts on the environment. Nevertheless, such technologies are important components in the renewable energy transition.

For applicable renewable technologies in the pharmaceutical industry, there are some technologies that are only economic or practical to apply at grid/off-site scale. Depending on the location, it may be possible to procure these renewable energy sources from generators/suppliers, or more accurately to procure a certificate that confirms that the facility's use of energy is renewable. The mechanisms for sourcing renewable electricity vary across regulated and deregulated markets. Small to midsized facilities can coordinate with other companies to scale renewable sourcing. However, challenges remain, including price uncertainty and volatility in virtual PPAs and minimum volume requirements for long-term PPAs. Adhering to GHG Protocol and SBTi guidelines [15] on purchased energy is crucial for accurately measuring and managing emissions, thereby enabling organizations to set and achieve science-based emission reduction targets.

3.4.1 Grid/Off-Site Scale Renewable Electricity

Solar PV: Solar Photovoltaic (PV) systems convert sunlight directly into electricity using semiconductor materials. Generation of power is intermittent (only during daylight hours) and dependent upon the intensity of sunlight at the location, which impacts the solar PV asset energy output. Solar PV operators consider balancing energy output and local electricity rates in determining the investment in project viability. Off-site solar PV systems are typically larger and located in remote areas. They can benefit from economies of scale, potentially lowering the cost per unit of electricity generated. However, they require extensive transmission infrastructure to deliver power to end users, which can result in energy losses and higher costs. Off-site systems also depend on the grid's reliability, making them less resilient during outages.

Wind: Wind electricity generation harnesses the kinetic energy of wind to produce electrical power using wind turbines. These turbines convert wind energy into mechanical power, which is then transformed into electricity by a generator. Off-site wind generation involves large wind farms located in areas with optimal wind conditions, often far from the point of use. While these systems benefit from economies of scale and can produce large amounts of electricity, they require extensive transmission infrastructure, which can lead to energy losses and higher costs.

Hydro: Hydroelectric generation harnesses the energy of flowing or falling water to produce electricity. This process typically involves a dam that stores water in a reservoir. When released, the water flows through turbines, spinning them to generate electricity. Hydroelectric systems are usually larger and can supply power to a broader area. While they benefit from economies of scale, they often face higher transmission losses and can be more vulnerable to grid disruptions. However, they can provide a more consistent and substantial power supply to meet regional demands. In some regions, hydroelectric systems are increasingly vulnerable to climate change, leading to reduced water availability in some seasons.

Tidal/Wave: There are several large scale tidal/wave energy generation schemes in operation globally using the predictable and reliable patterns of tidal flows and wave motion to drive turbines to produce electricity. Suitable sites are limited to coastal areas with strong tidal currents. However, the installation and maintenance costs of tidal energy schemes are high compared to other technologies, and they can cause environmental impact to marine ecosystems and sediment transportation.

Nuclear: While strictly not a renewable energy source, generation of electricity from nuclear fission is largely carbon emission-free during its operating lifetime. Nevertheless, construction of nuclear facilities requires decades to plan, construct and commission, and requires enormous quantities of concrete, a carbon-intensive commodity, thus, embedded carbon is considerable. There is also the issue of nuclear waste, which is a challenge from an environmental perspective, although improvements in technology promise to mitigate this long-term issue. There is growing interest in small, modular nuclear reactors that could be more scalable, quicker and less costly to deploy. This is still an emerging technology and is worth watching for the medium term.

3.4.2 Grid/Off-Site Scale Renewable Heat

Provision of renewable heat is gaining traction globally with global heat consumption.

“The rise in policy attention renewable heat is gaining worldwide is due not only to environmental considerations but to acute energy security concerns in the context of the current global energy crisis...Global heat consumption -excluding ambient heat harnessed by heat pumps- is projected to grow almost 14 EJ (+6%) during 2022-2027.”
(International Energy Agency, Analysis and Forecast to 2027 [36])

However, in many regions it is either non-existent or in early stages of deployment.

In contrast to renewable electricity, which can be distributed over large distances, renewable heat must be more locally sourced as distribution losses are very high relative to the energy transported. There are several examples of local renewable heat networks, mainly in European/Scandinavian countries and elsewhere. These use a range of technologies and techniques to generate renewable heat (or make use of otherwise wasted heat) and distribute to industry and domestic users in a relatively small geographical area. Examples include:

- Ground source or geothermal heat extracted and boosted to a more useful temperature using heat pumps
- Heat recovery from industrial processes or waste incineration that is distributed in a local heat network

Other possible sources of renewable heat:

- **Biomethane:** Renewable methane gas, generated from anaerobic digestion, has made the largest contribution to the increase in renewable heat globally. It can be generated anywhere, provided feedstock for the digestion process is available. It is injected into a natural gas distribution network to displace fossil natural gas use and combusted in the same way as fossil natural gas, requiring little need for equipment changes. In this way, organizations can gain access to renewable biomethane by procuring certificates of origin, similar to renewable grid electricity. There remain a number of sustainability challenges with this approach:
 - Anaerobic digestion feedstock is often sourced from crops grown by farmers. This can displace food production and drive-up crop prices, impacting vulnerable communities.

- Biomethane still emits greenhouse gases and other pollutants during combustion. Although a renewable source, it remains an environmental pollutant.
- It is costly compared to fossil natural gas – around 5–10 times the cost per unit of energy. However, costs are likely to reduce over time if demand drives production scale to increase.
- **Hydrogen:** To be a renewable heat source, the hydrogen gas should be generated by electrolysis using electricity from renewable sources. There are a number of problems to overcome to establish hydrogen as a reliable and efficient source of renewable heat:
 - Hydrogen gas needs to be transported to the point of combustion. Possible options include piping or liquefied transportation, normally by road. Hydrogen gas is approximately one quarter the energy density of natural gas and therefore four times the gas volume is required (at standard temperature and pressure) for the same energy demand; consequently, the distribution pipes may need to be larger.
 - By contrast, liquefied hydrogen gas requires complicated and expensive cryogenic storage and distribution facilities, also requiring renewable electricity.
 - Overall, the energy balance of using hydrogen as a fuel is highly inefficient, and therefore the cost is very high. Advances in technology may improve this over the long-term and it may then be applicable in specific circumstances where the economics are favorable [36].

3.4.3 On-Site Renewable Energy

On-site provision of renewable energy, both electricity and heat, is entirely viable and there are many examples of successful deployment on sites. The availability and applicability of each renewable energy source is very much dependent on the local site geographical, physical, and economic circumstances. The following section present an overview of each established technology.

3.4.3.1 Renewable Electricity

Solar PV: The cost of solar PV has been reduced substantially over the last decade. Geographical location, sunlight intensity and duration are considerations, although with increasing panel efficiency, most locations will benefit from this technology. The integration of battery banks with solar panels can help smooth and extend the energy supply profile, offering benefits as battery technology becomes more affordable. Additionally, they can ensure consistent backup power for critical systems.

Roof Mounted: Often building roof spaces are ideal locations for solar PV panels and can connect into building electrical services. Locating on roofs hides these from view and makes use of an otherwise redundant space. Consider extending the parapet around the roof higher than the typical 1.1 m height to protect the panels from wind uplift and the requirements of additional steel during installation. Some facility insurance companies require the roof structure to be fire rated where PV panels are fitted. For new facilities, these decisions should be considered early in the design process or could result in costly retrofit works.

Ground Mounted: If land is available, ground mounted solar arrays are cost-effective and can be quickly deployed. They may require local planning permission. They can be positioned at a level to permit daylight to the ground below to facilitate the regeneration of a site to its original habitat, resulting in biodiversity net gain. It is necessary to consider long-term land use, although ground mounted solar arrays can be relocated.

Car Parking Shading: The dual benefit of creating solar shading for car ports and generating electricity is often an attractive solution. However, it comes at a cost premium compared with roof mounting due to support structures required, but can have the added benefit of reducing the heat island effect from large areas of concrete or asphalt.

Wind: Several pharmaceutical facilities have successfully installed wind generation turbines on-site. They require reliable and predictable wind strengths, which can be location specific requiring long-duration studies (wind surveys) to remove uncertainty on investment. There are also challenges related to planning considerations with neighbor/local acceptance, aesthetic appeal, noise, and impact on the natural environment, all of which need to be considered carefully.

Fuel Cells (Hydrogen): A hydrogen fuel cell is a device that converts the chemical energy of hydrogen into electricity through an electrochemical reaction. It provides clean power and heat as a by-product. Downsides are availability of hydrogen and the very high capital cost of fuel cells, which results in a high cost for electricity produced.

3.4.3.2 Renewable Heating

Waste Heat Recovery and Heat Pumps: Many sites have a multitude of processes using heat or generating heat as a by-product. Often this heat is wasted or actively disposed of in cooling towers. The main problem is matching the wasted heat with a suitable heat need, and at a usable temperature. An efficient technology is to collect this waste heat in a suitable heat network to either reuse directly or use as a feed to a heat pump such that it can be elevated to a more useful temperature. A correctly designed heat network can make use of this wasted heat in a very economical way. It is also largely emissions-free.

Solar Thermal/Concentrated Solar Thermal: Solar thermal technology harnesses the sun's energy to generate heat, which can be used for various applications such as heating water, space heating, and even electricity generation. Unlike PV systems that convert sunlight directly into electricity, solar thermal systems convert sunlight into thermal energy. It is scalable from single panel domestic water heaters to large scale solar thermal collector to produce steam to generate electricity. However, it is an intermittent source of energy and most effective in locations with high solar intensity. Given this, coupling with thermal storage technologies may offer a more reliable source of heat on a 24-hour basis.

Hydrogen: Gas displaces the use of fossil natural gas and is burned in boilers to generate hot water or steam, with the combustion products being mainly water. It is only considered renewable if the hydrogen is generated using renewable electricity, commonly known as green hydrogen. Green hydrogen is currently not readily available anywhere. Due to the practical difficulties of generation, storage, and distribution and the associated costs of this complicated supply chain, it is very uncertain whether this technology will prove to be viable for pharmaceutical manufacturing sites in the long-term.

Biomass: Biomass is a combustion process where fuel comes from a renewable source. This renewable source can be waste materials, such as waste wood, sawdust, and even rice husks, but more common materials are wood chipping/pellets sourced from managed timber forests. There are concerns that extensive use of biomass can contribute to deforestation and careful sourcing and due diligence is needed. Transporting large amounts of biomass to sites, sometimes from the other side of the world or even locally by road, increases GHG emissions (Scope 3). Some facilities have used this solution based on their geographical location and set a 50-mile radius for supply of biomass material. It may not be accurate to describe biomass as a totally renewable/environmentally friendly option as it very much depends on how the fuel is sourced and how far it travels, and remains a combustion process emitting pollutants to the environment.

Biomethane: Biomethane is rapidly becoming the most common source of industrial renewable heat globally. It relies on combustion of methane, although the methane gas is provided from renewable sources (see the explanation in Chapter 2). On-site generation of biomethane is possible, but not common due to complexities of the supply chain of Anaerobic Digestion (AD) feedstock. More commonly, a purchasing arrangement is established to procure certificates of origin from the methane gas supplier. It requires little or no change to on-site combustion equipment, but availability is far from universal.

Geothermal: A huge amount of heat energy can be available from underground. Advances in drilling technologies are making this more accessible and available to users, although drilling remains hugely expensive with a high risk of failure to find suitable heat sources. Nevertheless, combined with heat pumps to boost to usable temperatures, even low heat grade geothermal sources are beneficial.

3.4.3.3 Renewable Cooling

Cooling Towers: Cooling towers are a well-established technology that uses the principle of evaporative cooling to remove heat from cooling water. They depend on ambient conditions being favorable for evaporation, although these vary seasonally, and desired temperatures may not be achieved during certain times of the year. The risk of *Legionella* contamination exists, but it can be effectively managed through proper water treatment. Utilizing recycled water as the cooling tower make-up water can be an effective strategy for conserving water resources, provided that the water meets the necessary quality requirements.

Water Source Cooling: If a site has access to a large water body (river, sea, large lake), it may take advantage of the cooling energy within this water. Water can be pumped through heat exchangers to provide cooling on-site. This should only be considered if there is no possibility of utilizing site waste heat anywhere else on-site (rarely the case). Discharge temperatures should be considered because high temperatures can impact the biodiversity within the water bodies.

Chillers/Refrigeration: When powered with renewable electricity and containing low Global Warming Potential (GWP) refrigerants (natural refrigerants, e.g., CO₂, propane, ammonia), these common technologies can provide renewable cooling energy.

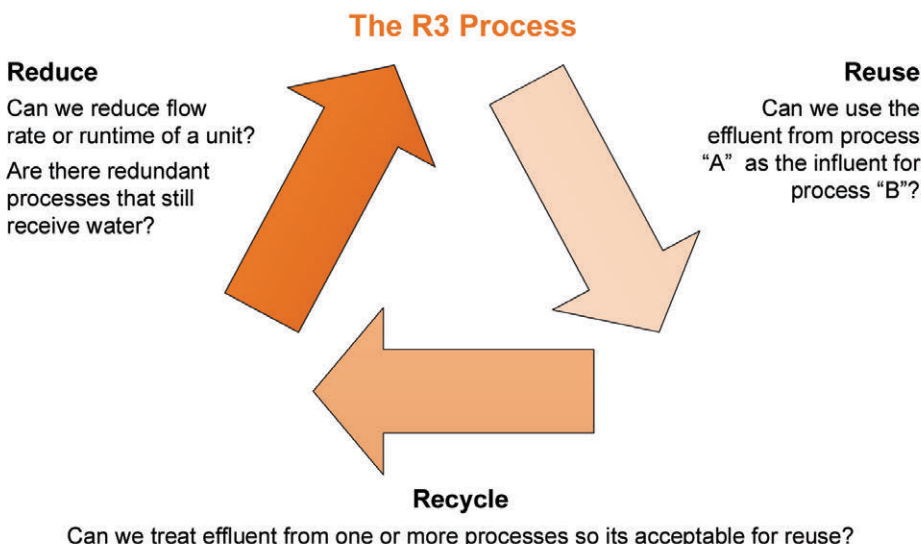
3.5 Water and Waste Reduction

In Chapter 2, the circular economy is introduced as a solution to cope with limited natural resources, one of which is water. In this section, the focus is on reducing water intake and consumption as well as reducing the waste footprint in the pharmaceutical supply chain.

Traditionally, water and waste management programs exist especially in the operations space, contributing to operational efficiencies and financial savings. Accelerated trends such as droughts, accumulation of plastic in the eco-system, and increasing legal regulations (e.g., extended producer responsibility) call for broader and accelerated thinking about water and waste. Existing operations that already have water and waste programs are, however, a good nucleus for wider water and waste reduction programs through connections to a wider group of value chain partners (internal and external), including customers and patients. The circular economy as an overarching principle is a good guidepost for this transition and in a lot of cases, initiatives offer new business opportunities or financial savings.

Access to the right quantity and quality of water is and will become more and more difficult in the decades to come [37]. In an era of ever-increasing frequency and magnitude of water scarcity, saving water, a vital production ingredient for almost any pharmaceutical product, is simply a business imperative.

A water conservation action plan is most likely to follow the process of: Reduce → Reuse → Recycle (R3). Figure 3.2 illustrates the R3 process.

Figure 3.2: Reduce-Reuse-Recycle (R3) Process*Used with permission from Arcadis N.V., www.arcadis.com.*

Fixing the basis is primordial: through setting up a water balance, the main consumers and leakages (expressed as gaps) can be identified. This forms the basis of improvement actions. Leakages should be identified and stopped, water consumptions understood and challenged. In a next step, water circularity and reuse options should be considered. Cooling uncontaminated water and using it for other processes or in gardening is an option. Water-saving process improvements (e.g., reducing/eliminating blowdown) and water-saving technology should be investigated and installed as part of regular obsolescence/replacement planning.

In manufacturing, reducing the consumption of water per batch and adjusting cleaning recipes should be evaluated and implemented with proper cleaning validation. Additionally, when managing the use of water for manufacturing cleaning processes, water quality considerations should be addressed. Water to be discharged after cleaning processes should comply with local regulations. Any introduction of active pharmaceutical ingredients or excipients, hazardous substances including substances of concern into the water discharge, should be minimized or even eliminated. Examples are introduction of dry before wet cleaning and segregation of water flows.

Reducing water consumption should also consider the environmental impact on surface water, runoff water, and floodwater on-site, and the impact to public water systems. With the rise in flooding events, occurrences that were once deemed 1 in 100 year events are now happening more frequently. The current aqueduct underground infrastructure is struggling to handle even regular rain events that surpass the capacity of such systems. Addressing water management not only helps with sustainability but also supports biodiversity and site regeneration efforts, highlighting the importance of holistic environmental considerations in construction and infrastructure planning.

Designing the location of utilities and buildings on a site should consider factors such as local history, proximity to bodies of water, for example, lakes or rivers, and accessibility. It is essential to assess known flood levels and anticipate potential changes due to environmental factors when determining the elevation of utilities, equipment, and structures. Taking worst-case flood scenarios into account helps prepare for resilience against flooding risks. Landscaping can be utilized to protect facilities by implementing measures such as:

- Creating natural barriers using strategically placed vegetation or berms
- Reducing the amount of hard paving and using permeable paving to facilitate water infiltration and reduce runoff

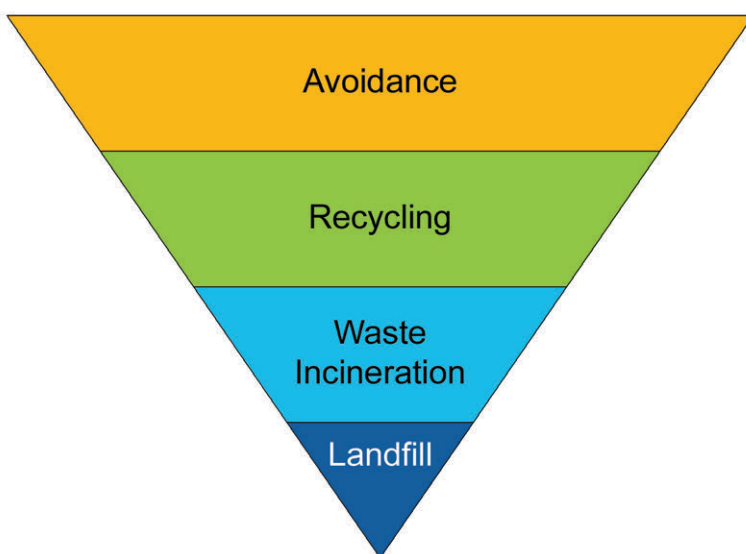
- Specifying xeriscape landscaping with exclusively local native plants that are drought-tolerant and do not require watering once established is a smart strategy for sustainable water usage. By utilizing native species adapted to the local climate, an eco-friendly and low-maintenance landscape can be created that conserves water and supports biodiversity.
- Utilizing rain gardens or bioswales to manage stormwater and prevent flooding
- Where green or living roofs are used, consider opting for vertical gardens; this frees up roof space that can then be utilized for solar panels, maximizing energy generation potential. Furthermore, vertical gardens can manage run off from roofs, offer insulation benefits, and help protect the building from solar gain, making them a multifunctional and sustainable choice in urban environments.
- Reusing surface water for various applications can be a sustainable practice in building management. By implementing filtering, RO, and UV treatment, recycled water can be used for closed-loop utilities, equipment cleaning, and toilet facilities. This can significantly reduce water consumption and minimize the demand for freshwater resources.

By integrating these landscaping strategies, the site can be better equipped to withstand and mitigate the impacts of floods while enhancing its environmental sustainability.

While conceptually similar to water conservation, waste and material reduction and conservation measures imply more value-chain integration, and technology innovation is also available with a wider network of suppliers and partners. The hierarchy of waste (Figure 3.3) helps to classify current waste disposal routes and sets the status quo. Waste diverted into landfills should be reduced to the absolute minimum or eliminated altogether. Equally, waste incineration (with or without thermal energy recovery) needs to be scrutinized and eliminated as much as possible. However, as the industry turns from a linear to a circular economy business model, recycling (and not only downcycling) and avoidance need to become the norm.

Figure 3.3 presents the transition from a linear to a circular economy, away from landfilling and waste incineration to recycling and avoidance schemes.

Figure 3.3: Waste Hierarchy



Both hazardous (e.g., chemical and pharmaceutical) and nonhazardous waste (e.g., packaging) is generated as part of the development and manufacturing process in the pharmaceutical industry. This waste may also be generated at hospital sites and patient dwellings, thus extending the reach of this waste generation.

Extended producer responsibility schemes and industry voluntary actions aim at reducing waste generation. Continual improvement processes along the value chain can help reduce waste quantities; however, product design plays a vital role and needs to be scrutinized for maximum effect.

Examples of successful pharmaceutical waste initiatives have been collated recently [38]. These include aspects such as solvent recycling, increased packaging recycling, and take-back schemes. In many cases, savings on material even deliver a favorable business case, which may make waste avoidance and the entry into the circular economy an economically attractive option.

While solutions within the scope of a single manufacturing site or at the company level are easier to implement, true cross-value chain partnerships (including downstream waste contractors) are essential as they offer the most significant opportunities for impact. It is crucial to assess the impact on GxPs and safety throughout the process and continuously maintain the quality of both the process and the product.

Ideally, a strategy to build a circular economy plan is to start measuring waste as much as possible along the value chain. First, identify and realize early quick wins and set up roadmaps on waste avoidance and recycling in manufacturing. Then start working on aspects of product design, scrutinizing single-use versus multiple-use, and focusing on secondary and tertiary packaging. Also integrate the circular economy into the product development cycle and have a strategy to optimize commercial products. Take-back schemes and patient education practices are a good start to work on end-of-life aspects. While typically end-of-life aspects are the hardest to solve, it is in this area where the highest impact for patients and buyers may be achieved.

3.6 Environmental Sustainability Metrics for Pharmaceutical Operations

3.6.1 *Sustainability Metrics and Their Role in Shifting Behavior*

Environmental sustainability metrics are crucial for the pharmaceutical industry for several compelling reasons. First, the industry has a notable environmental footprint, encompassing GHG emissions, water usage, and waste generation. By quantifying these impacts, companies can pinpoint areas needing improvement. Additionally, regulatory bodies are increasingly demanding sustainability reporting and adherence to environmental standards, making metrics essential for compliance. Stakeholders, including investors, customers, and employees, are calling for greater transparency and accountability in environmental practices, and metrics provide a clear, measurable way to communicate these efforts. Furthermore, tracking sustainability metrics can lead to more efficient resource use, cost savings, and enhanced operational performance.

These metrics can significantly influence behaviors and actions within the industry. By setting SMART goals, companies can chart a clear path toward sustainability. Regular tracking of these metrics allows companies to monitor their progress and identify areas needing additional effort. Applying lean and statistical process control principles to ongoing logged data and calculated metrics enables the identification of performance trends and potential improvement opportunities. The drive to meet sustainability targets can also spur innovation in processes, products, and technologies, such as adopting green chemistry practices to reduce the environmental impact of drug manufacturing.

Moreover, sustainability metrics create a framework for accountability. Publicly reporting sustainability performance enables stakeholders to hold companies accountable, driving continual improvement. Embedding these metrics into corporate performance reviews and incentive structures can shift organizational culture toward more sustainable practices.

3.6.2 Selected Sustainability Metrics for Pharmaceutical Operations

Several sustainability metrics are particularly recommended for the pharmaceutical industry due to their significant impact and the ability to drive meaningful change. These metrics are recommended because they address the most significant environmental impacts of pharmaceutical operations and should provide clear, actionable insights that can drive improvements. By focusing on these areas, pharmaceutical companies can enhance their sustainability performance, comply with regulatory requirements, and meet the growing expectations of stakeholders.

Measurement only adds value if it drives action or learning, and results in decision-making. Before establishing formal (and even informal) metrics and reporting, it is advisable to consider:

- How will the metric help the move toward the organization goals and objectives?
- How should the targets be determined?, which group sets these metrics and on what basis?
- How reliable, repeatable, and accurate do the metrics need to be?
- How can the metric be leveraged to engage and motivate employees to learn and innovate?
- Are there any balancing metrics required to guard against unintended consequences?

The following are some examples of metrics used in the industry:

- **Carbon Footprint:** Measuring total greenhouse gas emissions from operations

Carbon footprint is a critical metric because the pharmaceutical industry is one of the most carbon-intensive sectors. Tracking and reducing GHG emissions, particularly Scope 1 (direct emissions from the organization's facilities) and Scope 2 (indirect emissions from purchased energy), is essential for mitigating climate change. Many companies are also focusing on Scope 3 emissions, which include all other indirect emissions across the supply chain, as these often represent the largest portion of their carbon footprint.

- **Water Usage:** Tracking the amount of water used in manufacturing processes

Water usage is another vital metric. The pharmaceutical industry relies heavily on water for various processes, and with global water scarcity becoming a pressing issue, efficient water management is crucial. Monitoring water usage helps companies identify opportunities to reduce consumption and implement water-saving technologies.

- **Waste Generation:** Quantifying the volume of hazardous and nonhazardous waste produced

Waste generation is also a key metric. Pharmaceutical manufacturing can produce significant amounts of hazardous and nonhazardous waste. By tracking waste generation, companies can develop strategies to minimize waste, improve recycling rates, and ensure proper disposal methods, thereby reducing their environmental impact.

- **Energy Efficiency:** Monitoring energy consumption and efficiency improvements

Energy Efficiency is important for reducing operational costs and environmental impact. By monitoring energy consumption and implementing energy-efficient practices and technologies, pharmaceutical companies can lower their energy use and associated emissions.

- **Energy Use Intensity (EUI)** is a crucial metric for assessing the energy efficiency of pharmaceutical manufacturing facilities. It measures the amount of energy consumed per unit area of a facility, typically expressed in kilowatt-hours per square meter per year (kWh/m²/year) or British thermal units per square foot per year (Btu/ft²/year). EUI helps identify how efficiently a facility uses energy, highlights opportunities for energy-saving measures and allows pharmaceutical companies to benchmark their energy performance against industry standards or similar facilities.
- **Sustainable Sourcing:** Sustainable sourcing assesses the sustainability of raw materials and supply chain practices. This metric ensures that the materials used in pharmaceutical products are sourced responsibly, with minimal environmental impact and ethical considerations in mind.

3.6.3 Sustainability Metrics as Part of a Balanced Scorecard

The Balanced Scorecard (BSC) is strategic tool helping organizations translate their vision into actionable objectives across financial, customer, and internal processes, while learning and growth areas can help guard against unintended consequences of sustainability measures driving undesirable behaviors or actions.

Integrating sustainability metrics into the BSC, known as the Sustainability Balanced Scorecard (SBSC), enhances its relevance by including ESG factors. This integration provides a comprehensive view of performance, balancing short-term financial goals with long-term sustainability objectives. It aligns the corporate strategy with societal goals such as reducing carbon emissions and promoting social equality, thus enhancing reputation while driving forward compliance with regulatory requirements. Sustainability metrics also help identify and mitigate risks related to environmental and social issues, encourage innovation, and improve transparency. Reporting on these metrics builds trust with stakeholders and can lead to competitive advantages. By incorporating sustainability metrics into the BSC, organizations ensure their sustainability initiatives are aligned with overall business goals, driving long-term success and resilience.

3.7 Viewing CaPEX, OpEX, and Total Cost of Ownership (TCO)

Sustainable projects often have longer payback periods because they prioritize environmental and social benefits alongside financial returns. This longer payback period is necessary to account for the additional costs associated with sustainable features or technologies, but in the long run, these projects can lead to significant cost savings and a positive impact on the environment.

Broadening the scope of the financial investment to the entire lifecycle of a system or project often reveals that although there might be a higher upfront capital investment, the long-term operational savings over time can more than make up for the expense. This mindset shift can help in demonstrating the financial and energy-saving potential of sustainable solutions. Funding opportunities, grants, and incentives can further support these initiatives and help bridge the initial investment gap. The operational savings generated from adopting sustainable practices over the payback period can serve as a valuable source of funds to repay any initial capital investment or loans.

4 Environmental Sustainability of Pharmaceutical Products

4.1 Introduction

Sustainability by Design (SbD) as a framework aims to integrate sustainable practices throughout the entire business aspects of developing, launching, and managing a product in the market. It considers the environmental impact of a product throughout its lifecycle, from raw material extraction to final disposal, to avoid shifting environmental impact from one part of the lifecycle to another.

Innovation plays a crucial role in identifying environmental footprint reduction opportunities by adopting new technologies and methods, such as single-use technologies, continuous processing, and digitalization. Resource efficiency focuses on minimizing the use of materials and energy in the manufacturing process by selecting materials with a lower environmental footprint and optimizing processes to reduce resource consumption. This includes using intensified processes to increase efficiency and reduce the need for new facilities and equipment. Circularity aims to keep resources in use for as long as possible, extracting maximum value from them, and recovering and regenerating products and materials at the end of their life. Waste reduction strategies involve reducing waste generation and promoting reuse and recycling by designing processes that generate fewer waste streams and selecting materials that are easier to recycle. Implementing processes that allow for the reuse and recycling of materials can significantly reduce waste and resource consumption. Opting for consumables that are biodegradable or can be easily recycled reduces the environmental impact of waste disposal and supports circularity.

There are three distinct phases in the product concept lifecycle where sustainability impacts have different characteristics:

- Drug Research and Discovery (until candidate selection)
- Drug Development – Clinical Development (until launch)
- Marketed Product Life Cycle Management

Each of these phases presents unique opportunities for enhancing sustainability, as well as inherent limitations that must be addressed.

By understanding the unique characteristics and challenges of each phase, targeted strategies can be developed to enhance sustainability across the entire product lifecycle. In general terms decisions made on product characteristics and process routes during the early stages of development often have fundamental sustainability implications that cannot be easily reversed at later stages.

4.2 Drug Research and Discovery Sustainability Strategies

In the drug research and discovery phase, the focus is on identifying and developing new compounds. This phase offers significant potential for sustainability improvements through the adoption of a green chemistry framework that aims to design chemical products and processes that reduce or eliminate the use and generation of hazardous substances. The 12 Principles of Green Chemistry provide a roadmap for sustainable innovation, focusing on aspects such as waste prevention, atom economy, and the use of safer solvents [39]. To operationalize these principles, tools such as DOZN™, a quantitative green chemistry evaluator, enable companies to benchmark processes against metrics such as energy efficiency, toxicity, and resource use [40]. Additionally, conducting a LCA helps identify and quantify the environmental impacts at each stage of the product's lifecycle. However, the inherent uncertainty and complexity of this phase can pose challenges for consistently applying these principles.

4.2.1 Medicinal Chemistry/Biology

Reducing the daily API dose required can ultimately decrease the manufacturing volume and associated waste. Technological approaches, such as improved oral bioavailability for low solubility/high permeability compounds, aid this goal. Medicinal chemists play a key role in identifying high-potency compounds within optimal physicochemical parameters.

Medicinal chemists must generate numerous compounds quickly, often resulting in unoptimized routes with respect to yield, waste, and environmental impact. Implementing continuous synthesis from the outset can expedite access to structural analogs, enhancing efficiency and sustainability in drug discovery.

4.2.2 Target Selection and Optimization

Designing APIs with high specificity, low doses, high bioavailability, and minimal adverse effects benefits both patient health and environmental exposure. While environmental efforts often focus on manufacturing practices, intrinsic drug design also plays a crucial role in designing sustainability practices into the products.

When screening new drug targets and modalities, software is used to evaluate hundreds of potential structures and pharmacophores. Compounds are selected based on therapeutic potential and market approval likelihood. Early software screening identifies off-target toxicities (e.g., liver, kidney) and mutagenic activity (Lhasa Derek, Leadscape). [88, 89] Using screening software for exotoxicological endpoints can offer early detection before it becomes an issue.

Targeted treatments, such as monoclonal antibodies, are more efficient and reduce off-target toxicities, leading to lower doses and shorter treatment durations, although they consume more energy, water, and generate more waste per dose. Small-molecule manufacturing involves numerous chemical intermediates with potential environmental impact compared to biologics production. Biologics, being more biodegradable, pose less risk to aquatic environments. Ultimately, designing highly bioavailable, water-soluble drugs minimizes patient doses and environmental exposure.

Small-molecule drug compounds' overall quality is evaluated based on binding potency, selectivity, and Absorption, Distribution, Metabolism, and Excretion (ADME) properties. Predictive assessment of a molecule's target profile is crucial for clinical trial success.

4.2.3 Reduction of Hazards

Side effect and off-target toxicity assessments, typically addressed through animal studies, follow Organisation for Economic Co-operation and Development (OECD) guidelines. Both Good Laboratory Practice (GLP) and non-GLP studies elucidate a compound's mechanism of action and target organ toxicity, varying by molecule type and region.

In silico techniques offer multiple applications, including identifying toxic liabilities before testing, optimizing molecular design and binding fit to protein targets, and predicting overall physicochemical properties of drug compounds. By designing more potent and longer-lasting drug compounds, these techniques reduce dosage requirements, inherently contributing to green drug discovery. Additionally, using *in silico* tools minimizes the need for extensive laboratory experiments, thereby reducing the consumption of resources and generation of waste.

Persistent, lipophilic compounds are more likely to bioaccumulate. Thus, developing less lipophilic, water-soluble drug compounds aligns with reducing environmental impact.

Environmental Risk Assessments (ERAs), mandatory for new drug registrations by European Medicines Agency (EMA), evaluate API discharge into wastewater and sediments. These assessments, part of drug application submissions, follow OECD guidelines. Industry efforts to improve conventional toxicity testing include *in vitro*, *in silico* methods, and ML, some of which are regulator-approved [41].

4.2.4 Laboratory Research

4.2.4.1 Emerging Technologies

Staying updated with laboratory technological advances can significantly reduce our environmental impact. Digital tools such as AI and ML enable researchers to analyze vast datasets quickly, identifying potential drug compounds faster than traditional methods. This not only speeds up the drug development process but also reduces the environmental footprint. AI-driven technologies, such as graph neural networks, have been used to predict antibiotic activity and cytotoxicity for millions of compounds, identifying effective compounds against resistant bacteria [42]. By implementing smaller sample sizes, miniaturized assays, and greener methods, sustainability goals can be met while minimizing resource consumption and waste generation. AI algorithms, including support vector machines, random forests, and neural networks, enhance drug target discovery and analysis, streamlining therapeutic interventions and paving the way for personalized healthcare [42]. Optimizing resource use and reducing the need for physical experiments further helps in minimizing the environmental footprint of pharmaceutical research.

Optimizing method conditions to improve throughput and assay robustness while minimizing material inputs and method time is crucial for analytical sustainability. Project teams can explore a wide process design space early in development to identify optimal conditions. High throughput screening equipment can rapidly explore various conditions for a given transformation, and once proof-of-concept conditions are identified, it aids in completing product testing efficiently. Using a Design of Experiment (DOE) approach, rather than a one-variable-at-a-time method, reduces time and testing, effectively optimizing assay parameters.

4.2.4.2 Method Selection

Switching from solvent-intensive chromatographic methods to spectroscopic approaches can eliminate the need for extra solvents and can be used in manufacturing to minimize plant down time. At-line testing methods involve taking samples from the production process and analyzing them near the manufacturing line. This approach allows for more complex and precise measurements than in-line or online methods, as the samples can be prepared and analyzed under controlled conditions. By implementing at-line testing methods, manufacturers can obtain real-time insights into the production process, enabling quick adjustments to maintain quality and efficiency.

Whenever possible, High Performance Liquid Chromatography (HPLC) methods should be updated to Ultra Performance Liquid Chromatography (UPLC). This switch increases sample throughput, reduces operating expenses, and cuts buffer/solvent use and waste generation. Transferring USP compendial methods from HPLC to UPLC can reduce solvent consumption by over 90% [43]. Additionally, UPLC is more energy-efficient than HPLC, potentially reducing electricity usage and associated Scope 2 emissions.

Solvent concerns also exist in such areas as oligonucleotide biomanufacturing. Oligonucleotides are usually produced at-scale through solid-phase oligonucleotide synthesis. As currently applied, these employ large amounts of hazardous solvents and reagents and demonstrate a high carbon footprint. The need for greener reagents, solvents, and technologies was recognized by the FDA and the result was a request for better analytical methods and more stringent impurity specifications [44].

Integrated chromatography and spectroscopy techniques, such as Mass Spectrometry-Liquid Chromatography (MS-LC) and Mass Spectrometry-Gas Chromatography (MS-GC), offer significant sustainability advantages over using these methods individually. By combining the high separation efficiency of LC/GC with the precise molecular identification capability of MS, integrated systems enhance analytical throughput, reduce sample and solvent consumption, and minimize waste generation. This efficiency translates into lower energy consumption and resource usage, aligning with green chemistry principles. Additionally, integrated systems enable real-time process monitoring in pharmaceutical manufacturing, reducing the need for excessive batch testing and reprocessing, thereby improving process efficiency and sustainability.

Another key advantage is enhanced accuracy and specificity, reducing false positives and negatives, which minimizes unnecessary returns and resource waste. This is particularly critical in Quality Control and regulatory compliance, ensuring that pharmaceutical products meet stringent safety and efficacy standards while reducing material and energy-intensive validation steps. Furthermore, integrated techniques facilitate continuous manufacturing and Process Analytical Technology (PAT), supporting the industry's shift toward more sustainable and adaptive production processes. These improvements contribute to reducing the overall carbon footprint of pharmaceutical manufacturing while maintaining high product quality and safety.

Furthermore, the carbon footprint associated with the energy consumption of spectroscopic equipment and at-line testing methods should be considered (Scope 1 emissions). Cleanroom operations, which are essential for maintaining controlled environments, significantly impact energy consumption and emissions. The manufacturing equipment used in spectroscopic and chromatographic methods also contributes to Scope 3 emissions, considering their utilization and lifespan.

4.2.4.3 Use of Fetal Bovine Serum (FBS) in Cell Based Assays

The use of FBS in cell culture originated in the late 1950s, when it became integral for cell proliferation. However, FBS production is linked to the cattle industry. FBS production has significant environmental impacts, including global warming potential, eutrophication, acidification, land use, water consumption, and fossil-resource scarcity. Replacing FBS with serum substitutes offers reduced environmental impact, consistent quality, and reliability, although they come with higher costs. Reducing FBS concentration in culture media from 10% to 2% also leads to substantial environmental benefits. In addition, human platelet lysate has been shown to be an effective substitute for FBS in promoting adhesion, survival, and proliferation of human Mesenchymal Stem Cells (hMSCs).

While completely eliminating FBS may not be feasible, reducing its use is beneficial. Conversely, FBS being an animal by-product supports the sustainable use of cattle resources.

4.2.4.4 Plastics Usage and Recycling

Plastics, particularly single-use, are extensively used in the life sciences industry. Although completely eliminating plastic use is impractical, reducing its usage and environmental impact is essential. Some strategies to achieve this include:

- Replacing single-use plastics with low carbon glass, metal, or durable plastics
- Miniaturizing methods
- Bulk ordering plastic consumables
- Choosing products with eco-friendly packaging
- Finding alternative uses for plastic packaging

One of the key benefits of using glass, metal, and more durable materials is their reusability. These materials can be cleaned and sterilized for repeated use, which significantly reduces waste compared to single-use plastics.

In pharmaceutical R&D, significant efforts are being made to reduce plastic waste and enhance recycling options. For example, advanced recycling technologies are being developed to chemically break down used plastics into their base components, which can then be reconstituted into new laboratory-quality materials [46]. Additionally, the industry is exploring biodegradable and compostable materials for packaging and drug delivery systems [47, 48].

4.2.5 Data and Digital Technology

Leveraging digital technology to improve sustainability is referred to as sustainable digitalization. Examples of sustainable digitalization include green initiatives that digitize paper-intensive processes, such as electronic laboratory notebooks and digital libraries of documents for regulatory information management and quality document systems. Predictive manufacturing platforms help minimize resource consumption and waste, while augmented and virtual reality glasses may reduce travel requirements. Previously mentioned *in silico* methods and ML are already utilized to predict hazards, although regulatory approval of these methods is still under development.

Regulators are optimistic about digitized approaches, although more data is needed as these technologies mature. Early engagement with regulators is recommended to gather feedback and build trust in the approach. Regulatory acceptance is advancing, as seen in the FDA's 2025 draft guidance, but full approval as standalone tools requires robust validation and standardization [49]. Currently, these applications operate within hybrid frameworks, where computational insights inform – but do not replace – experimental and clinical workflows.

4.2.6 My Green Lab® Initiatives

Laboratories can foster a culture of sustainability by adopting eco-friendly practices, reducing waste, conserving energy and water, and engaging employees in sustainability initiatives. An example of such efforts is the My Green Lab certification, which aims to promote sustainability within scientific laboratories which are often resource-intensive environments.

My Green Lab is a leading organization in this field, working globally to integrate sustainable practices into the scientific community [50]. They offer various programs and tools, such as the Accountability, Consistency, and Transparency (ACT) label, which ranks the sustainability of laboratory products, and the Center for Energy-Efficient Laboratories (CEEL), which researches energy-saving opportunities for laboratory equipment.

Similarly, the National Institutes of Health (NIH) Green Labs Program focuses on increasing awareness and participation in sustainable practices among NIH laboratory personnel [51]. This program includes initiatives for energy conservation, water management, waste reduction, and green chemistry.

4.3 Sustainability by Design for Drug Development

During the drug development phase, the sustainability emphasis shifts to optimizing the formulation and manufacturing processes. This phase provides opportunities to implement resource-efficient practices, such as continuous processing and the use of single-use technologies, which can reduce energy and material consumption. Additionally, this phase focuses on ensuring the stability and efficacy of the drug, scaling up production from laboratory to commercial scale, and conducting extensive clinical trials to demonstrate safety and effectiveness. However, the need for rigorous testing, speed to market, and regulatory compliance can limit the extent to which these resource-efficient practices can be adopted.

The stringent requirements for clinical trials and regulatory approvals often necessitate the use of established processes and materials, which may not always align with the latest sustainability innovations. Furthermore, the pressure to bring new drugs to market quickly can sometimes prioritize speed over sustainability, making it challenging to fully integrate resource-efficient practices into the development process.

4.3.1 *Process Innovation and Improvement for Reducing Environmental Impact*

To address the pressing need for reducing environmental impact in pharmaceutical manufacturing, process innovation and improvement must be at the forefront of industry efforts. Transitioning from traditional batch production to continuous production presents a significant opportunity for environmental and operational benefits. Continuous production can optimize resources, reduce downtime, and offer tighter control over processes, potentially leading to lower overall energy consumption and waste generation. By streamlining processes and reducing the need for large scale batch facilities, continuous production can decrease the physical footprint of manufacturing plants. This reduction in land use can help preserve natural habitats and reduce the environmental impact associated with construction and land development. Additionally, continuous production may use less water for cooling and cleaning, potentially contributing to lower overall water consumption and reducing the strain on local water resources.

Solvent recovery represents another vital area for process improvement. Given that 80–90% of pharmaceutical processes involve the use of solvents [52], which are responsible for around 60% of the energy consumption in the manufacture of active ingredients, enhancing solvent recovery can lead to substantial per-product emissions reductions. Solvent recovery not only allows for the reuse of solvents, reducing the need for new solvent production, but also minimizes waste generation by adopting a circular economy approach. By recovering and reusing solvents, pharmaceutical manufacturers can significantly reduce the volume of hazardous waste that needs to be treated or disposed of, thereby lessening the environmental impact of waste management.

Another critical innovation is the adoption of biocatalysis and enzymatic processes, which utilize enzymes or whole-cell biocatalysts derived from biological sources to catalyze chemical reactions. These processes can replace traditional chemical synthesis methods that often involve harsh conditions, hazardous reagents, and the generation of significant waste and emissions. Recent advancements in enzyme purification have further enhanced the efficiency and sustainability of biocatalytic processes [53]. This innovative approach not only simplifies the purification process but also ensures sustained enzyme activity, crucial for the efficiency of biocatalytic reactions. These processes typically produce fewer by-products and hazardous waste, which reduces the impact on waste management systems and minimizes the risk of environmental contamination. Furthermore, the use of water-based solvents in biocatalysis can lead to more sustainable water management practices, as these solvents are generally less harmful to aquatic ecosystems compared to traditional organic solvents.

PAT plays a significant role in enhancing sustainability within pharmaceutical manufacturing. PAT is designed to monitor and control manufacturing processes through real-time measurements of Critical Quality Attributes (CQAs) and Critical Process Parameters (CPPs). PAT emphasizes Quality by Design (QbD), where product quality is engineered into processes rather than validated through post-production testing. This paradigm shift reduces reliance on end-product inspections and enables proactive adjustments during manufacturing, thus promoting the minimization of deviations, which could lead to waste reduction. By enabling real-time monitoring and control, PAT helps identify inefficiencies early in the production process, preventing the production of out-of-specification products and significantly reducing waste [54, 55].

The integration of digital tools and automation in pharmaceutical manufacturing can optimize processes and significantly reduce material and energy consumption. By leveraging advanced data analytics, real-time monitoring, and automated control systems, manufacturers can achieve greater precision and efficiency in their operations.

Aiming for shorter campaign lengths by using available equipment efficiently in multi-product facilities is another strategy to consider. By reducing the time dedicated to each campaign, facilities can potentially manufacture more products within a given period, thus optimizing resource use and reducing energy consumption per campaign. This approach not only increases the throughput of the facility but also allows for greater flexibility and responsiveness to market demands, ultimately contributing to a more sustainable manufacturing process. However, a case-by-case assessment is needed to ensure adverse impact is mitigated.

During the development phase, storage temperature requirements for a drug or product should be considered. Whenever possible, it is advisable to develop products that can be stored and shipped at ambient temperatures, as this eliminates the need for refrigeration or cold temperature packaging, such as coolers and gel packs. This approach not only enhances energy efficiency but also reduces packaging waste.

4.3.2 *Material Selection and Recycling for Packaging and Device Design*

Selecting sustainable materials and implementing recycling strategies for packaging and devices are critical for reducing the environmental impact of pharmaceutical products. The choice of materials can influence the entire lifecycle of a product, from production to disposal, especially when final products are distributed and packaged in multiple thousands of vials or blister packs and include secondary packaging.

Key considerations for material selection include choosing materials with a lower environmental footprint, such as those made from renewable resources or recycled content. For example, using bioplastics derived from corn starch or sugarcane for vials and blister packs can significantly reduce carbon emissions compared to conventional plastics. Second-generation biomass feedstocks, such as used cooking oil, offer a more sustainable alternative to first-generation biofuels, which are typically derived from food crops such as corn and sugarcane. Opting for materials that can be easily recycled or have existing take-back programs is also essential. An example is the use of Polyethylene Terephthalate (PET) for secondary packaging, which is widely recyclable. Additionally, ensuring that selected materials comply with relevant regulations and do not pose risks to human health or the environment is crucial. For example, ensuring that the materials used for blister packs are free from harmful chemicals such as Bisphenol A (BPA) is necessary to comply with health safety standards.

Additional sustainable material considerations include using wet-strength paper labels and ensuring that closures made of plastic and aluminum can be separated efficiently. Type 1 borosilicate glass for vials and blisters can be recycled effectively. Additionally, optimizing the distribution network, given that empty glass vials occupy a large volume and require significant packaging material, can notably reduce the environmental impact. The packaging material used for distribution should be recycled or reused whenever possible.

Strategies for recycling involve designing packaging and devices so that they can be easily disassembled for recycling. An example is creating modular designs for secondary packaging where each component can be separated and processed individually. It is noted that monomaterial packaging is best to enable recycling. Implementing systems for recovering and recycling materials from used products can significantly enhance sustainability efforts. For example, establishing take-back programs where patients can return used vials and blister packs for material recovery can reduce waste. Additionally, working with suppliers who have strong environmental credentials and offer recycling programs is vital. Partnering with suppliers that provide certified sustainable materials or have robust recycling initiatives can help ensure a more sustainable supply chain, such as suppliers offering closed-loop recycling for packaging materials.

Further opportunities to enhance sustainability in packaging include optimizing packaging weight for a compact footprint, using fiber from certified sources such as Forrester Stewardship Council (FSC)-certified paper, and preferring uncoated and unlaminated versions to simplify fiber digestion and avoid contamination. It is crucial to use adhesive applications that do not form problematic stickies during the recycling process. Transitioning packaging printing to use Cyan, Magenta, Yellow, and Key (Black) (CMYK) rather than Pantone® spot colors, in line with company sustainability programs, can also make a difference. Providing clear disposal information/symbols on packaging or product websites can guide proper recycling. Finally, the use of e-leaflets can eliminate the need for additional leaflet paper, thus further reducing the environmental footprint.

4.3.3 *Material Selection for Product Design*

Eventually, material usage in pharmaceutical packaging and device design cannot be meaningfully reduced any further. However, making smart choices in selecting input materials and incorporating lifecycle thinking into process development can significantly mitigate environmental impacts.

In the context of environmental sustainability, "lifecycle" refers to the environmental impact of a product throughout its entire lifecycle, including extraction and processing of raw materials, manufacturing, distribution, use, recycling, and final disposal. To avoid shifting environmental impact from one part of the product lifecycle to another, considering the full lifecycle during process development is essential. Lifecycle thinking provides a framework to consider tradeoffs that may result from decisions made during process development.

Traditionally, the scope considered during process development is narrow, focusing on the processes required to transform raw materials into API and drug products. Expanding this scope to include the production of consumables and raw material inputs, as well as waste streams produced during manufacturing and subsequent treatment and disposal, opens new avenues for reducing environmental impacts through process design.

To minimize lifecycle environmental impacts, several strategies should be considered. Selecting sustainably sourced, bio-based, renewable, recycled, or recyclable consumables is crucial, provided that an LCA is conducted to evaluate the potential for any unintended impacts. While consumables are often downcycled, choosing options that allow recycling or partnering with vendors with take-back programs is preferable over incineration or landfill disposal. For example, incineration with energy recovery can mitigate some greenhouse gas impacts compared to landfilling.

When selecting raw material inputs, it is important to consider their sources and impacts. Materials introduced to an API or Drug Product (DP) process carry their own environmental footprints. Nearly 99% of pharmaceutical feedstocks are derived from petrochemicals. Choosing lower-impact materials, such as cationic resins over anionic resins, and renewable materials such as plant-derived carbohydrates or organic acids, can reduce these impacts. Bio-based alternatives, such as tubes made from bioplastic, are also preferable.

Minimizing overall waste generation and creating fewer waste streams can reduce the environmental impact. Designing processes to reduce the number of individual waste streams decreases waste treatment and disposal requirements. Early in-process development, teams should map out waste stream components and evaluate treatment and disposal needs. Using biodegradable and low-toxicity buffers and additives can simplify waste treatment, and avoiding nitrogen and phosphorus-containing buffers reduces the risk of eutrophication.

Reducing the impacts of equipment cleaning is another critical factor. Designing processes with minimal cleaning requirements, such as converting a batch process to continuous processes or using single-use equipment, can significantly cut down on waste and cleaning material usage.

It should be noted that single-use technologies have become increasingly popular in the pharmaceutical industry due to their convenience and ability to reduce cross-contamination risks. However, their environmental impact is a subject of debate. On one hand, single-use systems can minimize the need for extensive cleaning and sterilization processes, thereby reducing water and energy consumption. They also offer flexibility and scalability, which can be beneficial for small-scale production and rapid response to market demands. On the other hand, single-use technologies generate significant plastic waste, which poses challenges for disposal and recycling. To address these conflicting views, it is essential to consider the specific context and application of single-use technologies and make an informed decision. For example, in scenarios where minimizing contamination risks and ensuring product safety are paramount, single-use equipment systems may be justified despite their environmental drawbacks. Conversely, in situations where sustainability goals prioritize waste reduction and resource conservation, alternative solutions such as reusable equipment systems should be explored.

4.4 Marketed Products Lifecycle Management

Finally, the marketed product lifecycle management phase involves the ongoing production, distribution, and post-market surveillance of the drug. This phase allows for the implementation of circular economy principles, such as promoting the reuse of materials. Additionally, it focuses on maintaining product quality, ensuring supply chain efficiency, and monitoring the drug's safety and efficacy in the market. However, the established infrastructure and practices in this phase can make it challenging to introduce significant changes without disrupting existing systems. Regulatory resubmissions and the need for continuous compliance with evolving regulations can further complicate the adoption of new sustainability practices. The requirement to maintain consistency in product quality and performance often necessitates adherence to established processes, which may limit the flexibility to implement innovative sustainability measures.

However, the high importance of bringing new medicines to the market, accelerating product development, and lowering operational costs often means that many potential sustainability improvements identified during drug development may not be implemented before launch. Therefore, the progress of product sustainability must also be embedded in the product's lifecycle management, from the product's initial launch until its final withdrawal. [110]

Product changes after launch are much more complex due to regulatory restrictions. Sustainability principles should be integrated into commercial product lifecycle management decisions, processes, and governing documents to ensure that commercial product strategies and processes align with corporate sustainability commitments in a measurable way.

Distinct opportunities for improving sustainability performance are feasible both before and after the regulatory approval and launch of the product, considering that patient centricity and quality attributes remain central to drug development and commercialization. Examples include yield improvement, solvent recycling opportunities, and packaging improvements. Additionally, device circularity has been successfully implemented through a mail-back program, which allows patients to return used devices by mail to a collection site, where they are autoclaved, shredded, and sent to a waste-to-energy facility for incineration with energy recovery.

Decisions to engage in retrofitting should consider factors such as the volume of product produced, process/energy intensity, and the stage of the product in lifecycle management. Including sustainability attributes and metrics in the development of stage-gating processes ensures that previously achieved sustainability improvements are not lost.

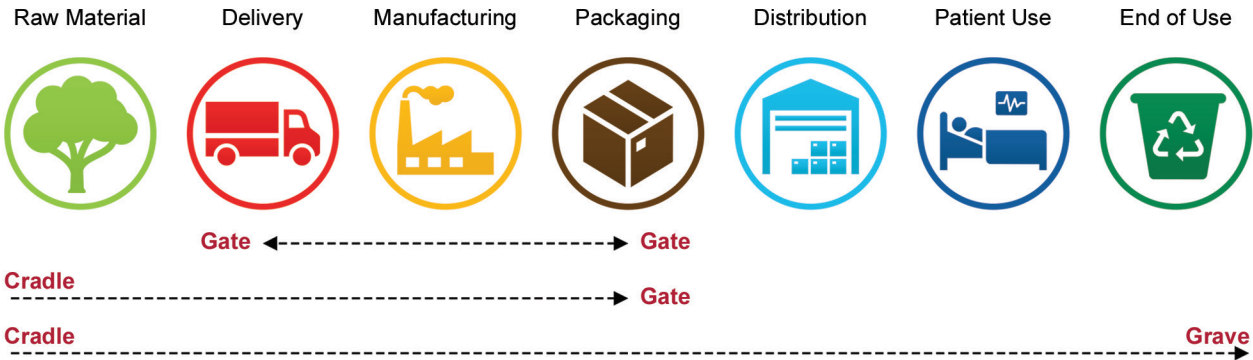
4.5 Life Cycle Assessment of Pharmaceutical Products

Pharmaceutical companies face rising demands for environmental sustainability data at the product level. Industry-wide efforts are underway to establish a robust, credible, and shared framework for measuring, reporting, and communicating product level environmental data. Ongoing work aims to create a pharmaceutical database of materials and processes to increase data quality and consistency of LCAs for various pharmaceutical modalities (e.g., small molecules, biologics, vaccines, plasma products, gene therapy).

LCA (ISO 14040 [121], ISO 14044 [122]) is methodology used to evaluate the environmental impacts of a product, process, or service throughout its lifecycle. LCA takes a wide range of environmental factors such as energy and material consumption, GHG emissions, water usage, waste generation and disposal, and other potential impacts on ecosystems and human health into account. It can help to identify opportunities for improving the environmental performance of a product (hotspots), as well as to compare different materials or alternative processes for the same product. LCA typically does not address the economic or social aspects of a product, but the lifecycle approach and methodologies described in ISO 14040 can be applied to these other aspects. By considering the entire lifecycle, LCA enables businesses to identify opportunities for improvement, optimize resource efficiency, and reduce waste.

LCA can encompass different boundaries, such as cradle-to-grave, cradle-to-gate, or gate-to-gate (see Figure 4.1).

Figure 4.1: Different Boundaries during Life Cycle Assessment (LCA)



Cradle-to-Grave: This approach considers the entire lifecycle of a product, from raw material extraction (cradle) to disposal (grave).

Cradle-to-Gate: This approach includes the stages from raw material extraction (cradle) to the factory gate, before the product is transported to the consumer.

Gate-to-Gate: This approach focuses on a specific part of the product's lifecycle, typically the manufacturing process within a factory.

An LCA can incorporate the **cradle-to-cradle** approach. Cradle-to-cradle is a variation of the cradle-to-grave model, but instead of ending with disposal, it focuses on recycling or upcycling materials to create new products, effectively closing the loop. This approach aligns with the principles of the circular economy.

These boundaries help in defining the scope of the LCA study and ensure that the assessment is comprehensive and relevant to the specific goals of the analysis.

LCA includes a variety of metrics to evaluate the environmental impacts of a product, process, or service. Some of the key metrics are:

- **Global Warming Potential (GWP):** Measures the impact on climate change, often expressed as CO₂ equivalents
- **Energy Consumption:** Total energy used throughout the lifecycle
- **Water Usage:** Amount of water consumed
- **Resource Depletion:** Use of non-renewable resources such as minerals and fossil fuels
- **Acidification Potential:** Impact on soil and water acidity
- **Eutrophication Potential:** Nutrient enrichment in water bodies leading to algal blooms
- **Ozone Depletion Potential:** Impact on the stratospheric ozone layer
- **Human Toxicity:** Potential harm to human health from chemicals
- **Ecotoxicity:** Impact on ecosystems from toxic substances
- **Particulate Matter Formation:** Contribution to air pollution and respiratory issues

These metrics help in understanding and mitigating the environmental impacts associated with different stages of a product's lifecycle.

Selecting appropriate metrics is crucial to drive meaningful actions. The key metric is GWP, which measures the impact on climate change and is often expressed as CO₂ equivalents. GWP is directly linked to GHG emissions, as it quantifies the potential of these gases to trap heat in the atmosphere, contributing to global warming.

4.5.1 LCA Tools

Several tools are available to help in calculating and modeling sustainability metrics for various pharmaceutical processes. Some notable LCA tools are:

- **SuperPro Designer® [58]:** This tool helps in modeling, evaluating, and optimizing processes, providing detailed insights into material and energy balances, economic evaluations, and environmental impacts.

- **SimaPro [59]:** One of the most well-known LCA software tools, SimaPro is used for detailed life cycle assessments. It allows users to model and analyze complex lifecycles, providing insights into environmental impacts across all stages.
- **BioSolve Process™ (BSP) [60]:** An Excel-based tool used industry-wide for economic and environmental modeling of processes and plants. BioSolve incorporates Process Mass Intensity (PMI) and operating CO₂ emissions, offering predictive information about environmental impacts.
- **PMI-LCA Tool by ACS-GCIPR [61]:** *“The PMI-LCA Tool is a high-level estimator of Process Mass Intensity (PMI) and environmental lifecycle information that can be customized to fit a wide variety of linear and convergent processes for synthesis of small molecule active pharmaceutical ingredients (APIs). An ecoinvent dataset is utilized as the source of the life cycle impact assessment (LCIA) data.”*

4.6 Environmental Sustainability Metrics for Pharmaceutical Products

In the pharmaceutical industry, the environmental footprint of drug development and manufacturing processes is increasingly under scrutiny. As companies strive to meet ambitious sustainability goals, it is essential to adopt metrics that can accurately measure and guide efforts to reduce environmental impacts. This chapter explores the key metrics for assessing the environmental sustainability of pharmaceutical products, focusing on waste, water use, hazardous material use, and carbon footprint.

4.6.1 Process Mass Intensity (PMI) and Waste

PMI is a mass-based metric that measures material efficiency by calculating the ratio of the mass of all material inputs to the mass of the isolated product. The simplicity of PMI allows it to be calculated using readily available lab-scale or production data.

Since waste is all input that is not product, PMI can easily be converted to the environmental factor (E-factor), which is PMI minus 1. The ideal E-factor is 0, indicating no waste generation. PMI is crucial for understanding the material efficiency of a process and identifying areas for improvement.

$$\text{Process Mass Intensity (PMI)} = \frac{\text{mass of all material inputs (kg)}}{\text{mass of product out (kg)}}$$

PMI calculation to assess waste impact has been used to assess the environmental footprint to produce a biotechnology products. [62, 63, 64]

Benchmarking PMI

The ACS GCI Pharmaceutical Roundtable [61] has collected blinded PMI data from member companies to benchmark typical PMI numbers for different modalities. For example, monoclonal Antibody (mAb) production has an average PMI of 7700, with water accounting for 94% of the PMI. This benchmarking helps companies understand where they stand relative to industry standards and identify opportunities for reducing material use.

Scope of PMI Calculation

For biologics, PMI includes raw materials, consumables, water, and excipients used throughout the process, including upstream, downstream, formulation, and cleaning operations. It excludes fixed equipment, waste treatment materials, and materials used in analytical methods. Breaking PMI down at different levels, such as production stages or specific unit operations, can help allocate development resources effectively.

4.6.2 Water Use

Carbon emissions from purifying water and treating wastewater account for a significant proportion of the total carbon footprint of biologic API and DP manufacturing. These water-related emissions can be modeled using the Water Related Impact of Energy (WARIEN) metric. WARIEN is a metric that couples carbon emissions from the energy needed to produce water for injection (WFI), purified water (PW) and pure steam/clean steam (CS – referring to Clean Steam), along with the energy needed to heat treat wastewater to the total mass of said materials (PMI).

For hot WFI skids (WFI produced via distillation of PW) it is calculated as follows:

$$\text{WARIEN} = \text{PMI}_{\text{WFI}} \times \text{CEQ}_{\text{WH}} + \text{PMI}_{\text{PW}} \times \text{CEQ}_{\text{PW}} + \text{PMI}_{\text{CS}} \times \text{CEQ}_{\text{CSH}}$$

Where: CEQ = Carbon Equivalent
WH = WFI Hot
CSH = Clean Steam Hot

For cold WFI skids it is calculated:

$$\text{WARIEN} = (\text{PMI}_{\text{WFI}} + \text{PMI}_{\text{PW}}) \times \text{CEQ}_{\text{WC}} + \text{PMI}_{\text{CS}} \times \text{CEQ}_{\text{CSC}}$$

Where: WC = WFI Cold
CSC = Clean Steam Cold

PW can be produced via membrane, such as RO-EDI or RO-RO, RO-RO-EDI, or it can be produced via vapor compression. WFI can be produced via membrane technology (RO-EDI-UF or RO-RO-EDI-UF or other combinations) and often is called “Cold WFI,” or via distillation via vapor compressor or multi-effect still (“Hot WFI”). In practice, the production of Cold WFI production skips the extra storage of PW. Pure steam/clean steam is in practice usually derived from either PW (usually for Hot WFI) or WFI (usually for Cold WFI). It is important to note that “Cold WFI” can officially be used according to the Ph. Eur., USP, and JP, with approval for China expected in the near future.

Strategies for Reducing Water Use

Companies can optimize processes to minimize water consumption, recycle and reuse water where possible, and implement single-use technologies that eliminate the need for cleaning and validation. These strategies not only reduce water use but also lower the associated energy demand and carbon footprint. However, the utilization of single-use technologies may require waste management whose overall environmental impact would need to be evaluated.

4.6.3 Carbon Footprint

The primary lifecycle impact tracked in the pharmaceutical sciences is the PCF. To calculate the cradle-to-gate carbon footprint of API and/or DP manufacturing, the following sources of carbon emissions must be estimated:

- Carbon emissions to produce material inputs (Scope 3)
- Carbon emissions from transporting raw materials and intercompany shipments (Scope 3)
- Carbon emissions from energy used in the plant during manufacturing (Scope 2)
- Carbon emissions from cleanroom operations (Scope 1)
- Carbon emissions from process energy, such as steam (Scope 1)
- Carbon emissions from capital goods, including manufacturing equipment [65] (Scope 3)

- Carbon emissions from treating and disposing of waste produced during manufacturing (Scope 3)

Carbon emission factors for some input materials are available in third-party databases such as Ecoinvent [120]. Missing emission factors must be modeled individually, either by selecting a proxy emission factor or by modeling the supply chain for the material.

To convert PMI into a carbon footprint, it is essential to calculate the GHG emissions associated with the production and usage of water, consumables, and materials. The following steps outline the methodology:

First, gather detailed data on the types and quantities of water, consumables, and materials used in the production process. This includes:

- Water: Mass of each clean water type consumed (kg)
- Consumables: Types and quantities of disposable items (e.g., filters, tubing) used
- Materials: Types and quantities of raw materials (e.g., chemicals, solvents) used

Second, identify the emission factors for each type of resource. Emission factors represent the amount of GHG emissions produced per unit of resource consumed. These factors can be obtained from databases such as the EPA [8] or the Intergovernmental Panel on Climate Change (IPCC) [123]. Examples include:

- Emission factor for water: kg CO₂e per kg (utilizing the WARIEN)
- Emission factor for consumables: kg CO₂e per unit
- Emission factor for materials: kg CO₂e per kilogram

Third, calculate the GHG emissions for each resource category by multiplying the quantity of resource used by its respective emission factor. This can be expressed as:

- GHG emissions (water) = Volume of water consumed × Emission factor (water)
- GHG emissions (consumables) = Quantity of consumables × Emission factor (consumables)
- GHG emissions (materials) = Quantity of materials × Emission factor (materials)

Finally, sum the GHG emissions calculated for water, consumables, and materials to obtain the total carbon footprint of the process. This can be expressed as:

- Total Carbon Footprint = GHG emissions (water) + GHG emissions (consumables) + GHG emissions (materials)

In addition to these resource categories, facility energy consumption is a critical factor in carbon footprint analysis. Energy emissions in clean rooms are significantly influenced by HVAC systems, which are essential for maintaining controlled environments. These systems often represent a substantial portion of the facility's energy consumption due to the stringent requirements for air quality, temperature, and humidity. Additionally, the energy demand of devices within clean rooms contributes to the overall carbon footprint, especially when considering the emissions from boilers used for heating and the specific grid mix of the location.

By optimizing process times and increasing yields, facilities can effectively reduce campaign lengths, allowing more products to be produced within the same infrastructure. This not only enhances productivity but also minimizes the energy required per unit of product, thereby reducing the overall carbon footprint. In facilities where energy consumption is the primary contributor to emissions, these efficiency improvements can have a profound impact on sustainability goals. Moreover, transitioning to greener grid mixes and integrating renewable energy sources can further mitigate the environmental impact.

In summary, careful management of HVAC systems, optimized process efficiencies, and strategic energy sourcing are pivotal in reducing the carbon footprint associated with clean room operations.

4.6.4 Hazardous Material Use

In the pharmaceutical industry, the use of hazardous materials is a critical consideration for sustainability. Substances of Concern (SoC) [66] and Substances of Very High Concern (SVHC) are chemicals that pose significant risks to human health and the environment. In the EU, applicable legislation includes REACH (Registration, Evaluation, Authorization and Restriction of Chemicals) [111], the CLP Regulation on Classification, Labeling and Packaging of Substances and Mixtures [112], and the Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment (RoHS) [113].

Table 4.1 shows a list of applicable legislations. These substances are identified based on their inherent properties, such as toxicity, persistence, and bioaccumulation. Moreover, the new definition of SoC, particularly in the context of product circularity and recyclability, has been evolving to address the challenges of a circular economy under the Eco-design for Sustainable Products Regulation (ESPR). SoCs are now defined as substances that not only pose risks to human health and the environment but also impede the recyclability and reusability of products [67].

Table 4.1: Hazardous Material Applicable Legislations

Montreal Protocol	List of Controlled Ozone-Depleting Substances
Stockholm Convention	List of Persistent Organic Pollutant (POPs)
WHO International Agency for Research on Cancer	List of Classified Carcinogens
Canada	Toxic Substances List and the Virtual Elimination List
European Chemicals Agency (ECHA)	• Candidate List of Substances of Very High Concern for Authorization • Substances classified as substances with Carcinogenic, Mutagenic or Reproductive (CMR) Toxicity Properties 1a or 1b under Annex VI of Classification, Labelling and Packaging (CLP) Regulation
US Environmental Protection Agency (EPA)	Toxics Release Inventory's Persistent, Bioaccumulative and Toxic (PBT) Chemicals List and PBT Chemicals under the Toxic Substances Control Act (TSA) Section 6(h)
US National Toxicology Program	Report on Carcinogens
State of California	Proposition 65 List

To substitute hazardous substances, companies should first create an inventory of chemicals used in processes and assess their hazards. They should then consider less hazardous chemicals, alternative techniques, or different product designs. Focusing on the function of the substance and exploring broad alternatives, including changes in production processes and product design, is crucial. Involving suppliers and customers in the substitution process helps identify concerns and opportunities. Finally, implementing the chosen alternatives and continuously monitoring their performance and impact ensures the effectiveness of the substitution.

Reducing the use of hazardous substances can lower regulatory compliance costs. Substitution supports sustainable chemistry and safe-by-design principles, contributing to a healthier environment and society.

Regarding hazardous material use, tracking and seeking alternatives for hazardous materials, such as Triton™ X-100 and Tri-n-butyl phosphate (TnBP), are essential to minimize risks to human health and the environment. Emerging SoC, such as Per- and Polyfluoroalkyl Substances (PFAS) and materials used in bacterial endotoxin testing (LAL-Test), are also gaining attention. Companies should proactively seek alternatives to these materials and adopt more sustainable practices.

Resources for sustainable chemical substances substitution:

- The OECD Guidance on Key Considerations for the Identification and Selection of Safer Chemical Alternatives provides a comprehensive framework for identifying and selecting safer chemical alternatives, focusing on sustainability and minimizing risks to human health and the environment [68].
- The ECHA Strategy to Promote Substitution to Safer Chemicals through innovation, outlines the European Chemicals Agency's approach to promoting the substitution of hazardous chemicals with safer alternatives, emphasizing innovation and sustainable production [69].

5 The Alignment of Sustainability, Quality, and Regulatory Aspects

5.1 Introduction

The pharmaceutical industry increasingly recognizes the importance of sustainability in drug development and manufacturing processes. As companies respond to the urgency of reducing environmental impact and adopting sustainable practices, integrating sustainability with quality and regulatory requirements becomes both essential and complex. Quality and regulatory frameworks are designed to protect patient safety, product integrity, and compliance with global standards, while sustainability introduces additional goals, such as reducing carbon footprints, minimizing resource consumption, and limiting waste. Achieving alignment among these three elements – sustainability, quality, and regulatory – requires balancing priorities without compromising product safety, efficacy, or regulatory compliance. Although synergistic in nature, integrating these practices presents challenges that require innovative approaches, cultural adaptability, and strong collaboration within and between organizations.

Sustainability goals in the pharmaceutical sector align with broader societal and healthcare priorities, such as those outlined in the EU Taxonomy and the UN Sustainable Development Goals [70, 71]. These include reducing GHG emissions, minimizing water and energy consumption, optimizing supply chain processes, and promoting circular economy practices. Such goals must align with existing quality and regulatory requirements to ensure that patient safety and product integrity remain uncompromised. The challenge lies in embedding sustainability into the pharmaceutical lifecycle while balancing technical feasibility, business economics, and compliance obligations. For example, adopting energy-efficient equipment or reducing cleanroom air change rates must demonstrate clear benefits without increasing product quality risk or regulatory challenges.

5.1.1 Quality Landscape

Quality in pharmaceutical development and manufacturing is fundamentally concerned with safeguarding the patient through product integrity and consistency. This focus is deeply embedded in the pharmaceutical quality culture and in frameworks such as ICH Q8, which describe QbD for the development of pharmaceutical products [72].

QbD aims to develop quality products and processes through active management of development knowledge to build a “design space” that uses scientific data to establish both Normal Operating Ranges (NOR) and a “control space,” which supports scale up and operation. In this way quality is “built-in” to the product and process. ICH Q8 [72] recognizes that “quality cannot be tested into products;” it needs to be a part of the development process and therefore considered at the very start of the product lifecycle. To be successful, QbD requires scientific data to support knowledge of what is quality critical and how that may change as a process is scaled up commercially. The more knowledge, the broader the design space and the more scope for operating within different parts of the design space while remaining compliant. The quality risk assessment is a vital part of this process.

The focus on the whole product lifecycle aims to build-in sustainability and in that regard, it is analogous to the approach taken in ICH Q8 [72]. As sustainability becomes an increasing focus, quality professionals face the challenge of integrating eco-friendly practices into these rigorous frameworks while maintaining existing standards for product quality, process control, quality risk management, and regulatory compliance.

5.1.2 Regulatory Landscape

Regulatory frameworks establish the “what” in pharmaceutical development and manufacturing – what companies must achieve to ensure compliance with global regulations to protect patients. These frameworks define acceptable risk thresholds within operational limits and standards. Companies, however, determine the “how” through their systems of work: validated processes, procedures, and technologies. While these systems of work are typically conservative to minimize risk, they also offer opportunities for improvement. As sustainability objectives emerge, regulatory functions must adapt to support innovative solutions that align with environmental goals while ensuring compliance with existing regulations.

5.1.3 Roles

The integration of sustainability into pharmaceutical operations requires a multidisciplinary approach involving key roles across the organization. Each function brings unique expertise to the process. Engineering and production disciplines play a critical role in designing and implementing sustainable solutions that are technically viable, balancing cost, feasibility, and operational impact. Product development teams ensure that sustainable practices are aligned with product development strategies and quality attributes. Regulatory affairs acts as a bridge between interpreting compliance requirements and facilitating regulatory acceptance of novel sustainability initiatives. Legal teams ensure alignment with global environmental and regulatory laws, while Quality Management provides the oversight needed to maintain product safety and integrity. The Health, Safety, and Environment (HSE) department focuses on minimizing environmental impact and ensuring workplace safety while adopting sustainable practices.

5.1.4 Synergy of Roles

In the pharmaceutical environment, decision-making around processes, equipment, and materials involves input from multiple functions, including production, quality, engineering, procurement, and HSE. Therefore, integrating sustainability into pharmaceutical operations requires cross-functional collaboration where each department brings unique expertise and priorities. Each function operates with distinct goals: production aims to produce quality products efficiently, quality ensures product safety and compliance, engineering balances cost and feasibility, and HSE protects employees and the environment. Harmonizing these priorities to achieve sustainable outcomes requires cross-functional collaboration, scientific understanding, and trust among stakeholders.

The principles of QbD, which emphasize building quality into processes from the outset, align closely with the lifecycle approach to sustainability, where active knowledge management and risk-based decision-making enable robust process design. Similarly, integrating sustainability into Chemistry, Manufacturing, and Controls (CMC) processes demands data-driven assessments to evaluate risk, impact, and feasibility, ensuring quality remains “built-in” throughout the product lifecycle. This approach ensures that quality and compliance remain robust while enabling innovation and environmental responsibility.

The introduction of sustainability often prompts novel approaches to familiar systems of work, challenging the status quo and practical challenges often arise when balancing sustainability with existing quality and regulatory frameworks. HVAC systems, for example, have traditionally relied on conservative airflow rates to maintain cleanroom standards. While such setups are effective, they are also energy-intensive. By adopting a sustainability approach, less intrusive and more energy-efficient solutions – such as optimized control systems or pretreatment technologies – can be implemented without compromising quality. Similarly, sterilization processes that rely on excessive overkill margins can be refined through improved risk assessments, reducing energy consumption while maintaining sterility assurance.

Packaging innovations also illustrate this dynamic. Standardizing materials and reducing secondary packaging can minimize waste, but such changes must consider risks associated with extreme storage conditions, especially in regions with challenging climates. Regulatory departments play a key role in assessing and facilitating such changes, ensuring they align with global compliance requirements. The integration of sustainability into manufacturing frequently reveals inefficiencies and opportunities for improvement. These discoveries may lead to additional controls, which could increase costs. Alternatively, it can prompt innovative thinking and actions that lower costs, enhance quality, and reduce environmental impact.

However, success requires a cultural shift within organizations. Teams must embrace change, prioritize knowledge-sharing, and build trust through collaborative decision-making. Quality professionals, supported by subject matter experts, must lead this change by balancing innovation with compliance. Management has a critical role in fostering this culture, creating an environment where sustainability is viewed not as a competing priority but as a means to drive innovation and resilience. By aligning sustainability with quality and regulatory goals, pharmaceutical companies can not only meet their environmental responsibilities but also drive innovation, reduce costs, and ensure compliance, ultimately creating greater value for patients, stakeholders, and the environment.

5.2 Interaction Between Sustainability and Quality Management

In the pharmaceutical industry, quality and sustainability share a foundational commitment to protecting human health, safety, and well-being. While quality ensures that pharmaceutical products meet rigorous standards for safety, efficacy, and continuous composition and high-quality content of the drugs, sustainability focuses on minimizing environmental impact and ensuring the long-term availability of resources. Both disciplines aim to reduce risks, improve efficiency, and foster sustainable innovation, which creates a strong basis for collaboration.

Sustainability should be seen as a natural extension of the Quality team's mission to safeguard patients, uphold product integrity, and strengthen operational resilience. Sustainable practices often complement Quality objectives by addressing risks such as supply chain instability and improving the sourcing of raw materials. In addition, it can reduce the likelihood of product recalls due to particulate and microbial contaminants.

To integrate sustainability into Quality, organizations should prioritize education and engagement through targeted training programs. It should be ensured that these training and information efforts are targeting the whole organization, including managers and senior management. These initiatives can feature case studies, regulatory updates, and hands-on workshops that demonstrate how sustainable practices contribute to quality outcomes. Training should focus on areas such as waste minimization, green chemistry, energy-efficient equipment, and the adoption of sustainable materials and practices, key strategies for reducing environmental impact while maintaining regulatory compliance. In addition to that, Quality teams should participate in discussions with regulators through industry forums, regulatory feedback sessions, or pilot programs that explore sustainable practices.

The adoption of digital Quality Management Systems (QMS) and advanced data analytics is essential to achieving these goals. By monitoring sustainability metrics in real time, organizations can identify opportunities where sustainable practices align with regulatory requirements and cost-efficiency. This data-driven approach empowers teams to make informed decisions that benefit both quality and the environment.

5.2.1 Opportunities for Synergy

1. Optimal Performance

Defining qualification and process boundaries, understanding target operating ranges, and implementing precision-driven adjustments are crucial to achieving sustainable operations. Examples include:

- **Building Management Systems (BMS) Optimization:** Precision in building management systems enables fine-tuned environmental control and significant energy savings. For example, implementing defined heating and cooling set points – creating a range within which no active heating or cooling occurs – can minimize energy consumption while maintaining critical environmental conditions.
- **Qualification and Operational Ranges:** Qualification ranges are often significantly broader than operational ranges. However, operational ranges can be extended as long as they remain within the defined qualification limits. Clearly defining and carefully monitoring these operational ranges is essential to ensure consistent product quality while maximizing resource efficiency.
- **Air Change Rate Reduction:** Reducing air change rates based on precise calculations and facility needs minimizes energy consumption while maintaining controlled environments.
- **Setback of Airflow and Temperature:** Implementing setback strategies for airflow and temperature can significantly enhance energy efficiency in pharmaceutical facilities without compromising product quality or regulatory compliance. During non-operational hours or periods of reduced activity, airflow rates and temperature set points can be adjusted to less stringent conditions while still maintaining environmental control.

For example, lowering air change rates or expanding temperature ranges during unoccupied periods reduces energy demand from HVAC systems. These adjustments are carefully calibrated to ensure that critical conditions are quickly restored when full operations resume. Setback strategies, when combined with precise monitoring and control systems, contribute to operational sustainability and cost savings while maintaining compliance with industry standards.

2. Control

Understanding and managing variability enables proactive decision-making, reducing defects and resource waste. For example, real-time monitoring of water systems can optimize sanitization processes, reducing their frequency or intensity as needed. This approach not only leads to energy savings but also supports broader sustainability goals.

The development of six sigma performance in processes supports the optimal control of processes through eliminating defects and decreasing variation.

3. Continual Improvement

Ongoing asset upgrades, facility improvements, and process refinements drive better sustainability and quality outcomes over time. For example:

- Facilities and Equipment Upgrades: Replacing legacy systems with modern, efficient alternatives enhances energy efficiency and improves compliance with engineering standards
- Process Improvements: Streamlining workflows and adopting cleaner technologies contributes to reduced environmental impacts and improved operational reliability
- Product Management: Minimizing product write-offs and overproduction through optimizing inventory

4. Sustainable Supply Chains

Diversifying and localizing supply chains can reduce environmental impact and enhance supply chain resilience, supporting sustainability and uninterrupted product quality. Engaging suppliers in sustainability efforts is essential but can be challenging due to differing priorities, capabilities, and levels of commitment. To address this, companies can develop clear sustainability criteria for supplier selection and establish long-term partnerships with suppliers who share their sustainability goals. Providing training and resources to suppliers can also help them improve their sustainability practices. Sustainable procurement and sustainable supply chains are discussed in Chapter 6.

5. Packaging Innovations

Using renewable or lightweight materials in packaging can reduce waste and environmental impact without compromising functionality or compliance.

6. Circular Economy Practices

Practices such as solvent recovery, water reuse, and waste recycling support sustainability while maintaining operational control and cost-efficiency.

7. Digital Transformation

Real-time monitoring and analytics streamline quality assurance and track sustainability metrics, enabling proactive improvements that benefit both areas.

8. Cross-Functional Collaboration

Collaborative efforts between Quality and sustainability teams drive innovation by integrating environmental considerations into process design while ensuring safety and efficacy remain priorities. When investigating energy-saving opportunities in manufacturing equipment, the greatest efficiencies are often found by examining subsequent process steps.

For example, to reduce water usage in a facility, it is essential to analyze where and how the water is consumed. In the case of pure water, such as WFI, which incurs the highest costs and energy demands, cleaning processes typically account for the largest share of consumption. Optimizing these processes can lead to substantial reductions in water usage.

Effective optimization requires cross-functional collaboration, as expertise from multiple departments is critical to success. For example, engineering's "water experts" may identify reduced water consumption as a key achievement. These savings might stem from process improvements made by "cleaning experts." To ensure alignment and motivation, such achievements must be recognized and rewarded as a team effort by management.

Furthermore, Quality representatives play a vital role in evaluating and managing the risks associated with such changes. For example, transitioning from hot WFI (70°C–80°C) to ambient-temperature WFI (20°C) for the final rinse in equipment cleaning could impact the dissolution rate of certain impurities. Thorough assessments must be conducted to confirm the process remains safe and effective before implementation. As emphasized in Section 5.1, all departments must collaborate and agree on any process changes. For additional guidance, refer to the *ISPE Good Practice Guide: Membrane-Based Water for Injection Systems* [124], which offers insights into optimizing WFI usage while maintaining compliance and efficiency.

By leveraging these synergies, pharmaceutical organizations can align sustainability with quality objectives, paving the way for innovations that enhance patient care, operational resilience, and environmental stewardship. Together, these efforts demonstrate the industry's commitment to delivering safe, effective products while reducing its environmental footprint.

5.2.2 Risk Management

Sustainability does not inherently equate to increased risk. By leveraging established risk management principles within Regulatory CMC – such as QbD – Quality teams can systematically evaluate and mitigate potential compliance challenges associated with sustainability initiatives. These robust frameworks ensure that sustainable practices are implemented in a controlled, scientifically sound manner, safeguarding both product quality and patient safety while achieving environmental objectives.

A critical focus should be placed on validating CQAs that directly impact product safety, efficacy, and compliance, rather than treating all requirements as equally critical. For example, cleanroom air change rates may be optimized to reduce energy consumption, provided that essential CQAs, such as particulate control and sterility assurance, remain uncompromised. This targeted validation approach enables the implementation of sustainable solutions while maintaining strict adherence to core quality standards.

Rigorous testing and validation are vital to ensuring that sustainable innovations align with regulatory requirements and quality expectations. LCAs can further support these efforts by quantifying the long-term environmental and operational benefits of sustainability measures, often demonstrating that the associated initial investments and validation efforts are well justified.

Emerging technologies such as Computational Fluid Dynamics (CFD) analysis, AI, and digital twins offer valuable tools for identifying and controlling risks. These technologies enable predictive modeling and real-time monitoring, enhancing both the efficiency and reliability of sustainable practices.

To foster broader adoption, organizations should actively participate in industry collaborations to establish standardized approaches for validating and implementing sustainable practices. Such initiatives not only help mitigate risks but also promote consistency, innovation, and alignment across the pharmaceutical sector, driving progress toward a more sustainable future. There are also benefits from joining Public-Private Partnerships (PPP) that are available to the industry. Good risk management is effectively supported by good change management and effective leadership.

Together, people and processes can mitigate risk to patients and meet sustainability goals. Changes in processes and operating practices can impact people in several ways, which need to be managed. For example, there may be employee resistance to changes in working practices unless the rationale is effectively communicated by leadership and employees have the opportunity participate in the change development.

5.2.3 Conflict Management

It can be perceived that in ensuring product quality, operational and sustainability ambitions always create a conflict or misalignment between them. Even though there can be occasions where this is true, it is often the result of a difference in viewpoint, language, or communications and a failure to take a system-level approach to the problem.

An example in biotechnology is the decision between the use of single-use (disposable) process equipment and standard stainless reusable/cleanable equipment. Both solutions have advantages and disadvantages in each of the key functional areas of quality, operations, and sustainability. Which is better over the long-term is dependent on many factors and a judgment will need to be made. Whichever choice is made, it is essential that all parties understand why the decision was reached at that time and work together to mitigate any potential negative impact.

Organizations should ensure that all strategic, operational, and functional aims are fundamentally, at a base level, aligned – i.e., everyone is striving to make the “system” better to deliver the optimal outcomes for the patient. Often this is portrayed in the organization’s core value statements. One of the reasons why conflicts may arise can be the lack of system-level thinking. When a system-level approach to problems is taken, it becomes easier to see how entrenched positions or fixed views of the right answer may not be optimal. Compromise can be sought, issues can be resolved or mitigated through creative problem solving, allowing long-term planning to drive toward a sustainable future.

Conflicts may occur in various scenarios, for example:

- **Local Codes versus Sustainability Objectives:** Local building codes in some regions may mandate the use of specific materials for pharmaceutical manufacturing facilities that are not environmentally friendly (e.g., concrete with high carbon emissions), conflicting with a company’s sustainability goal to use low carbon or recycled materials in construction.
- **Quality – Sterility Goals versus Environmental Goals:** Ensuring the sterility and stability of a pharmaceutical product often requires multi-layered plastic and foil packaging, which conflicts with sustainability objectives to reduce waste and move toward recyclable or biodegradable materials.
- **Quality – Purity Goals versus Environmental Goals:** Producing high-purity APIs may involve processes that generate significant chemical waste or require high water usage, clashing with the goal to minimize environmental impact through green chemistry. In this case there may be opportunities to review recycling of water to minimize water use, for example.

Strategies that may be helpful to employ in managing conflict:

- Multidisciplinary teams, collaboratively working to create the right solutions
- Embed sustainability system-level thinking in all business decisions

- Propose questions that help frame the problem – e.g., how can “this goal” be achieved and also “that goal?”
- Actively seek to understand other perspectives
- Avoid compromising on the values of the organization

In sustainability, as in many other areas, achieving absolute perfection is not feasible. However, this does not diminish the importance of striving for continual improvement. The goal should be to achieve meaningful progress today and aim for incremental advancements tomorrow. This approach requires effective communication, a willingness to seek compromises, and skillful negotiation, all while being guided by the core values of the organization. These principles serve as a compass for navigating the complexities of sustainability and driving consistent, value driven progress.

5.2.4 Sustainable Practices with the QMS

1. QMS Considerations

A QMS is integral to achieving and balancing both quality and sustainability objectives within the pharmaceutical industry. By embedding lean practices, robust supplier management, and effective auditing, QMS can align operational excellence with environmental and social responsibility. Regardless of the contents it may have, the QMS has a role in supporting sustainable practices alongside its essential goal in Quality management.

Efficient Practices

- **Streamlined Processes:** QMS promotes process optimization and waste reduction through lean methodologies, minimizing resource consumption (e.g., energy, water, raw materials) while maintaining high-quality outputs.
- **Continual Improvement:** Tools such as Six Sigma or Kaizen can drive ongoing improvements, reducing inefficiencies that contribute to environmental and operational waste.
- **Efficient Documentation:** Transitioning to digital QMSs reduces paper usage and enhances real-time data tracking.

Supplier Management

- **Sustainability-Driven Procurement:** QMS integrates sustainability criteria into supplier selection, ensuring vendors meet environmental and ethical standards alongside quality requirements.
- **Risk-based Approaches:** Regular risk assessments of suppliers promote compliance with fair-trade and sustainable practices, reducing the likelihood of environmental or social violations in the supply chain.
- **Partnerships for Improvement:** QMS encourages collaboration with suppliers to adopt green practices, such as reducing packaging waste or sourcing sustainable raw materials.

Auditing and Compliance

- **Integrated Audits:** QMS enables comprehensive audits that assess both quality and sustainability performance, ensuring alignment with regulatory and environmental standards. These audits require a multifunctional team to work in collaboration to cover all aspects of compliance.
- **Proactive Issue Resolution:** Early detection of noncompliance through audits minimizes disruptions and supports preventive action, reducing environmental and quality risks.
- **Alignment with Standards:** Certifications such as ISO 14001 [114] and ISO 9001 [115] can be integrated, streamlining efforts toward dual objectives.

Data-Driven Decision-Making

- **Monitoring and Reporting:** QMS provides robust tools for tracking quality and sustainability metrics, helping organizations identify improvement areas and make informed decisions.
- **Lifecycle Assessments:** By incorporating lifecycle thinking, QMS supports product designs and processes that minimize environmental impact while ensuring quality.

A well-implemented QMS not only safeguards product quality but also advances sustainability goals by driving efficiency, enforcing ethical practices in the supply chain, and fostering a culture of continual improvement. By integrating lean principles, effective supplier management, and rigorous auditing, QMS becomes a powerful enabler of sustainable and high-quality operations. There can be resistance to change within the organization, as employees and stakeholders may be accustomed to existing processes. Overcoming this resistance requires effective change management strategies and clear communication. Companies can foster a culture of sustainability by incorporating sustainability goals into their corporate values and providing ongoing education and training to employees. Recognizing and rewarding sustainable practices can also motivate employees to embrace change.

2. Quality Control (QC) Practices

Analytical Methods

Points to consider:

- Use of materials such as solvents and reference standards that may be difficult to recycle should be considered when thinking about sustainability and connecting it with quality aspects.
- The time-to-result of the analytical methods that are used and how it affects efficiency and use of materials.

The faster a product can be tested, and a result can be achieved, whether for QC purposes or for in-process control purposes, the better. Achieving faster testing results for in-process controls means that, if an intermediate must be rejected, the error can be found faster, leading to less value generation before scrapping the product and thus less waste. Achieving faster results during QC product release testing means shorter lead and delivery times, less required storage space, and smaller facility area footprint. Consequently, there is also a longer shelf life left for the patient and less product to be discarded at home.

The pharmaceutical industry still relies heavily on end-product testing to release products for the intended markets. This is due to various factors, such as existing registration dossiers, change management processes, and often also a conservative approach by the industry to keep perceived patient and financial risk low. However, many past and future projects exist with regards to in-process control testing, such as using Near-Infrared Spectroscopy (NIR) in granulators, raw material identification, and many other processes.

To summarize, shortening the duration of any decision to be taken in the process leads to improvements in terms of cost, material usage, and final carbon footprint.

5.3 Interaction Between Sustainability and Regulatory Management

Regulatory affairs and quality assurance play a crucial role in ensuring that sustainable medicines meet all necessary standards and regulations. Regulatory affairs tasks include preparing, compiling, and submitting regulatory applications for various regions, a significant effort for global products that may have marketing authorizations in as many as 100 countries.

5.3.1 Global Regulatory Trends

Regulatory authorities worldwide are increasingly prioritizing sustainability as part of their frameworks, reflecting the growing recognition of environmental and social impacts in healthcare and pharmaceuticals. Initiatives such as the European Green Deal [125], the US FDA's sustainability-focused guidelines [126], and global adoption of the UN Sustainable Development Goals (SDGs) [14] are driving stricter regulations on waste management, carbon emissions, and the environmental impact of drugs and packaging. Authorities are also emphasizing transparency through reporting requirements, such as sustainability disclosures and carbon accounting. This shift encourages the pharmaceutical industry to adopt greener practices, such as using eco-friendly materials, reducing resource consumption, and improving supply chain sustainability, while maintaining compliance with evolving standards.

5.3.2 Regulatory Flexibility

Changes in product design can significantly impact regulatory compliance. Validation and qualification processes may be affected, requiring that the new design does not compromise the product's safety, efficacy, or quality. Compliance with both internal and external requirements must be ensured, involving updates to documentation and adherence to all relevant standards and guidelines. The impact on quality must be assessed, including any changes to the manufacturing process, equipment, or materials, to ensure that the product continues to meet all quality standards. A thorough risk assessment is necessary to identify any new risks introduced by the design change, evaluating the potential impact on the product's safety, performance, and regulatory compliance. Changes in product design often require updates to regulatory submissions, providing detailed information about the changes, their impact, and any additional testing or validation that has been conducted.

Quality assurance is essential to ensure that sustainably produced medicines meet all regulatory and quality standards. Changes in packaging design often require updates to product labeling and artwork, ensuring that the labeling meets global requirements for readability, prominence, and accuracy. Regulatory guidelines for sustainable packaging focus on reducing environmental impact while maintaining product safety and efficacy. This includes using renewable or recycled materials (whenever local markets allow it), optimizing materials and energy use, and ensuring effective recovery and utilization in a circular economy. Regulatory changes can influence packaging design, material choices, sterilization methods, and labeling techniques. For example, Extended Producer Responsibility (EPR) laws require packaging producers to manage the entire lifecycle of their products, emphasizing end-of-life management. Ensuring that packaging changes do not compromise product quality or safety is crucial, involving rigorous testing and validation of new packaging designs to meet regulatory standards.

At times, regulatory requirements may conflict with sustainability goals. Regulatory frameworks increasingly provide flexibility to encourage the adoption of innovative, sustainable methods without compromising product safety or efficacy. For example, agencies such as the US FDA and EMA support the use of green chemistry principles in pharmaceutical manufacturing. These agencies may approve novel processes, such as continuous manufacturing, which reduces waste, energy consumption, and production costs compared to traditional batch processes. Another example is the acceptance of recyclable or biodegradable packaging materials, provided they meet safety and stability requirements. This regulatory flexibility fosters innovation, allowing companies to align sustainability goals with compliance obligations.

5.3.3 Dossiers, Regulatory Aspects, and Product Lifecycle

All the abovementioned circumstances can and will help any pharmaceutical enterprise on its road to sustainability. Nevertheless, all pharmaceutical products need a registration dossier for each country in which they are marketed. In such dossiers, detailed information is laid down on how to manufacture and how to test those products. The pharmaceutical facility needs to adhere to what is written in those documents, and this is regularly inspected by authorities. The level of detail described in a dossier may vary slightly between different products and dosage forms, but typically, the information level is quite detailed.

It is generally recommended to keep descriptions in product details rather generic, if possible, and to avoid patented technologies or trademarks. Moreover, unnecessary detailed process descriptions may eliminate the possibility for energy savings. For example, the description of a “unidirectional flow with an air flow speed of 0.45 m/s” in a sterile filling unit will not only lead to challenges in manufacturing but also eliminate the possibility to operate the air flow at, e.g., 0.4 m/s, which would be acceptable according EU GMP Annex 1 [127] while still maintaining “unidirectional properties.”

Other examples include the use of “distilled WFI” or the use of nitrogen as an “excipient” due to gas overlay in sterile vials. Additional information can be found in the *ISPE Good Practice Guide: Process Gases (Second Edition)* [129].

During changes to manufacturing processes, it is advisable to involve representatives from operations, engineering, and quality in the creation of data provided to regulatory affairs for new or modified registration dossiers.

5.3.4 Regulatory Innovation

5.3.4.1 ICH and Global Alignment for Sustainability

ICH [92] supports sustainable practices by promoting harmonized guidelines that reduce redundancies in pharmaceutical development and manufacturing. For example, ICH Q8 to Q12 emphasize QbD, continual manufacturing, and lifecycle management, which improve resource efficiency, minimize waste, and enhance environmental sustainability. Global alignment through ICH reduces duplication of testing and regulatory submissions across regions, lowering the environmental impact associated with lengthy approval processes and repetitive studies. By fostering collaboration and streamlined practices, ICH enables the industry to adopt innovative, sustainable methods while maintaining consistent global quality standards.

5.3.4.2 Sustainability-Driven Innovation

Managing changes in CMC processes is inherently complex, and introducing sustainability adds another layer of complexity. This requires careful planning, coordination, and communication across various departments and stakeholders. To overcome this, organizations can establish cross-functional teams dedicated to sustainability initiatives, ensuring that all relevant parties are involved in the decision-making process. Implementing robust change management frameworks can also facilitate smoother transitions. The collaborative approach facilitates predictable high performance and better alignment with sustainability goals.

One of the primary challenges in integrating sustainability into CMC change management is ensuring that initiatives comply with stringent regulatory requirements. Changes to manufacturing processes, materials, or packaging must meet regulatory standards, which can be complex and time-consuming. To address this, companies can engage with regulatory bodies early in the process to seek guidance and ensure that sustainability goals align with regulatory expectations. Additionally, maintaining transparent communication and documentation can help streamline the approval process.

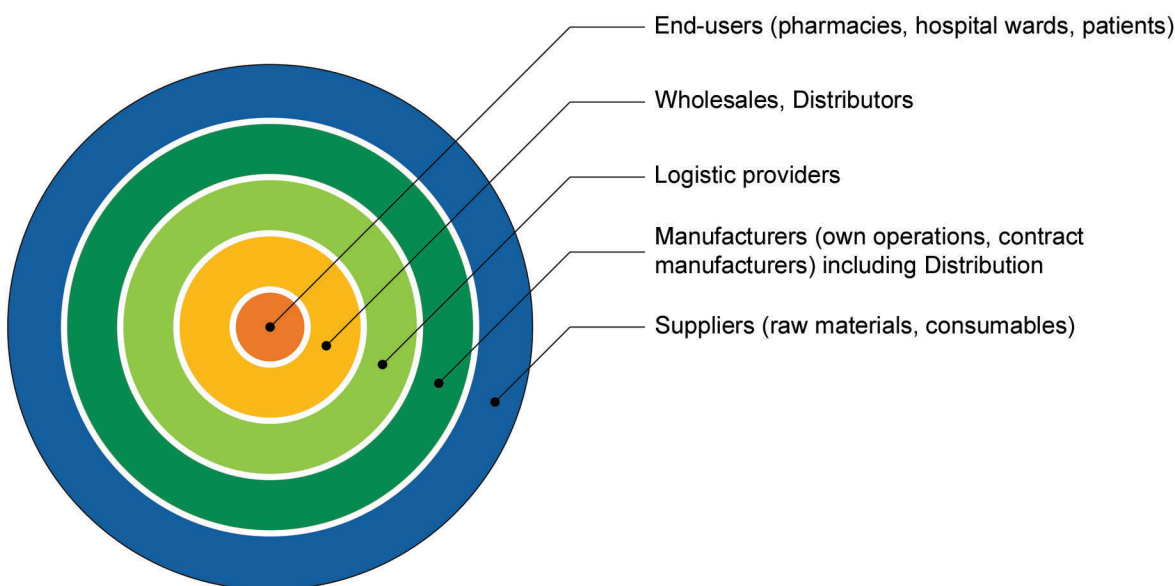
6 Sustainability within the Supply Chain

6.1 Introduction

Environmental sustainability within the pharmaceutical supply chain is crucial for mitigating environmental impacts and ensuring long-term viability. According to McKinsey's research, the supply chain is responsible for 90% of companies' environmental impact [73]. By far, the biggest factor impacting a company's carbon footprint is Scope 3 emissions – those embedded both upstream and downstream in a company's supply chain [74].

Sustainability in the supply chain refers to efforts made by organizations to identify, assess, and improve their environmental, financial, and social aspects, ensuring supply chain integrity to prevent counterfeits and uninterrupted delivery of medicines. This includes a comprehensive approach involving all stakeholders: suppliers (raw materials, consumables), manufacturers (own operations, contract manufacturers, and distribution network of materials within the manufacturing steps), logistic providers for the pharmaceutical product last mile, wholesalers, distributors, and end users (pharmacies, hospital wards, patients) (see Figure 6.1).

Figure 6.1: Stakeholder Landscape Influencing Supply Chain Sustainability



6.2 Sustainability-Driven Supply Chain Optimization

In the context of sustainable supply chain management, several practices can be implemented to enhance sustainability in the following essential areas: procurement, distribution and logistics, as well as use of medicines. Sustainable procurement refers to reducing emissions of harmful gases while producing and transporting goods and ensuring suppliers and partners do the same. Distribution and logistics sustainability practices can focus on implementing less carbon-intensive transportation methods, such as sea freight instead of air freight, and optimizing distribution routes. A more complex solution involves using less energy-intensive refrigeration distribution, which must ensure the quality of the medicines. Additionally, reducing packaging where feasible is another aspect of sustainable distribution. However, the most significant impact in the supply chain may be achieved through optimizing the use of medicines.

6.2.1 Sustainable Sourcing and Procurement

6.2.1.1 Sustainable Sourcing and Contractual Obligations

Sourcing materials and products from sustainable sources involves prioritizing suppliers that utilize renewable resources, maintain low carbon footprints, and practice sustainable sourcing within their own supply chains. Embedding sustainability clauses in contracts with suppliers ensures they are legally obligated to meet specific sustainability targets. These clauses can specify GHG reduction targets, waste management practices, and sustainable packaging requirements, thereby fostering a more environmentally responsible supply chain.

6.2.1.2 Collaborative Initiatives and ESG Reporting

Collaborative initiatives such as the Pharmaceutical Supply Chain Initiative (PSCI) [18] play a pivotal role in promoting sustainability within the supplier landscape. PSCI provides member companies with shared resources and best practices to enhance environmental performance across the supply chain. Additionally, guidance from the World Business Council for Sustainable Development (WBCSD) [31] aids pharmaceutical companies in adopting Science Based Targets for GHG reduction, ensuring transparent and accountable practices. Research indicates that publicly traded companies tend to have higher maturity in ESG target-setting compared to privately and family-owned companies. Despite challenges with standardized data use and reporting within the ESG framework, ESG reporting can improve supply chain transparency, reduce risks, and enhance efficiency [75].

6.2.1.3 Incorporating Sustainability into Procurement Policies

Procurement policies play a crucial role in ensuring that sustainability criteria are met. These policies require suppliers to demonstrate their commitment to sustainability by adhering to environmental regulations, using eco-friendly materials, and minimizing energy consumption, water usage, and waste generation. Suppliers can commit to various carbon reduction targets, such as providing Scope 1, 2, and 3 carbon footprints, setting baselines for reducing GHG emissions, and committing to increasing the use of renewable electricity. Key procurement practices include incorporating sustainability criteria into supplier evaluation and selection processes, ensuring that suppliers meet environmental standards by assessing their environmental management systems, GHG emissions, energy use, and waste management practices.

6.2.2 Distribution and Logistics Sustainability Practices

In the context of Life Cycle Analysis [61], distribution refers to the stage where materials and products are transported from the suppliers or own manufacturing site to various distribution centers or retailers, encompassing all intermediate transportation steps necessary for the pharmaceutical product's existence. This stage includes the energy requirements and emissions generated by transportation among subsystems, as well as the disposal of waste. Transportation data is reported in miles or kilometers shipped, which are then converted into units of ton-miles or ton-kilometers, reflecting the weight of the shipment and the distance shipped.

Logistics, particularly last mile logistics, involves the final stage of delivery where products are transported from a local distribution center to the end customer, focusing on efficient and timely delivery to the final destination. Distribution and logistics sustainability often involves optimizing delivery routes and utilizing sustainable transportation methods.

Pharmaceutical companies can optimize their distribution footprint by consolidating orders for raw materials, APIs, and packaging supplies. For example, placing a single, large order with a supplier, rather than multiple smaller ones, can reduce the frequency of shipments, thereby decreasing the number of vehicles, aircraft, and ships in operation, and subsequently lowering GHG emissions.

For companies managing their own distribution networks, emissions can be minimized by improving the efficiency of product shipments to warehouses, pharmacies, and healthcare facilities. Utilizing advanced transportation management systems to consolidate freight loads into fewer vehicles can significantly cut emissions. Additionally, companies can explore the use of electric vehicles, alternative fuels such as biofuels, using cargo ships instead of planes, and optimizing delivery routes to further reduce their carbon footprint.

Pharmaceutical firms outsourcing their distribution can select logistics partners that prioritize sustainable transportation practices. Adopting Less-Than-Truckload (LTL) shipping, which combines shipments from multiple clients to maximize truck capacity and optimize delivery routes, can also contribute to emission reductions compared to Full-Truckload (FTL) shipping, where trucks may not be fully utilized.

A more complex sustainability improvement area involves using less energy-intensive refrigeration in distribution and logistics, while still maintaining the stringent quality standards required for medicines.

6.2.3 Reducing Environmental Impact Through Improved Product Management

The most significant impact on the pharmaceutical supply chain can be achieved through optimizing product management by improving inventory management, minimizing recalls, extending expiration dates, and ensuring proper disposal of drugs. Unused medicines significantly affect both the environment and public health. Improper disposal can lead to environmental contamination, while the production and distribution of these unused medicines result in unnecessary carbon emissions.

The return of unused medicines is crucial to reduce risks associated with medication accumulation in domestic environments. This system aims to mitigate dangers such as non-adherence, accidental ingestion, and improper disposal. Patients or their representatives are responsible for returning unused medications to pharmacies, which can be burdensome without a well-designed system. Enhancements such as home collection or community drop-off points can increase patient engagement and compliance, ensuring safe disposal and minimizing pharmaceutical waste.

Additionally, packaging and distribution systems can be adjusted to sell medicines in quantities that match required treatment dosages, and public information should be updated to provide accurate dosing methods. Efficient dosing systems can be used, whether administered by healthcare professionals or for domestic use. Manufacturers should also provide information on proper disposal methods.

Optimizing the pharmaceutical supply chain involves leveraging advanced technologies such as AI, the Internet of Things (IoT), and blockchain to achieve real-time visibility, predict demand fluctuations, monitor transportation conditions, and ensure product authenticity. These innovations can significantly enhance the efficiency and sustainability of the supply chain.

6.2.3.1 The Role of Supplier Partnerships in Optimization of the Pharmaceutical Supply Chain

It can be challenging to assess and track the climate and nature impact on the value chain that is out of the direct control of a pharmaceutical company due to the numerous parties and processes involved. Attention should initially be given to first-tier suppliers to identify the biggest opportunities upon which to focus improvement efforts. After this, second- and third-tier suppliers can be prioritized where they have a particular role to reduce climate impact and improve nature associated with a finished pharmaceutical product.

The influence that an individual pharmaceutical company has on a supplier can be very limited when they represent only a small proportion of that supplier's business. In these situations, it is worthwhile to engage in collaborative initiatives with other companies that have aligned interests to increase leverage on that supplier (pending compliance with anticompetition laws and suitable confidentiality agreements).

6.2.3.2 Scorecards – Benchmarks – Improvement Tools to Optimize Supply Management

The EcoVadis' ESG platform [101], with its standard performance scorecards, benchmarks, and performance improvement tools, is a good example of how many companies are settling on a common approach. The use of such platforms and supporting tools makes processes more cost-efficient for suppliers too, reducing the number and variety of assessments received and progress reports requested from different customers, and for different supplier manufacturing sites related to the same customer.

6.2.3.3 Strategic Partnerships to Optimize Supply Management

Strategic partnerships provide another means to help suppliers that might otherwise not be able to access green initiatives. For example, the Energize initiative involving Schneider Electric and 10 pharmaceutical companies is aimed at facilitating access to the green energy market for PPAs for hundreds of small to medium-sized suppliers that might not otherwise be readily available to them [15]. The program is a first-of-its-kind effort to leverage the scale of a single industry's global supply chain in a precompetitive fashion to drive system-level change.

6.2.3.4 Collaborative Sustainability Integrated Approach – Human Rights-Quality-Health-Safety

Environmental sustainability, of course, is just one factor considered when engaging suppliers. Many pharmaceutical companies are integrating sustainability as one of a number of topics, such as human rights and quality, health, and safety, to be evaluated and managed together within an integrated approach. A single supplier audit may cover multiple topics instead of having a separate audit for each topic, and the results of that audit shared within a customer community with appropriate confidentiality. The procurement department needs to work closely with the business, including the quality department and other relevant functions, to understand the consequences of serious deficiencies and agree on action plans.

Some changes can take a long time to implement, especially if regulatory approvals are needed. Ultimately, if a supplier is unable to meet a customer's sustainability expectations or requirements for another topic, and they cannot address the root cause behind that, then an alternative supplier might be sought. Care must be taken to ensure selected alternatives do not bring other more pressing problems to the supply chain, such as compromised product quality. Early decisions are therefore needed on critical supplier relationships so that actions can be completed in a timely manner without compromising the pace of overall sustainability transformation.

6.3 Clinical Supply Chain

The healthcare sector significantly contributes to the global carbon footprint, and clinical trials are no exception. One of the first steps in making clinical trials more sustainable is to assess their carbon footprint. Tools such as the Clinical Trial Carbon Calculator [116], developed by the Sustainable Healthcare Coalition, provide indicative numbers on the carbon emissions associated with specific activities related to a clinical trial.

Areas to assess in a clinical trial include activities at the clinical trial site, patient travel to the site, the planning and monitoring activities from the sponsor, activities related to tissue sampling, shipment, evaluation, storage, and destruction and, finally, the Investigational Medicinal Products (IMPs) and other materials used in a clinical trial as well as their supply chain activities such as storage and recycling/destruction.

Many of these activities are related and dependent on each other, thus all strategies must consider all effects and side-effects. While focusing on the supply chain-related activities in this Guide, it needs to be realized that any decision has an impact on many other activities. So, a full analysis of the carbon footprint of the entire clinical trial is needed to use meaningful levers in the setup strategy for a clinical trial.

It is important to note that sustainability considerations are usually an additional aspect to be included when balancing the cost incurred to meet the clinical trial's robustness, stability, patient and site centrality, and overall quality during the design and the conduct of a clinical trial (especially its IMP and other materials supply chain). In many cases, environmental benefits go along with cost benefits, and sometimes with patient centric aspects, and in most situations a careful balance is needed.

Collaboration across departments and with external partners is essential for developing a robust framework and strategy for measuring and improving the sustainability performance of clinical trials.

6.3.1 Main Contributors to the Carbon Footprint in a Clinical Trial

Across clinical trials, from a single site Phase I (trials with few participants over few days) to large Phase III trials (multiple years with many sites and countries with thousands of patients), the share of the activities varies greatly.

Activities associated with patient travel and sponsor's planning, traveling, monitoring, and DP including shipment may all have shares up to 50% of the trial budget, depending on the nature of the trial and the DP. Detailed data is not yet widely available; however, efforts are currently in process for analyzing completed clinical trials or survey/monitor ongoing trials for patient travel, DP and materials used.

The DP contribution depends very much on the product itself. For example, chemical API have in their manufacture on average an emission of about 1 ton of CO₂ equivalents per kg API manufactured while biological API (e.g., mAB) have emissions in the range of 10–100 ton CO₂ equivalents per kg API manufactured [117]. Therefore, any strategy for a clinical trial should consider the specifically determined value or at least proximity data for a DP type to take the right levers.

A dedicated tool for the more specific determination of the carbon footprint of a clinical trial is essential to balance all aspects and determine a successful strategy already in the planning phase of a clinical trial, possibly with designing various scenarios.

6.3.2 Tool Development for the Carbon Footprint

Key features for a tool include but are not limited to:

- Reporting of GHG emissions (CO₂e)/carbon footprint/LCA-data
- Declaration of hot spots for CO₂e to determine reasonable levers to lower the footprint
- Allow inclusion of trial specific data for API and DP in use
- Include typical data for packaging configuration, transportation mode, shipping lanes
- Consider waste destruction
- Allow scenario discussions for clinical trial planning teams to optimize the carbon footprint in balance with patient centricity, costs, trial robustness and other business needs
- Graphical display of results and scenarios

6.3.3 Strategies for More Sustainable Clinical Trial Supply Chains

Optimizing the design and execution of clinical trials can lead to significant environmental benefits and cost savings. Key strategies include:

- **For Patient and Trial Site Staff Travel:** Utilizing decentralized trial models and telemedicine can minimize the need for travel and should be considered especially for long distances; however, this needs to be balanced with the need for dispensing trial drug and other materials to the patient.
- **For Sponsor Staff Travel:** Implementing electronic data capture and remote monitoring reduces sponsor employee travel. Even in the planning phase, virtual and hybrid meetings minimize travel, especially in large multinational trials. This approach may be applied to investigator kick-off meetings as well, while maintaining a diligent balance between sponsor/investigator engagement and travel savings.

- **For the DP:** An essential consideration is the balance of amounts being produced for a clinical trial and the amounts shipped to the sites or regional/local depots. For high carbon footprint biologicals, which typically are expensive and unique, it can be meaningful to consider a packaging and distribution strategy with many single shipments to trial sites. Chemicals with a low carbon footprint, possibly well available, can be shipped efficiently in larger amounts to local depots, if not to sites.
- **For the Transportation of DP and Other Materials:** The mode of transportation (e.g., air, ship, or road), the distance of transport and the conditions (e.g., ambient, cold, or frozen) are essential for any strategy, and should be in line with the overall considerations mentioned above. Strategies should involve evaluation and balance of compilation of shipments across trials, accepting longer shipping times via sea freight by upfront timeline setting and use of advanced IRT-systems, possibly including simulation software, for optimized distribution schemes. Attention needs to be paid to the cold and frozen shipments with their need for insulating material and the possible return shipment/reuse of materials/containers.
- **For Eco-Friendly Packaging:** The avoidance of mixed material and especially any kind of plastic needs to be considered. Low weight and low volume are a goal in packaging design to reduce packaging materials and the transport and storage impact (which goes along with trial site interests). Substituting PFAS in devices and packaging is essential.
- **The Reduction of Labeling:** This not only avoids mixed materials but also reduces costs and workload for packaging operations. The reduction of labeling to the regulatory/legal minimum and the addition of electronic provision of further data and information via QR-codes or any other means supports this and may offer further opportunities in patient focus, e.g., via further languages or additional information to caregivers.
- **For Sampling and Sample Logistics:** Strategies need to consider the source of materials (local or central) as well as location of laboratories (local or central). An option can be the compilation of sample shipments across trials or, from a site perspective, across sponsors, for the same central laboratory. It is important to minimize the frequency of sample taking as it reduces material use and the volume of samples to be stored and shipped. Further, it allows remote patient visits if no other intervention/dispensing at site occurs.
- **For many other materials, including consumables, background medication and devices,** strategies should include local sourcing on demand unless the trial specifics require a central source. For materials such as monitoring devices at sites or with patients, wearables, smartphones, etc., reuse should be part of the strategy. The same applies for background and other commercially available medication, which should be considered. For example, donation programs may be considered if it is not possible to use the material in other trials.

6.4 Ethical Considerations in Supply Chain Sustainability

Regulatory pressures are driving companies to more comprehensively report their sustainability impacts. Similarly, there is increasing attention on corporate governance and human rights alongside environmental sustainability. Governments are enacting legislation to curb environmental impacts, resulting in restrictions on GHG emissions and urging companies to integrate these issues into supply chain management.

6.4.1 Transparency in Emissions Reporting Across the Supply Chain

Concrete reporting requirements, such as the Non-financial Reporting Directive [118], HGB §289c Abs. I and II [119], and EC Guidelines on non-financial reporting in Europe [83], have focused on GHG emissions. In the US, the Securities and Exchange Commission (SEC) will impose more stringent sustainability disclosure requirements by 2025. The EU's Corporate Sustainability Reporting Directive (CSRD) [7], implemented in 2024, mandates target-setting disclosure. Additionally, countries in the Asia-Pacific region, such as New Zealand, India, and Thailand, are tightening regulations on sustainability impact disclosure [75].

Transparent emission reporting is crucial for accountability and stakeholder trust. Companies are required to publicly set and commit to SBTi [15] for reducing their GHG emissions. Obtaining SBTi verification for near-term and long-term net-zero targets underscores their commitment to mitigating climate change.

Challenges persist in standardizing data usage and ensuring comparability and meaningfulness of sustainability ratings. However, efforts toward digitalization, such as the adoption of Pharma™ 4.0 [128] technologies and blockchain, can enhance transparency and improve reporting accuracy. Transparency in ESG efforts can ultimately lead to reduced supply chain risks and improved overall efficiency.

6.4.2 Supplier Audits and Ethical Compliance

Supplier audits play a critical role in ensuring ethical compliance within the pharmaceutical supply chain. These audits evaluate suppliers against established standards for labor practices, human rights, environmental sustainability, and regulatory adherence. Ethical compliance includes verifying the absence of modern slavery, child labor, and unsafe working conditions, while also ensuring suppliers follow sustainable practices such as waste management and responsible sourcing of raw materials. Regular, risk-based audits help identify and address noncompliance issues, fostering accountability and driving improvements across the supply chain. By embedding ethical standards into supplier relationships, companies promote transparency, sustainability, and social responsibility.

6.4.3 Human Rights and Fair-Trade Practices

In the context of trade, the UN Guiding Principles on Business and Human Rights provides an international framework that outlines how businesses must respect human rights across their operations and supply chains [77]. This includes, for example, ensuring freely chosen employment, prohibiting child labor, promoting nondiscrimination, and upholding freedom of association [78].

One of the key expectations is to conduct human rights due diligence. This involves identifying, preventing, and mitigating potential human rights adverse impacts. To effectively manage these risks and comply with regulatory requirements, companies should integrate continual risk management, conduct regular audits, and embed human rights clauses into contractual agreements with suppliers to ensure that human rights are respected throughout the value chain.

Evolving regulatory requirements are also shaping the landscape of human rights and fair-trade practices. For example, the EU Corporate Sustainability Due Diligence Directive (CSDDD) requires companies to conduct human rights and environmental due diligence [79]. This is complemented by sector-specific regulations, such as the European Union Deforestation Regulation (EUDR), which mandates that companies verify their products do not come from areas contributing to deforestation or biodiversity loss, protecting both the environment and affected communities [80]. Additionally, the Carbon Border Adjustment Mechanism (CBAM) introduces new reporting and cost mechanisms to control the carbon intensity of imported goods, while the EU Import Ban Regulation prevents the import and export of goods produced with forced or child labor [81, 82].

These evolutions along with guidance from initiatives such as the PSCI [18] and the World Business Council for Sustainable Development (WBCSD) [31] supports companies to set Science Based Targets for reducing GHG emissions while maintaining transparency and accountability in their practices.

6.4.4 Corporate Social Responsibility (CSR) and Stakeholder Engagement

CSR in the pharmaceutical industry reflects a company's commitment to societal well-being, environmental stewardship, and ethical business practices. Effective CSR initiatives align with sustainability objectives by addressing key issues such as improving global health equity, reducing environmental impact, and fostering community development. For example, programs to expand access to essential medicines in underserved regions not only enhance public health but also reinforce a company's ethical responsibilities.

Stakeholder engagement is central to successful CSR efforts, as collaboration with patients, employees, regulators, suppliers, and local communities ensures that initiatives address real-world needs. This engagement is often closely tied to other sustainability strategies, such as emissions reporting and ethical supplier audits, which enhance transparency and accountability. By integrating stakeholder feedback into policies, companies can strengthen trust, build partnerships, and align their CSR goals with broader societal priorities.

Pharmaceutical companies have linked CSR with their supply chain sustainability by embedding human rights principles into supplier contracts and supporting local healthcare systems. These efforts demonstrate how CSR can create a positive social impact while maintaining operational efficiency and compliance. Ultimately, robust CSR strategies, reinforced by transparent stakeholder engagement, help companies foster resilience, enhance reputation, and contribute meaningfully to global sustainability goals.

In Europe, corporate responsibility in this context is already legally enforced by Directive (EU) 2024/1760 [79], the CSDDD, which aims to foster sustainable and responsible corporate behavior in companies' operations and across their global value chains.

7 Appendix 1 – References

1. Whitford, A., and Barle, A.L., “Rethinking Overall Sustainability: A Systems-Based Approach,” *Journal of Sustainability Studies*, 2023, Issue 12, No. 3, pp. 45–67, <https://journalofsustainability.net/>.
2. Draft of Seventh Assessment Report 2024, Intergovernmental Panel on Climate Change (IPCC), United Nations (UN), www.ipcc.ch.
3. “Circular Economy Introduction,” Ellen MacArthur Foundation, www.ellenmacarthurfoundation.org.
4. Atkins, P.A., Gerber, M.S., and King, K.J., “Directors’ Oversight Role Today: Increased Expectations, Responsibility and Accountability—A Macro View,” *Harvard Law School Forum on Corporate Governance*, 10 May 2021, <https://corpgov.law.harvard.edu/2021/05/10/directors-oversight-role-today-increased-expectations-responsibility-and-accountability-a-macro-view/>.
5. Financial Reporting Council, “The UK Corporate Governance Code,” April 2016, www.frc.org.uk/getattachment/ca7e94c4-b9a9-49e2-a824-ad76a322873c/UK-Corporate-Governance-Code-April-2016.pdf.
6. European Commission, “Corporate Sustainability Due Diligence, How to integrate human rights and environmental concerns in value chains,” Proposal for EU Directive, October 2023, www.europarl.europa.eu.
7. Directive (EU) 2022/2464 of the European Parliament and of the Council of 14 December 2022 amending Regulation (EU) No 537/2014, Directive 2004/109/EC, Directive 2006/43/EC and Directive 2013/34/EU, as regards corporate sustainability reporting, *Official Journal of the European Union*, <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32022L2464>.
8. GHG Reduction Programs & Strategies, US Environmental Protection Agency (EPA), www.epa.gov.
9. SSBJ issues Inaugural Sustainability Disclosure Standards to be applied in Japan, Sustainability Standards Board of Japan (SSBJ), Accounting Standards Board of Japan (ASBJ), Financial Accounting Standards Foundation (FASF), www.ssb-j.jp/en/ssbj_standards/2025-0305.html.
10. The State Council, “China charts path to unified sustainability disclosure by 2030,” The People’s Republic of China, 27 May 2024, Updated 28 May 2024, https://english.www.gov.cn/news/202405/28/content_WS6655152bc6d0868f4e8e78b8.html.
11. “National Greenhouse and Energy Reporting Scheme,” Australian Government, Clean Energy Regulator, 24 July 2025, <https://cer.gov.au/>.
12. EMA, Committee for Medicinal Products for Human Use, “Guideline on the Environmental Risk Assessment of Medicinal Products for Human Use,” EMEA/CHMP/SWP/4447/00 corr 2, 1 June 2006, European Medicines Agency (EMA), www.ema.europa.eu/en/documents/scientific-guideline/guideline-environmental-risk-assessment-medicinal-products-human-use-first-version_en.pdf.
13. Wingate, G.A.S., Ek, Y., and Melton, T., “Sustainability: Corporate Ambition, Governance, and Accelerated Delivery,” *Pharmaceutical Engineering*, March/April 2023, Vol. 43, No. 2, pp. 30–36, www.ispe.org.
14. UN, “Transforming Our World: The 2030 Agenda for Sustainable Development,” Department of Economic and Social Affairs: Sustainable Development, United Nations (UN), October 2015, <https://sdgs.un.org/2030agenda>.
15. Science Based Targets Initiative, www.sciencebasedtargets.org.

16. Science Based Targets Initiative, The Corporate Net-Zero Standard, Version 1.2, March 2024, www.sciencebasedtargets.org.
17. Hargreaves, L., "Schneider Electric: Decarbonizing the Healthcare Industry," *Scope 3 Magazine*, 30 October 2024, <https://scope3magazine.com/transportation-and-logistics/airbus-omv-how-biomass-can-make-aviation-sustainable>.
18. PSCI, "Decarbonisation playbook for the Pharmaceutical Industry," October 2024, Pharmaceutical Supply Chain Initiative (PSCI), <https://pscinitiative.org/>.
19. Endangered Species Act, U.S. Fish & Wildlife Service, www.fws.gov.
20. "Federal Government partnership strengthens fight against deforestation in 70 priority Amazon municipalities," 22 March 2025, updated 26 March 2025, Secretaria de Comunicação Social, Brazil, www.gov.br/planalto/en.
21. National Biodiversity Authority, Ministry of Environment, Forest and Climate Change, Government of India, <http://nbaindia.org/>.
22. The Convention on Biological Diversity, Text of the Convention, 13 May 2016, <https://www.cbd.int/convention/text>.
23. Taskforce on Nature-related Financial Disclosures (TNFD), <https://tnfd.global/>.
24. GRI 101: Biodiversity 2024, Amsterdam, the Netherlands, Global Reporting Initiative (GRI), 2024, www.globalreporting.org.
25. Directive (EU) 2022/2464 of the European Parliament and of the Council of 14 December 2022 amending Regulation (EU) No 537/2014, Directive 2004/109/EC, Directive 2006/43/EC and Directive 2013/34/EU, as regards corporate sustainability reporting, *Official Journal of the European Union*, <https://eur-lex.europa.eu/eli/dir/2022/2464/oj/eng>.
26. European Financial Reporting Advisory Group (EFRAG), European Sustainability Reporting Standard E4: Biodiversity and Ecosystems, Brussels: EFRAG, 2023, www.efrag.org/sites/default/files/sites/webpublishing/SiteAssets/ESRS%20E4%20Delegated-act-2023-5303-annex-1_en.pdf.
27. Kalundborg Symbiosis, www.symbiosis.dk/en/.
28. Task Force on Climate-Related Financial Disclosures, "Annex: Implementing the Recommendations of the Task Force on Climate-related Financial Disclosures," October 2021, <https://tnfd.global/>.
29. Task Force on Nature-Related Financial Disclosures, "The TNFD Nature-Related Risk and Opportunity Management and Disclosure Framework, Version v.03 Beta Release. November 2022, <https://tnfd.global/>.
30. Wingate, J., Lee, S., and Chen, M., "Corporate Governance in the Post-Pandemic Era," *Journal of Business Ethics*, Vol. 178, No. 2, 2023, pp. 245–268.
31. World Resources Institute and World Business Council for Sustainable Development. "The Greenhouse Gas Protocol: A Corporate Accounting and Reporting Standard," March 2004, <https://ghgprotocol.org/sites/default/files/standards/ghg-protocol-revised.pdf>.
32. *ISPE Good Practice Guide: Project Management for the Pharmaceutical Industry*, International Society for Pharmaceutical Engineering (ISPE), First Edition, November 2011, www.ispe.org.
33. PRINCE2® (PRejects IN Controlled Environments), www.prince2.com.

34. Program Management Institute (PMI), <https://pmi.org>.
35. Wrampe, J., and Carlson, C., "Benefits of membrane filtration systems for ambient WFI production," CRB, Accessed 26 July 2025, www.crbgroup.com.
36. "Renewables 2022: Analysis and Forecast to 2027," International Energy Agency (IEA), Revised version January 2023, www.iea.org.
37. Aqueduct Water Risk Atlas, World Resources Institute (WRI), www.wri.org.
38. EFPIA, White paper on circular economy, European Federation of Pharmaceutical Industries and Associations (EFPIA), 26 April 2024, www.efpia.eu/news-events/the-efpia-view/efpia-news/white-paper-on-circular-economy/.
39. ACS, 12 Principles of Green Chemistry, American Chemical Society (ACS), www.acs.org.
40. DOZN™ Quantitative Green Chemistry Evaluator Tool, Millipore Sigma, www.sigmaaldrich.com/US/en.
41. "Test No. 249: Fish Cell Line Acute Toxicity – The RTgill-W1 Cell Line Assay," *OECD Guidelines for the Testing of Chemicals*, Section 2, Organisation for Economic Co-operation and Development (OECD), 18 June 2021, www.oecd.org.
42. Manzano, T., and Whitford, W., "Artificial Intelligence in Drug Target Discovery," *Pharmaceutical Engineering*, January/February 2025, Vol. 45, No. 1, pp. 64–73, www.ispe.org.
43. Cielecka-Piontek, J., et al., "Solid-State Stability Study of Meropenem-Solutions Based on Spectrophotometric Analysis," *Chemistry Central Journal*, 08 June 2013, Vol. 7, No. 98, <https://link.springer.com/content/pdf/10.1186/1752-153x-7-98.pdf>.
44. Cell Culture Dish Inc., "Advances in Sustainable Oligo Manufacturing," *cellculturedish.com*, 24 January 2024, <https://cellculturedish.com/advances-in-sustainable-oligo-manufacturing/>.
45. Guiotto, M., Raffoul, W., Hart, A.M., et al., "Human platelet lysate to substitute fetal bovine serum in hMSC expansion for translational applications: a systematic review," *J Transl Med*, Vol. 18, Article No. 351, 15 September 2020, <https://doi.org/10.1186/s12967-020-02489-4>.
46. Plastics Technology, "Top 10 Breakthrough Technologies for Plastic Recycling", plastics-technology.com, Accessed 28 July 2025, www.plastics-technology.com/articles/top-10-breakthrough-technologies-for-plastic-recycling.
47. Schug, K. A., and Kaplitz, A., "Plastic pyrolysis – chemists explain a technique attempting to tackle plastic waste by bringing the heat," *The Conversation*, 18 March 2025, Updated 20 March 2025, <https://theconversation.com/plastic-pyrolysis-chemists-explain-a-technique-attempting-to-tackle-plastic-waste-by-bringing-the-heat-234453>.
48. Kosana, P., Živanović, S., and Jevtić, I., "Biopolymer Packaging Materials in the Pharmaceutical Industry," *AIDASCO Reviews*, 2024, Vol. 2 No. 1, pp. 46–56, www.pharmaexcipients.com/wp-content/uploads/2024/02/Biopolymer-Packaging-Materials-in-the-Pharmaceutical-Industry.pdf.
49. FDA Draft Guidance: Considerations for the Use of Artificial Intelligence to Support Regulatory Decision-Making for Drug and Biological Products, January 2025, US Food and Drug Administration (FDA), www.fda.gov.
50. My Green Lab®, My Green Lab's Converge Initiative: Driving Collaborative Supply Chain Engagement, <http://mygreenlab.org>.
51. NIH Green Labs Program, US National Institutes of Health (NIH), www.nih.gov.

52. Constable, D.J.C., Jimenez-Gonzalez, C., and Henderson, R.K., "Perspective on Solvent Use in the Pharmaceutical Industry," *Org. Process Res. Dev.*, 14 December 2006, Vol. 11, Issue 1, pp. 133–137, <https://pubs.acs.org/doi/10.1021/op060170h>.
53. Xinyi, L., Kuchinsky, L.M., et al, "Enzyme Purification and Sustained Enzyme Activity for Pharmaceutical Biocatalysis by Fusion with Phase-separating intrinsically Disordered Protein," *Biotechnology and Bioengineering*, Vol. 121, Issue 10, pp. 3155–3168, <https://doi.org/10.1002/bit.28787>.
54. "Revolutionizing Pharmaceutical Manufacturing: The Role of Process Analytical Technology in Minimizing Waste," Tablets and Capsules, 01 April 2024, www.tabletscapsules.com/3641-Technical-Articles/612086-Revolutionizing-Pharmaceutical-Manufacturing-The-Role-of-Process-Analytical-Technology-in-Minimizing-Waste/.
55. "Process Analytical Technology: Enhancing Pharma Development", Adragos Pharma, 06 September 2024, <https://adragos-pharma.com/>.
56. Berkeley, C., Berridge, J., and Manley, J.B., "PAT and Green Chemistry: The Intersection of Benign by Design and Quality by Design," *Pharmaceutical Engineering*, March/April 2009, Vol. 29, No. 2, pp. 8–20, www.ispe.org.
57. Barle, L.E., Melton, T., and Eamon, J., "Sustainability by Design for Pharmaceutical Products," *Pharmaceutical Engineering*, March/April 2023, Vol 43, No. 2, pp. 14–20, www.ispe.org.
58. "SuperPro Designer Overview," Intelligen, Inc., www.intelligen.com/products/superpro-overview/.
59. "LCA software for informed changemakers," Simapro, <https://simapro.com/>.
60. Biopharm, BioSolve *Process*™, <https://biopharmservices.com/biosolve-software/biosolve-process/>.
61. ACS Green Chemistry Pharmaceutical Roundtable, PMI Life Cycle Assessment, American Chemical Society (ACS), <https://acsgcipc.org/tools/pmi-life-cycle-assessment/>.
62. Budzinski, K., et al., "Introduction of a process mass intensity metric for biologics," *N Biotechnol*, 25 Mar 2019, Vol. 49, pp. 37–42, <https://pubmed.ncbi.nlm.nih.gov/30121383/>.
63. Cataldo, A.L., Sissolak, B., et al., "Water related impact of energy: Cost and carbon footprint analysis of water for biopharmaceuticals from tap to waste," *Chemical Engineering Science: X*, 2020, Vol. 8, www.sciencedirect.com/science/article/pii/S2590140020300290.
64. Cataldo A.L., et al., "Economics and ecology: Modelling of continuous primary recovery and capture scenarios for recombinant antibody production," *J Biotechnol*, 20 January 2020 Jan 20;Vol. 308, pp. 87–95, <https://pubmed.ncbi.nlm.nih.gov/31809781/>.
65. EPA, Scope 3 Inventory Guidance, US Environmental Protection Agency (EPA), www.epa.gov.
66. ECHA, "Substances of Concern: Why and how to Substitute?" European Chemicals Agency (ECHA), <https://echa.europa.eu/>.
67. Regulation (EU) 2024/1781 of the European Parliament and of the Council of 13 June 2024 establishing a framework for the setting of ecodesign requirements for sustainable products, amending Directive (EU) 2020/1828 and Regulation (EU) 2023/1542 and repealing Directive 2009/125/EC, *Official Journal of the European Union*, <https://eur-lex.europa.eu/eli/reg/2024/1781/oj/eng>.
68. OECD, *Guidance on Key Considerations for the Identification and Selection of Safer Chemical Alternatives*, OECD Series on Risk Management of Chemicals, Organisation for Economic Co-operation and Development (OECD), 4 March 2021, www.oecd.org.

69. ECHA, Strategy to promote substitution to safer chemicals through innovation, European Chemicals (ECHA), January 2018, <https://data.europa.eu/doi/10.2823/99862>.
70. European Commission, Tools and Standards, “EU taxonomy for sustainable activities,” Accessed 28 July 2025, https://finance.ec.europa.eu/sustainable-finance/tools-and-standards/eu-taxonomy-sustainable-activities_en.
71. “Sustainable Development Goals: 17 Goals to Transform our World,” United Nations, (UN), Accessed 28 July 2025, www.un.org.
72. ICH Harmonised Guideline: *Pharmaceutical Development – Q8(R2)*, Step 5, August 2009, International Council for Harmonisation (ICH), www.ich.org.
73. Bove, A.T., and Swartz, S., “Starting at the source: Sustainability in the supply chain,” 11 November 2016, www.mckinsey.com/capabilities/sustainability/our-insights/starting-at-the-source-sustainability-in-supply-chains#.
74. Mourad, T., “Supply chain sustainability is key to achieving climate goals: Here’s how organizations can get started,” World Economic Forum, 20 April 2023, www.weforum.org.
75. Bade, C., et al., “Sustainability in the pharmaceutical industry—An assessment of sustainability maturity and effects of sustainability measure implementation on supply chain security,” *Corporate Social Responsibility and Environmental Management*, 11 July 2023, Vol. 31, Issue 1, pp. 224–242, Correction 02 July 2024, <https://doi.org/10.1002/csr.2564>.
76. Schneider-Electric Energize Program, <https://hub.zeigo.com/energize>.
77. Guiding Principles on Business and Human Rights, United Nations (UN), 2011, www.ohchr.org/sites/default/files/documents/publications/guidingprinciplesbusinesshr_en.pdf.
78. ILO, “International labour standards and human rights,” 30 June 2025, International Labour Organization (ILO), www.ilo.org.
79. Directive (EU) 2024/1760 of the European Parliament and of the Council of 13 June 2024 on Corporate Sustainability Due Diligence and Amending Directive (EU) 2019/1937 and Regulation (EU) 2023/2859, *Official Journal of the European Union*, <http://data.europa.eu/eli/dir/2024/1760/oj>.
80. Official Journal of the European Union, Regulation (EU) 2023/1115 of the European Parliament and of the Council of 31 MAY 2023 on the making available on the Union market and the export from the Union of certain commodities and products associated with deforestation and forest degradation and repealing Regulation (EU) No 995/2010, <https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32023R1115>.
81. European Commission, Carbon Border Adjustment Mechanism, https://taxation-customs.ec.europa.eu/carbon-border-adjustment-mechanism_en.
82. European Union, The European Parliament, PE-CONS 67/24, REGULATION (EU) 2024/ OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL of on prohibiting products made with forced labour on the Union market and amending Directive (EU) 2019/1937, *Official Journal of the European Union*, <https://data.consilium.europa.eu/doc/document/PE-67-2024-INIT/en/pdf>.
83. Communication from the Commission — Guidelines on Non-Financial Reporting (Methodology for Reporting Non-Financial Information), European Commission, C 215, July 5, 2017, pp. 1–20, https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=oj:JOC_2017_215_R_0001.
84. Maxwell, J., *The EDGE: How competition for resources is pushing the world and its climate to the brink – and what we can do about it*, 2023, Nicholas Brearley Publishing, ISBN 978-1-39981-0 845.

85. *ISPE Good Practice Guide: Heating, Ventilation, and Air Conditioning*, International Society for Pharmaceutical Engineering (ISPE), Second Edition, September 2024, www.ispe.org.
86. *ISPE Baseline® Pharmaceutical Engineering Guide, Volume 4 – Water and Steam Systems*, International Society for Pharmaceutical Engineering (ISPE), Third Edition, September 2019, www.ispe.org.
87. *ISPE Good Practice Guide: Ozone Sanitization of Pharmaceutical Water Storage and Distribution Systems*, International Society for Pharmaceutical Engineering (ISPE), Second Edition, October 2024, www.ispe.org.
88. Lhasa Limited, "How Derek Nexus and Sarah Nexus Meet the 5 OECD Principles," Last modified 30 November 2021, www.lhasalimited.org/blog/how-derek-nexus-and-sarah-nexus-meet-the-5-oecd-principles/.
89. *Leadscope Model Applier User Guide, Version 2025.0.x*, Leadscope, Inc., April 2025, www.leadscope.com/lis-manuals/.
90. Investment Theme – Sustainability, S&P Global, www.spglobal.com.
91. Company ESG Risk Ratings, Sustainalytics, www.sustainalytics.com.
92. International Council for Harmonisation (ICH), www.ich.org.
93. WHO GMP Guidelines, World Health Organization (WHO), www.who.int/medicines/areas/quality_safety/quality_assurance/production/en/.
94. *ISPE Guide Series*, International Society for Pharmaceutical Engineering (ISPE), www.ispe.org.
95. Biodiversity Net Gain, UK Department for Environment, Food & Rural Affairs 21 February 2023, Last updated 16 June 2025, www.gov.uk/government/collections/biodiversity-net-gain.
96. CDP Worldwide, www.cdp.net.
97. The Greenhouse Gas Protocol, <https://ghgprotocol.org/>.
98. International Organization for Standardization (ISO), www.iso.org.
99. ISO 26000:2010 Guidance on Social Responsibility, International Organization for Standardization (ISO), www.iso.org.
100. UN, *Global Water Report 2023: Accelerating Action*, 2023, www.unwater.org/publications/un-world-water-development-report-2023.
101. EcoVadis, The Global Standard for Resilient, Sustainable Supply Chains, 2025, <https://ecovadis.com/>.
102. Directive 2006/43/EC of the European Parliament and of the Council of 17 May 2006 on Statutory Audits of Annual Accounts and Consolidated Accounts, amending Council Directives 78/660/EEC and 83/349/EEC and repealing Council Directive 84/253/EEC, *Official Journal of the European Union*, <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32006L0043>.
103. Directive 2013/34/EU of the European Parliament and of the Council of 26 June 2013 on the annual financial statements, consolidated financial statements and related reports of certain types of undertakings, amending Directive 2006/43/EC of the European Parliament and of the Council and repealing Council Directives 78/660/EEC and 83/349/EEC Text with EEA relevance, *Official Journal of the European Union*, <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32013L0034>.

104. UN, Paris Agreement. What is the Paris agreement?, United Nations (UN), <https://unfccc.int/process-and-meetings/the-paris-agreement>.
105. European Commission, "Energy Performance of Buildings Directive," accessed 23 July 2025, https://energy.ec.europa.eu/topics/energy-efficiency/energy-performance-buildings/energy-performance-buildings-directive_en#documents.
106. EU Regulation (EU) 2024/3110 of the European Parliament and of the Council of 10 April 2024 laying down harmonised conditions for the marketing of construction products and repealing Regulation (EU) No 305/2011, *Official Journal of the European Union*, https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=OJ:C_202501296.
107. CIBSE, *TM65: Embodied Carbon in Building Services – A Calculation Methodology*, Chartered Institution of Building Services Engineers (CIBSE), 2021, www.cibse.org.
108. "Water-Efficient Technology Opportunity: Steam Sterilizer Condensate Retrofit Kit," Federal Energy Management Program (FEMP), US Department of Energy, www.energy.gov/femp/water-efficient-technology-opportunity-steam-sterilizer-condensate-retrofit-kit.
109. ISO 50001:2018 Energy Management Systems – Requirements with guidance for use, International Organization for Standardization (ISO), www.iso.org.
110. Communication from the Commission, Ecodesign for Sustainable Products and Energy Labelling Working Plan 2025–2030, European Commission, 16 April 2025, <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A52025DC0187>.
111. Regulation (EC) No 1907/2006 of the European Parliament and of the Council of 18 December 2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH), establishing a European Chemicals Agency, amending Directive 1999/45/EC and repealing Council Regulation (EEC) No 793/93 and Commission Regulation (EC) No 1488/94 as well as Council Directive 76/769/EEC and Commission Directives 91/155/EEC, 93/67/EEC, 93/105/EC and 2000/21/EC, *Official Journal of the European Union*, <https://eur-lex.europa.eu/eli/reg/2006/1907/oj/eng>.
112. Regulation (EC) No 1272/2008 of the European Parliament and of the Council of 16 December 2008 on classification, labelling and packaging of substances and mixtures, amending and repealing Directives 67/548/EEC and 1999/45/EC, and amending Regulation (EC) No 1907/2006, and Amending Regulation (EC) No 1907/2006, *Official Journal of the European Union*, <https://eur-lex.europa.eu/eli/reg/2008/1272/oj/eng>.
113. Directive 2002/95/EC of the European Parliament and of the Council of 27 January 2003 on the restriction of the use of certain hazardous substances in electrical and electronic equipment, *Official Journal of the European Union*, <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32002L0095&qid=1753839702518>.
114. ISO 14001:2015 Environmental management series, International Organization for Standardization (ISO), www.iso.org.
115. ISO 9001:2015 Quality management systems – Requirements, International Organization for Standardization (ISO), www.iso.org.
116. Pistoia Alliance, Clinical Trial Carbon Calculator, Pistoia Alliance, Inc., www.pistoiaalliance.org.
117. My Green Lab®, 2023 Carbon Impact of Biotech & Pharma Report: Collective Action Accelerating Progress to the UN Race to Zero, www.mygreenlab.org.

118. EU Directive 2014/95/EU of the European Parliament and of the Council of 22 October 2014 amending Directive 2013/34/EU as regards disclosure of non-financial and diversity information by certain large undertakings and groups, *Official Journal of the European Union*, <http://data.europa.eu/eli/dir/2014/95/oj>.
119. Germany, *Handelsgesetzbuch* [Commercial Code], § 289c Abs. 1–2. Last amended by Art. 1 G v. 11.12.2023 I 3516, Accessed 19 July 2025, www.gesetze-im-internet.de/hgb/_289c.html.
120. Wernet, G., et al., "The ecoinvent database version 3 (part I): Overview and methodology," *The International Journal of Life Cycle Assessment*, Vol. 21, No. 9, pp. 1218–1230, <https://doi.org/10.1007/s11367-016-1087-8>.
121. ISO 14040:2006 Environmental management — Life cycle assessment — Principles and framework, International Organization for Standardization (ISO), www.iso.org.
122. ISO 14044:2006 Environmental management — Life cycle assessment — Requirements and guidelines, International Organization for Standardization (ISO), www.iso.org.
123. Intergovernmental Panel on Climate Change (IPCC), United Nations (UN), www.ipcc.ch.
124. *ISPE Good Practice Guide: Membrane-Based Water for Injection Systems*, International Society for Pharmaceutical Engineering (ISPE), May 2022, www.ispe.org.
125. The European Green Deal, European Commission, https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/european-green-deal_en.
126. US Food and Drug Administration (FDA), www.fda.gov.
127. EudraLex Volume 4 – Guidelines for Good Manufacturing Practices for Medicinal Products for Human and Veterinary Use, Annex 1: Manufacture of Sterile Medicinal Products, August 2022, http://ec.europa.eu/health/documents/eudralex/vol-4/index_en.htm.
128. Pharma™ 4.0, International Society for Pharmaceutical Engineering (ISPE), www.ispe.org.
129. *ISPE Good Practice Guide: Process Gases*, International Society for Pharmaceutical Engineering (ISPE), Second Edition, May 2022, www.ispe.org.

8 Appendix 2 – Glossary

8.1 Acronyms and Abbreviations

ACT	Accountability, Consistency, and Transparency
AD	Anaerobic Digestion
ADME	Absorption, Distribution, Metabolism, and Excretion
AHU	Air Handling Unit
AI	Artificial Intelligence
AMR	Antimicrobial Resistance
API	Active Pharmaceutical Ingredient
BMS	Building Management System
BNG	Biodiversity Net Gain (UK)
BPA	Bisphenol A
BSC	Balanced Scorecard
CBAM	Carbon Border Adjustment Mechanism
CCF	Corporate Carbon Footprint
CDP	Carbon Disclosure Project
CEEL	Center for Energy-Efficient Laboratories
CFD	Computational Fluid Dynamics
CLP	Classification, Labeling and Packaging
CMC	Chemistry, Manufacturing, and Controls
CMR	Carcinogenic, Mutagenic or Reproductive
CMYK	Cyan, Magenta, Yellow, and Key (Black)
CPP	Critical Process Parameter
CQA	Critical Quality Attribute
CS	Clean Steam
CSDDD	Corporate Sustainability Due Diligence Directive
CSR	Corporate Social Responsibility
CSRD	Corporate Sustainability Reporting Directive
DOE	Design of Experiment
DP	Drug Product
ECHA	European Chemicals Agency
EDI	Electrodeionization
EED	Energy-Efficient Design
EMA	European Medicines Agency

EP	Environmental Protection Agency (US)
EPD	Environmental Product Declaration
EPR	Extended Producer Responsibility
ERA	Environmental Risk Assessment
ESG	Environment, Social, and Governance
ESPR	Eco-design for Sustainable Products Regulation
ESRS	European Sustainability Reporting Standards
EUDR	EU Deforestation Regulation
EUI	Energy Use Intensity
EU	European Union
FBS	Fetal Bovine Serum
FSC	Forrest Stewardship Council
FTL	Full-Truckload
GC	Gas Chromatography
GHG	Greenhouse Gas
GLP	Good Laboratory Practice
GMP	Good Manufacturing Practice
GRI	Global Reporting Initiative
GWP	Global Warming Potential
HEPA	High-Efficiency Particulate Air
hMSC	human Mesenchymal Stem Cells
HPLC	High Performance Liquid Chromatography
HSE	Health, Safety and Environment
ICH	International Council for Harmonization
IDP	Integrated Design Process
IMP	Investigational Medicinal Product
IoT	Internet of Things
IPCC	Intergovernmental Panel on Climate Change
KPI	Key Performance Indicator
LC	Liquid Chromatography
LCA	Life Cycle Assessment
LCA	Life Cycle Analysis
LCCA	Life Cycle Cost Analysis
LCIA	Life Cycle Impact Assessment
LTL	Less-than-Truckload
mAb	monoclonal Antibody

ML	Machine Learning
MS	Mass Spectrometry
MSA	Mean Species Abundance
MS-GC	Mass Spectrometry-Gas Chromatography
MS-LC	Mass Spectrometry-Liquid Chromatography
NIH	National Institutes of Health (US)
NIR	Near-Infrared
NOR	Normal Operating Range
OECD	Organisation for Economic Co-operation and Development
OpEX	Operational Excellence
PAT	Process Analytical Technology
PCF	Product Carbon Footprint
PET	Polyethylene Terephthalate
PFAS	Polyfluoroalkyl Substance/Perfluoroalkyl Substance
PMI	Process Mass Intensity
POP	Persistent Organic Pollutant
PPA	Power Purchase Agreement
PPP	Public-Private Partnerships
PSCI	Pharmaceutical Supply Chain Initiative
PV	Photovoltaic
PW	Purified Water
QbD	Quality by Design
QC	Quality Control
QMS	Quality Management System
R&D	Research and Development
REACH	Registration, Evaluation, Authorization and Restriction of Chemicals
RO	Reverse Osmosis
RoHS	Restriction of the use of certain hazardous substances in electrical and electronic equipment (EU)
SbD	Sustainability by Design
SBSC	Sustainability Balanced Scorecard
SBT	Science-Based Target
SBTI	Science-Based Target Initiative
SBTN	Science-Based Target Network
SDG	Sustainable Development Goal
SEC	Securities and Exchange Commission
SMART	Specific, Measurable, Achievable, Relevant, and Time-bound

SoC	Substances of Concern
SVHC	Substances of Very High Concern
TCFD	Task Force on Climate-Related Financial Disclosures
TCO	Total Cost of Ownership
TNFD	Taskforce on Nature-related Financial Disclosures
TSA	Toxic Substances Control Act (US)
UF	Ultrafiltration
UN	United Nations
UPLC	Ultra Performance Liquid Chromatography
USP	United States Pharmacopeia
UV	Ultraviolet
VAV	Variable Air Volume
WARIEN	Water Related Impact of Energy
WBCSD	World Business Council for Sustainable Development
WFI	Water for Injection
WHO	World Health Organization



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